The Price of the Fence: Land Rationing around China's Export Processing Zones

Kai Gu*

Abstract

Place-based policies concentrate activity in chosen places, and their costs are usually sought in land prices and wages. China's export processing zones took land from local development zones that allocate industrial land by administrative rule at fixed prices. With customs and manufacturing-survey records on 0.1-degree cells for 1998–2007, we compare the surroundings of each zone with those of special customs zones opened after 2007. Near the zones, output and wages show no detectable increase, land prices did not rise around later zones, and entry of new domestic establishments fell in a step across productivity. A model in which the parent zone admits applicants by coarse tiers reads the price of the fence from that step: the entry turned away was worth 3.5 billion yuan a year, 22.3 percent of applicants' value and close to random rationing. A price for the same land would recover about half.

1 Introduction

Place-based policies concentrate economic activity in chosen places, and whether the gains from concentration exceed the distortions it creates is the classic question about them (Kline and Moretti 2014). The distortions are usually sought in prices: in land values that capitalize the subsidies, in wages that rise for local workers, in firms that move from one place to another (Busso et al. 2013; Gaubert 2018). This paper studies a policy whose main distortion leaves no trace in prices. China's export processing zones (EPZs) concentrated foreign assembly behind customs fences inside local development zones, and they took their land from development zones that allocate industrial land by administrative rule at prices that do not clear the market. When land is rationed rather than priced, the cost of a zone falls on the firms that

*University of Wisconsin–Madison. Job market paper.

are turned away, and it shows up only in who is admitted. We measure that cost, the price of the fence, and set it against the zone's gains.

The setting makes the question sharp. Between 2001 and 2007, China opened 52 EPZs. Their firms imported inputs free of duty, assembled them and exported the output; in 2006, 588 fenced firms exported about US$54 billion, 5.6 percent of China's exports. Behind the fence they bought and sold little locally, the mark of an enclave rather than an engine of local development (Rodríguez-Clare 1996). Each fence sits inside a parent development zone run by the host prefecture, which allocates its remaining industrial land to applicants one by one, under minimum standards of investment and output per hectare and at administered prices (Section 2). The fence therefore took land that the parent zone would otherwise have allocated to other firms, and the parent zone decided who would bear the loss.

We place customs records and the annual survey of manufacturing firms on a grid of 0.1-degree cells and estimate how manufacturing outside the fence changes after the first opening within 150 km, at each distance from the zone. Zones were placed in places that were already industrial (Figure 1), so a comparison with places that never received a zone would credit the zones with the faster growth of industrial places. We compare instead the rings around each zone with the same rings around special customs zones of the same program that opened after 2007, with effects for each ring and year (Section 3).

The first set of findings concerns the land next to the fence. Within 10 km, output outside the fence changed by 0.04 (0.05) over the first three years, with flat leads. Employment rose by 0.15 (0.06), but the wage per worker did not, and employment rose by similar amounts farther away. Around later zones of the same program, industrial land did not become dearer. What changed was who entered. Entry of new domestic establishments in the three lowest productivity fifths fell by -0.65, -0.22 and -0.61 log points, while entry of the top fifth did not fall, 0.23 (0.20). A fall in demand would have cut entry from the bottom in a ramp; a step of this kind is what rationing by coarse tiers produces.

The growth that accompanied the zones lay farther away. Output rose by 0.13 (0.06) 40 to 60 km from the zones and by 0.10 (0.04) 60 to 100 km away. The design locates this growth but does not attribute it to the zones. One channel that we can trace connects it to them: exporters 30 to 100 km away entered products related to those that fenced firms had exported the year before, more than products related to what fenced firms would export the year after (Section 5.3).

To turn the step in entry into a cost, we model the parent zone as a matchmaker. Firms choose where to produce, given the land they can obtain. The parent zone values a firm by the taxes and output it brings and observes only a coarse signal of its productivity. It admits a high tier for sure and the rest at random until its land is full, as the colleges of Fu (2014)

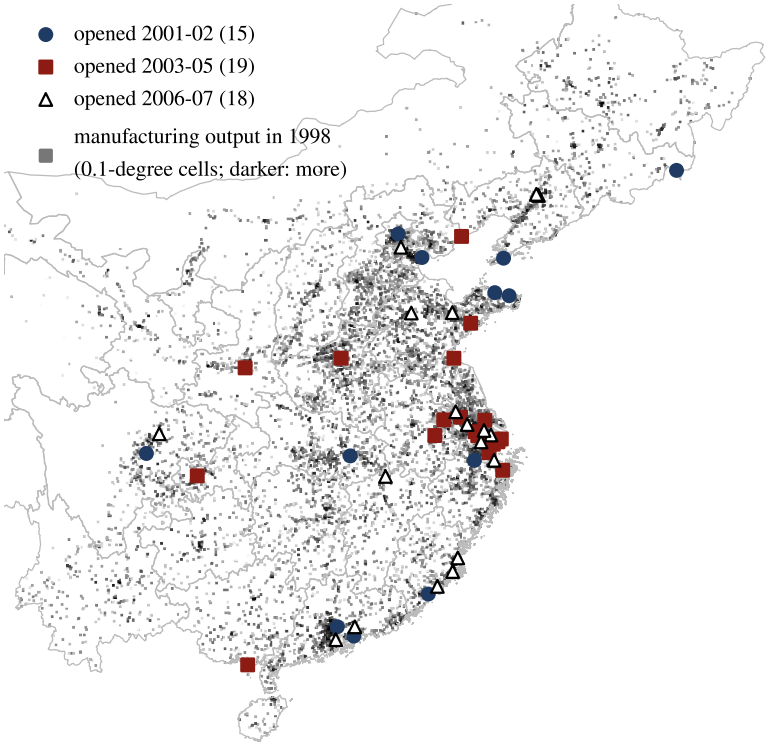


Figure 1: Export processing zones open by 2007. Symbols mark the year of customs acceptance (approval year plus two where acceptance is not recorded). Shading shows manufacturing output in 1998 in 0.1-degree cells with manufacturing above the survey threshold; darker cells produce more. The shading describes the places before the openings; it is not an estimated effect.

admit students when tuition does not clear the market. The multiplier on land then works as a price (Azevedo and Leshno 2016), and the composition of entry reveals it. Workers choose where to work, and the fence's establishments hire, pay taxes and may raise the profits of their neighbors. The model delivers three results: who gets land, the price of the fence, and the incidence of the zone's gains and costs (Section 6). Ten moments, entry and retention by productivity fifth, identify the rule up to a set; the set includes coarse tiers, a noisy score and ranking by first-year sales, while ranking by realized four-year value is rejected (Section 7).

The price of the fence is large because the rule is coarse. The applicants the parent zones turned away would have created 22.3 percent of the four-year value of all applicants, 3.5 billion yuan a year, and between 2.0 and 5.6 billion across the rules the data accept. An allocation of the same land by value would have lost only 3.7 percent, a Harberger triangle; a random allocation would have lost 29.4 percent. The estimated rule lies close to the random benchmark, because it treats the low tier as interchangeable and turns away firms of every size, including those that would have grown fastest. Selling the same land to the applicants that bid most per hectare would have recovered about half of the price, 52 percent under the best-fitting rule, and between 0 and 61 percent across the rules the data accept. The cost

falls entirely on firms that never entered: under the rule the admitted firms keep the scarcity rent, which a price would have transferred to the parent zone (Section 8).

Against this cost stand the gains of concentration. The fence's firms paid 6.1 billion yuan a year in taxes and employed about 1.25 million workers. For the area around the zones, the account is close to zero if those jobs paid no more than the workers' alternatives and positive if they paid more. For the country, it is negative unless much of the fence's activity was new to China: the zones pay for the land they took only if more than 0.72 of their activity would not have located elsewhere. The agglomeration gain to nearby firms is imprecise, with a point estimate as large as the fence's taxes and a confidence interval that includes zero.

The paper contributes to four literatures. Studies of place-based policy measure its costs through prices, such as land values, wages and the location of firms (Busso et al. 2013; Kline and Moretti 2014; Gaubert 2018); we measure a cost that arises because a price is missing, and we recover it from the composition of entry. Studies of school choice and college admissions recover the price of a seat from who is admitted when tuition does not clear the market (Fu 2014; Agarwal 2015; Azevedo and Leshno 2016). We apply that logic to a government that allocates land, and we show that, absent rationing before the fence, the rule rather than the scarcity of land accounts for up to 83 percent of the cost. Studies of land and misallocation in China document administered land prices and their consequences (Tao et al. 2010; Hsieh and Klenow 2009; Tian et al. 2024), and the political allocation of export quotas (Khandelwal et al. 2013); we value the entry that an administrative allocation of land turned away and split the cost into scarcity and the rule. Finally, for the literature on China's zones (Wang 2013; Alder et al. 2016; Lu et al. 2019), we separate the fence's own activity from its surroundings with customs identifiers and measure each channel through which a fenced zone could reach its neighbors.

2 Institutional Background

An export processing zone separates its firms from the economy around it in goods and leaves them connected through workers, ideas and land. This section describes the rules that do this, the land that the zones took and how it was allocated, and the scale of the fenced firms' activity.

An EPZ is a customs-supervised area inside an existing economic and technological development zone. Its firms import materials and components without duty or value-added tax, process or assemble them, and export the output; sales into China are treated as imports, and domestic goods sold into the zone are treated as exports and declared at customs. We call the customs boundary of an EPZ its fence and the firms inside it fenced firms. Transfers

of intermediate goods to processing firms elsewhere in China (deep-processing carry-over), piloted in two zones in 2002, were extended to all zones in 2005.

EPZs opened in three waves. Fifteen pilot zones, the first wave, were approved together in 2000 and opened in 2001–02 in the country's leading industrial and port cities. Later approvals extended EPZs to other provincial cities, with a second wave opening in 2003–05 and a third in 2006–07, and by 2007, 52 zones had opened (Figure 1).

Four channels connect an EPZ to the economy around it. Through *goods*, fenced firms could buy from local suppliers and sell to local customers; the customs rules restrict this channel by design. Through *workers*, fenced firms hire from the local labor market, which in the 2000s was fed by an elastic supply of migrants. Through *ideas*, fenced firms know products, buyers and standards that neighboring firms may acquire. Through *land and admission*, the zone occupies industrial land and draws on the local government's infrastructure budget and admission decisions. Local governments supply industrial land; the conversion of rural land to construction use is limited by annual construction-land quotas set by higher levels of government, and the zone is served from the same quotas. Admission to development zones is governed by minimum investment per unit of land. A zone can therefore change which firms are able to enter near it without changing what anyone pays for labor.

2.1 Fenced exports and the local economy

Fenced exports were large and new. In 2006 fenced firms exported about US$54 billion, 5.6 percent of China's exports; five zones account for 91 percent of fenced exports, and 87 percent of fenced export value in 2006 came from customs identifiers with no earlier customs record (Table 1, panel A; the series is in Appendix Figure I1).

The separation shows in what fenced firms sold and in what they bought (Table 1, panel B). Among fenced firms linked to the survey of manufacturers (Section 3), which account for 15.5 percent of fenced export value, the median firm sells 3 percent of its output in China, against 22 percent for processing exporters outside zones. When deep-processing carry-over was extended to all zones in 2005, the domestic-sales share of fenced firms shows no detectable change.

Sales say little about purchases, so the relevant measure is the domestic value added ratio (DVAR), domestic value added over gross exports. Following Kee and Tang (2016), the DVAR of a processing exporter is one minus its processing imports over its processing exports. It counts everything the exporter adds or buys in China, directly or through domestic suppliers, including its own wages and profits, and is therefore an upper bound on its domestic purchases. Computed from customs records for all fenced firms over the first four years of each zone,

the DVAR is 0.30 in the median zone and 0.34 weighted by exports; in the most generous variant, which drops firms that import more than they export, it is 0.57 in the median zone. Wages of fenced firms matched to the survey are about 6 percent of those firms' exports in the median zone.

In the same-product comparison the fence does not detectably lower domestic content: within six-digit product and year, the DVAR of fenced firms differs from that of processing exporters outside the fence by 0.026 (0.058). That comparison still allows the fence's procedure to add a cost of up to 0.35 in ad valorem terms to domestic inputs. Domestic content also rises with a zone's age, from 0.21 in the opening year to 0.52 in the sixth among the zones observed that long, so the low average reflects what fenced firms make, mostly computers assembled from imported parts, and how new their operations were.

2.2 The land around a zone

The industrial land next to a fence belongs to the parent development zone, and the government of the host prefecture allocates it. Land, investment promotion and infrastructure for an EPZ are the host prefecture's responsibility, so the fence occupies land and admission capacity that the prefecture could otherwise have allocated to other firms (Appendix A). The prefecture sells industrial plots at administered prices. For most of the period they were negotiated plot by plot, and from 2007 industrial land had to be sold by tender, auction or listing at no less than national minimum prices (Tian et al. 2024). Prices set this way do not clear the demand for sites, and local governments have used cheap industrial land to attract investment (Tao et al. 2010). Who obtains a plot is decided instead by admission. Parent zones screen applicants on investment and output per hectare and on the floor-area ratio of the plot, and from 2004 the central government imposed a minimum investment per hectare on development zones. The fence therefore tightens a quantity that is allocated by rule rather than by price. Where it binds, the zone can change which firms enter next to it without changing what anyone pays for land or labor; Section 4 looks for this change, and Sections 6 and 8 ask what it cost.

3 Data and Design

How does manufacturing change at each distance from an export processing zone once it opens? We answer with customs records and the annual survey of industrial firms placed on a grid of 0.1-degree cells (Section 3.1), and with an event-study design that compares the cells around each zone with cells at the same distance from zones of the same kind that opened later (Section 3.2).

Table 1: Fenced exports and their separation from the local economy

Panel A. Scale of fenced exports	
Fenced exports, 2006	US$54 billion (5.6% of China's)
Share of fenced exports in the five largest zones	91%
Share from customs identifiers new to customs	87%
Panel B. Domestic sales and domestic content: fenced firms against processing exporters outside zones	
Median domestic-sales share (fraction): fenced firms / processing exporters outside zones	0.03 / 0.22
Difference in domestic-sales shares, regression-adjusted (industry × year effects)	−0.24 (0.03)
Change after the 2005 extension of deep-processing carry-over	−0.007 (0.022)
DVAR of fenced exports (Kee and Tang 2016), median zone / export-weighted	0.30 / 0.34
Most generous variant (firms importing more than they export dropped), median zone	0.57
Fenced minus unfenced processing firms, same six-digit product and year	0.026 (0.058)
Local wages per unit of fenced exports, median zone (survey-matched firms)	0.06

DVAR is one minus processing imports over processing exports of all fenced firms, first four years of each zone, 32 zones (customs). The same-product comparison bounds each firm's DVAR to $[0, 1]$. Sales shares and wages from fenced firms matched to the survey (15.5 percent of fenced export value); the regression-adjusted difference is a mean difference conditional on industry and year, not the difference between the two medians. Standard errors in parentheses.

3.1 Data

Customs transactions (2000–2016) record every exporter's products, destinations and trade regime, and the imports and exports of the firms inside each fence. Those firms are identified by their customs identifiers (the enterprise codes under which customs records a firm's trade), which carry a code for EPZs. The annual survey of industrial firms (1998–2007) covers all state-owned firms and all non-state firms with sales above 5 million RMB (the survey threshold); it reports output, value added, employment, wage bill, fixed capital, profits, taxes, export deliveries (the value of output delivered for export), ownership and founding year. Its reporting unit is the legal-person firm; we refer to a firm at its location as an establishment.

We locate each firm by its exact coordinates where available and otherwise by the centroid of its town or county; the share of firm-years located by exact coordinates rises from 33 percent in 1998 to 90 percent in 2007 (Appendix Table D1). We assign each firm to one of 19,149 cells of 0.1 degrees and remove fenced firms, so that outcomes are cell-year totals of manufacturing (two-digit industries 13–43) outside the fence.

A domestic firm is one without foreign or Hong Kong, Macao and Taiwan capital. A *new establishment*, or entrant, is a domestic firm that first appears in the survey within a year of its founding, and entry in a cell and year is the number of its new establishments. A domestic *exit* is a firm present in year $t-1$ and absent in t, assigned to its cell in $t-1$; a firm that falls below the survey threshold also exits. Firms that first appear in 2004 but were founded earlier are *census additions*, existing firms that the 2004 economic census brought into the survey (22 percent of firm-years in 2004–07); none of the results depends on them. A zone's opening year is the year it passed customs acceptance, after which bonded operations may

begin; for zones without a recorded acceptance year (23 of the 59 EPZs in the directory, 52 of which opened by 2007), it is the approval year plus two, the median lag among the others.

3.2 Design

Zones were not placed at random. They went to places that were already industrial: nine in ten cells within 10 km of a future zone had manufacturing above the survey threshold in 1998, against fewer than three in ten of the cells never near a zone. A comparison with places that never received a zone would attribute to the zones the faster growth of industrial places in the 2000s. We therefore compare the surroundings of each zone with the surroundings of zones that the same national program placed in similar places later. China continued to open special customs zones after 2007, mostly comprehensive bonded zones, the type into which EPZs were later converted. The 69 that opened after the survey ends in 2007 (55 comprehensive bonded zones, 7 EPZs, 6 bonded ports and one bonded logistics park) were sited by the same approval process, in the same kinds of prefectures, and their surroundings are untreated in our years.

Let g_c be the year in which the first EPZ within 150 km of cell c opens and $r(c)$ the cell's distance ring from that zone: 0–10, 10–20, 20–30, 30–40, 40–60, 60–100 and 100–150 km (Figure 2). Cells within 150 km of a later zone are assigned to rings in the same way. For each outcome y_{ct}, a cell-year total of manufacturing outside the fence, we estimate

$$\mathbb{E}[y_{ct}] = \exp\left(\alpha_c + \delta_{r(c),t} + \sum_{r}\sum_{k=-4}^{3} \beta_{rk}\, \mathbb{1}\big[r(c) = r,\ t - g_c = k\big]\right) \tag{1}$$

by Poisson pseudo-maximum likelihood (Santos Silva and Tenreyro 2006), with event year $k = -1$ as reference. The cell effects α_c absorb permanent differences between places. The ring-by-year effects $\delta_{r,t}$ are shared by treated cells and cells at the same distance from a later zone, so every coefficient compares a ring around a zone that has opened with the same ring around a zone that has not yet opened. Cells that lie within 150 km of neither enter only through their own effects and do not identify the coefficients. We summarize each ring by the mean of β_{r0}, β_{r1} and β_{r2} minus half of $\beta_{r,-2}$, which nets out the trend in the two years before the opening, and report the mean of the leads $\beta_{r,-4}$ to $\beta_{r,-2}$ as a test of pre-opening trends. Coefficients are changes in logs: 0.10 is a rise of about 10 percent. Standard errors are clustered by prefecture, and statistics that combine several outcomes stack the prefecture-level influence functions of the separate regressions.

The later-treated comparison asks less of the data than a comparison with never-exposed places, and it gives up some precision. Its identifying assumption is that, absent the earlier

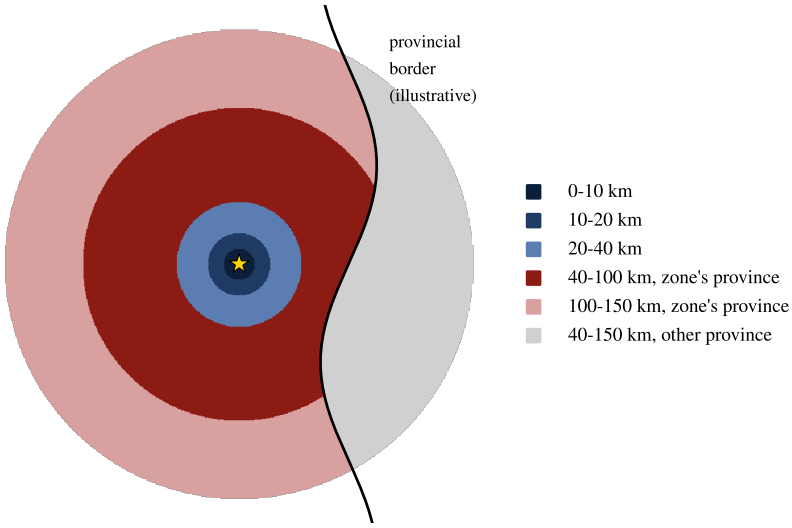


Figure 2: Distance rings around a zone (schematic). Rings around the first zone to open within 150 km of a cell; cells within 150 km of a special customs zone opened after 2007 form the same rings around that zone and serve as the comparison. Rings are discrete categories, not points on a continuous distance function; the schematic combines adjacent rings.

zone, the rings around it would have followed the same path as the rings around a zone that opened later. Two features support it. Leads are flat for output and entry within 10 km (Section 4.1), and the step in entry across productivity fifths, the central fact of the paper, compares fifths within the same cells and is unaffected by any trend common to them. Appendix G reports the main results on a comparison of matched never-exposed cells, which is more precise. The step in entry and the flat wage are common to both comparisons; the rise in output within 10 km is larger on the matched comparison and concentrated around the zones that opened into places already growing.

4 Effects at the Zone

Within 10 km of a fence, manufacturing outside it did not grow. Firms there hired more workers at unchanged wages, and the parent development zones admitted fewer new domestic establishments, with the losses concentrated in the less productive four fifths of entrants while the price of industrial land did not move. Output rose instead 40 to 100 km away. This section documents these facts on the later-treated comparison of Section 3.2: each estimate compares cells at a given distance from a zone with cells at the same distance from a special customs zone that opened only after 2007.

4.1 Within 10 km

Table 2 reports the change in each outcome after the opening, by distance from the zone. Within 10 km, output changed by 0.04 (0.05) and value added by 0.04 (0.06), with no movement before the opening: the mean lead for output is 0.00 (0.02). Fixed capital and taxes paid show no detectable change either. Production near the fence, outside it, therefore shows no growth that the comparison can separate from zero; at the upper end of its confidence interval it rose by about a tenth. The estimate stays between 0.02 and 0.07 when the zones of any one province are left out (Appendix F).

Employment rose, but not because the zone bid up wages. Employment within 10 km rose by 0.15 (0.06) and the wage bill by 0.12 (0.06), so the wage per worker shows no detectable change. The rise is not specific to the land near the fence: employment rose by similar amounts 10 to 100 km away, and within 10 km it had been falling relative to the comparison before the openings, with a mean lead of −0.11 (0.04). Measured against the mean of the years before the opening, the rise within 10 km is 0.19 (0.08), but it is fragile: treating the year before the opening as treated lowers it to 0.09 (0.06), and it does not survive any deviation from parallel trends (Appendix F). We therefore read it as a regional rise in employment, not as an effect of the zone. An elastic supply of migrant workers, which met new labor demand at the going wage, fits this pattern; Section 7 uses it to bound the elasticity of labor supply.

Timing. The leads are flat for output and for the entry of new establishments, so the estimates do not pick up growth that was already under way where zones opened. On a matched comparison, which compares near-zone cells with never-exposed cells in the same strata of region, port access and 1998 size, output within 10 km rose by about a seventh, almost entirely around the zones that opened in 2001–02 into places already growing (Appendix G). The later-treated comparison removes that difference, because the comparison cells were themselves chosen, a few years later, for the same kind of zone.

4.2 Who Enters

The zones changed who entered more than how many. The number of new domestic establishments within 10 km fell by −0.21 (0.16), about 19 percent, an estimate that cannot be distinguished from zero. Split by productivity, the change takes the form of a step. Entry of the three lowest productivity fifths fell by −0.65, −0.22 and −0.61 log points, the fourth fifth by −0.17 and the top fifth not at all, 0.23 (0.20) (Table 5). Among incumbents present in 1999–2000, the same step appears at a smaller scale: the share keeping their plots fell by −0.14 in the lowest fifth and changed by 0.03 in the top fifth. The contrast between the three lowest fifths and the top fifth, −0.72 (0.20), stays between −0.80 and −0.49 when any one

Table 2: Manufacturing outside the fence, by distance from the zone

	Distance from the zone, km					
	0–10	10–20	20–30	30–40	40–60	60–100
A. Production						
Output	+0.04	+0.01	−0.01	+0.06	+0.13	+0.10
	(0.05)	(0.04)	(0.06)	(0.06)	(0.06)	(0.04)
Value added	+0.04	+0.01	+0.01	+0.12	+0.19	+0.13
	(0.06)	(0.05)	(0.07)	(0.06)	(0.07)	(0.08)
Fixed capital	+0.02	+0.11	+0.05	−0.05	+0.20	+0.10
	(0.04)	(0.07)	(0.05)	(0.12)	(0.09)	(0.04)
Taxes	+0.05	−0.01	+0.06	+0.22	+0.21	+0.14
	(0.07)	(0.07)	(0.10)	(0.10)	(0.08)	(0.09)
B. Labor						
Employment	+0.15	+0.16	+0.06	+0.07	+0.18	+0.12
	(0.06)	(0.05)	(0.05)	(0.08)	(0.06)	(0.05)
Wage bill	+0.12	+0.10	+0.02	+0.09	+0.13	+0.06
	(0.06)	(0.06)	(0.05)	(0.08)	(0.07)	(0.05)
C. Firms and exports						
Number of firms	+0.04	+0.06	−0.05	−0.01	+0.11	+0.06
	(0.07)	(0.06)	(0.05)	(0.10)	(0.06)	(0.04)
New domestic establishments	−0.21	−0.21	−0.24	−0.15	+0.03	−0.14
	(0.16)	(0.19)	(0.11)	(0.17)	(0.19)	(0.09)
Domestic exits	+0.06	−0.03	−0.31	−0.19	−0.15	−0.03
	(0.12)	(0.10)	(0.11)	(0.13)	(0.13)	(0.09)
Export deliveries	+0.06	+0.00	+0.02	−0.03	+0.09	+0.14
	(0.11)	(0.09)	(0.07)	(0.13)	(0.08)	(0.09)
Exporting firms	+0.19	+0.11	−0.07	−0.06	+0.03	+0.10
	(0.10)	(0.09)	(0.09)	(0.20)	(0.10)	(0.06)
Mean lead, output	+0.00 (0.02)					
Mean lead, employment	−0.11 (0.04)					

Notes: Each entry is the change in the log of a cell's total after the first opening within 150 km: the mean of event years 0–2 minus half of event year −2, from equation (1) estimated by Poisson pseudo-maximum likelihood with cell effects and ring-by-year effects. The comparison is cells at the same distance from special customs zones opened after 2007. Totals exclude fenced firms. Mean lead: mean of event years −4 to −2. Standard errors clustered by prefecture in parentheses. 89 cells lie within 10 km of a zone.

province is left out and halves to −0.36 (0.14) when the year before the opening is treated (Appendix F). Its leads are not flat, so Section 7 also reports the model on the alternative baseline that subtracts the mean of the years before the opening.

The step is consistent with rationing by coarse tiers and not with a fall in demand. A fall in local demand would raise the profit a potential entrant needs to cover its fixed cost and would therefore cut entry from the bottom in a ramp, the lowest fifth falling most and each higher fifth less. Instead, the three lowest fifths fall by similar amounts and the top fifth rises. That pattern is what an allocation of land that admits a high tier for sure and rations the rest produces, the mechanism Section 6 models. Because applications and refusals are not observed, the model reads entry within 10 km as admission to the parent zone, a measurement assumption that the compactness of parent zones supports.

4.3 Land

Industrial land near the zones did not become dearer or more abundant. Land transactions are recorded only from 2007, so we test this on the 49 special customs zones that opened on new sites in 2008–2016, the zone type into which EPZs were later converted (Appendix Figure Q1). Three to 10 km from these zones, the area of industrial land sold changed by −0.19 (0.16) and its price per hectare by −0.02 (0.06), relative to land 40 to 100 km away in the same prefecture. Had a price rationed the land the fence took, the price of the remaining plots would have risen; with administered prices, the parent zone rations by admission and the transaction price need not move (Proposition 2).

4.4 Beyond 10 km

Output rose 40 to 100 km from the zones, not next to them. Figure 3 plots the estimates of Table 2 against distance. Between 10 and 40 km, output shows no detectable change. Between 40 and 60 km it rose by 0.13 (0.06) and value added by 0.19 (0.07); between 60 and 100 km output rose by 0.10 (0.04). Beyond 100 km the estimate falls back to 0.03 (0.04). This is the most robust fact in the paper: pooled over 40 to 100 km, output rose by 0.11 (0.04), between 0.08 and 0.15 when any one province is left out, and the post-opening trend would have to deviate from parallel trends by 7.9 times the largest pre-opening deviation to overturn it (Appendix F).

The design locates this growth; it does not attribute it to the zones. Cells 40 to 100 km from a zone lie in the same provinces as the zones and at the distance of the provinces' secondary industrial towns. The growth there could come from the zone's exporters, from provincial policy that accompanied the zones, or from activity that moved from elsewhere in

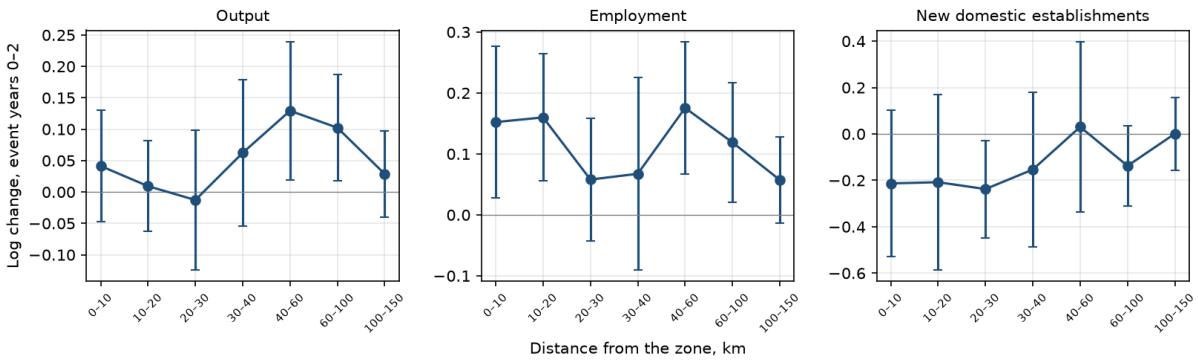


Figure 3: Output, employment and entry by distance from the zone. Estimates of Table 2 with 95 percent intervals: the mean of event years 0–2 minus half of event year −2, later-treated comparison.

China. On the matched comparison, the growth comes mostly from non-state firms expanding capacity and shows no sign of relocation within the province (Appendix G). Section 5 asks which channels could connect it to the zones; one that our data can trace is the entry of exporters into products related to those of the fenced firms, which concentrates 30 to 100 km away.

5 Mechanisms

What connected the zones to the places around them? The customs fence restricts one channel, trade in goods, and leaves three open: hiring, learning, and the land the zone occupies. We take the channels in turn. Purchases and hiring leave no distinct trace within 10 km (Sections 5.1 and 5.2); learning appears in related products 30 to 100 km away (Section 5.3); and the change in entry near the fence fits rationing of land rather than a price (Section 5.4). Section 5.5 then asks how the growth 40 to 100 km away came about. The analysis of learning uses customs records at the level of products and does not depend on a comparison of places; the finer decompositions of goods and of the hinterland's growth need the precision of the matched comparison and are reported in Appendix G.

5.1 Goods: little crosses the fence

Purchases leave no distinct trace near the fence. The customs rules make local purchases costly, and fenced firms made few: among fenced firms matched to the survey, the median sells 3 percent of its output in China, and a dollar of fenced exports carries about a third of a dollar of domestic value added (Section 2). The output of the fence's supplier industries is higher around zones with a thick base of those industries than around zones with a thin one.

A similar contrast appears when zones are split by the size of their unrelated industries, so the data do not separate purchases by fenced firms from the advantages of a thick industrial base (Appendix G.4.1).

5.2 Workers: more jobs, no detectable change in pay

If the labor demand of the zone's firms raised what neighboring employers pay, the wage per worker of manufacturing firms outside the fence within 10 km should rise. Employment there rose by 0.15 (0.06) and the wage bill by 0.12 (0.06); estimated jointly, the wage per worker changed by −0.035 (0.052). Employment rose by similar amounts 10 to 100 km away, and the wage per worker did not rise at any distance. This is the pattern of an elastic supply of migrant workers: new labor demand was met at the going wage. The measure is an average over surveyed firms, so it does not rule out higher pay for particular workers or in firms below the survey threshold. For welfare, a flat wage means that workers gained through wages neither from the fence nor from its neighbors; whether the new jobs were worth more than the workers' alternatives is a separate question, which Section 8 addresses.

5.3 Learning: related products, not copied markets

Ideas are the channel that the customs rule leaves widest open. Fenced firms hold buyer relationships, product know-how and compliance experience, and a neighboring exporter could acquire some of it in two ways: by following fenced firms into the markets they serve, or by moving into products close to what they make. The evidence favors the second, at a distance. Exporters enter products close to those that fenced firms nearby exported the year before more than products close to what fenced firms export the year after, mostly 30 to 100 km from the zones; they enter the markets that fenced firms serve in the same years as those firms do, as a common market shock would make them do. These results describe a time-ordered association consistent with learning, not a causal gain, so the welfare account of Section 8 reports this channel apart from its totals. Appendix S gives the full design and the further results.

Fenced firms were foreign assemblers in a few products, and few neighbors were directly exposed to their markets: in 2006, 99 percent of fenced firms were foreign-invested, one product line made up 61 percent of fenced exports, and only 0.17 percent of the product–destination entries open to non-fenced exporters had a fenced firm within 20 km exporting the same pair the year before.

Copied markets. Following fenced firms into their markets implies entry into the product–destination pairs that they served the year before. In a panel of non-fenced exporters, six-digit

products, destinations and years, 2002–06, entry into a pair is 2.9 percentage points (0.8) higher when a fenced firm within 20 km exported it the year before, against a baseline of 6.0 percent (Table 3, panel A). Adding the same pair's fenced exports in the following year lowers the previous-year coefficient to 0.9 points (1.1) and gives the following year 3.5 points (0.9). Neighbors and fenced firms enter the same markets at the same time, as a common market shock would make them do; we find no evidence that neighbors follow fenced firms in.

Related products. Acquiring capability implies entry into products close to what fenced firms make, wherever those products are sold. For exporter i, a four-digit product p that i has not exported since 2000, and year t, we estimate

$$E_{ipt} = \sum_{\ell} b_{\ell}^{\text{prev}} \mathbb{1}\left[\kappa_{ip,t-1} \in \ell\right] + \sum_{\ell} b_{\ell}^{\text{next}} \mathbb{1}\left[\kappa_{ip,t+1} \in \ell\right] + \sum_{m} c_m \mathbb{1}\left[o_{ip,t-1} \in m\right] + \alpha_{it} + \alpha_{pt} + u_{ipt}, \tag{2}$$

where E_{ipt} marks entry, $\kappa_{ip,t-1}$ is the proximity of p to the products that fenced firms of zones within 100 km exported in $t-1$ (the past portfolio), $\kappa_{ip,t+1}$ the same for $t+1$ (the next-year portfolio), $o_{ip,t-1}$ the proximity of p to the exporter's own products, and α_{it} and α_{pt} firm-year and product-year effects. The next-year portfolio is a timing test: regional trends that move fenced firms and their neighbors together would make it predict entry as well as the past portfolio does, while capability acquired from what fenced firms already make runs from the past portfolio.

Entry rises with proximity to the past portfolio, by 17, 51, 95 and 134 percent of the baseline rate across the four proximity bins, against −3, −4, +19 and +36 percent for the next-year portfolio (Table 3, panel B; Figure 4a). The excess of the first over the second, the asymmetry, is 76 (36) and 98 (38) percent of the baseline rate in the two highest bins.

Three qualifications bound this reading. Products that fenced firms will export for the first time next year, which the past portfolio cannot contain, also predict entry into related products, so part of the gradient reflects trends that fenced firms and their neighbors share; for fenced firms' own products, however, the past portfolio exceeds these new products by 111 percent of the baseline (27). The association holds for products that fenced firms keep exporting, 142 (39) percent of the baseline, not only for those they drop, so it does not run only through orders handed over from fenced firms. And among exporters with a zone within 100 km, the same-product asymmetry is 51 (28) percent, about half the full-sample estimate, with the point estimates largest 30 to 100 km from the zones rather than next to them (Appendix S).

The clearest evidence therefore concerns fenced firms' own products: what fenced firms exported the year before predicts entry into them better than what they will export next,

among products they keep as well as those they drop, mostly 30 to 100 km away, the range in which the hinterland grows. That points to diffusion within the region more than to learning between neighbors.

Table 3: Entry of non-fenced exporters into fenced firms' markets and into related products

Panel A. Markets: entry into a product–destination pair a fenced firm within 20 km exported (percentage points; baseline 6.0)				
	Previous year, alone	Previous year, given next year	Next year, given previous year	
Fenced exports of the pair	+2.9 (0.8)	+0.9 (1.1)	+3.5 (0.9)	
Panel B. Related products: entry into a new product, by its proximity to the fence's portfolio within 100 km (percent of baseline)				
	0.05–0.10	0.10–0.20	≥0.20	Same product
Fenced firms' exports, previous year	+17 (7)	+51 (20)	+95 (45)	+134 (39)
Fenced firms' exports, next year	−3 (3)	−4 (6)	+19 (15)	+36 (14)
Difference (percent of baseline)			+76 (36)	+98 (38)
Wild cluster bootstrap p-value			0.077	0.011
Fences' new products of the next year	−1 (5)	+29 (12)	+62 (25)	+7 (26)
Baseline entry rate (percent)	0.146			

Panel A: entry into a six-digit product–destination pair, non-fenced exporters, 2002–06; a fenced firm within 20 km exported the pair in the previous (next) year; firm × product × year and product × destination × year effects; standard errors clustered by firm. Panel B: equation (2), entry into a new four-digit product, entry years 2003–05, percent of the baseline entry rate (0.146 percent); standard errors clustered by nearest zone (36 clusters). Wild cluster bootstrap: null imposed, Rademacher weights, 9,999 draws. Last row: products that fenced firms of zones within 100 km first export in $t+1$ (not exported in $t-1$ or t), in a regression that also includes both full portfolios. The panels differ in design: panel A uses a 10 percent sample of exporters, six-digit product–destination pairs and fenced firms within 20 km; panel B a quarter of exporters, four-digit products and zones within 100 km. Other portfolios, the split by province and persistence: Appendix Table I3; checks of the comparison group, inference, timing and distance: Appendix Table R2.

5.4 Admission rather than price

The channel that the fence leaves open by construction is land. The zone sits inside a parent development zone, which allocates its industrial land establishment by establishment at administered prices, under minimum standards of investment and output per hectare (Section 2.2). When the fence takes part of that land, a price would ration the rest by raising what entrants pay; administered prices leave the rationing to the admission decision. Two facts from Section 4 fit admission rather than price. Entry fell in a step across productivity fifths, the lower fifths alike and the top fifth not at all (Section 4.2), and the price of industrial land did not move (Section 4.3). Section 6 builds the model of admission that turns these facts into the value of the entry turned away.

5.5 How the hinterland grew

The growth 40 to 100 km from the zones was shared by domestic and foreign-invested firms. On the later-treated comparison, output of domestic firms there rose by 0.09 (0.04) and

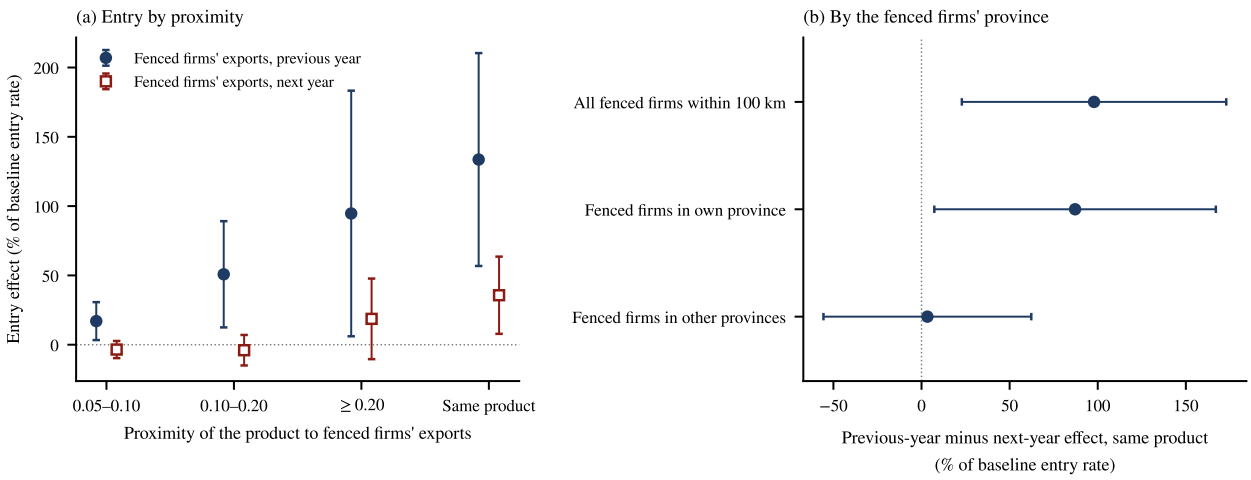


Figure 4: Entry of non-fenced exporters into new products. (a) By the product's proximity to what fenced firms of zones within 100 km exported in the previous year and in the next year: percent of the baseline entry rate. (b) Previous-year minus next-year portfolio for the same product, by the province of the fenced firms. Equation (2); the baseline entry rate is 0.146 percent; 95 percent confidence intervals, clustered by nearest zone. Splits by trade regime and ownership: Appendix Figure I4.

output of foreign-invested firms by 0.10 (0.07), and employment rose about as much as output (Table 2): the hinterland grew by adding plants and workers, not by raising the output of each worker. On the matched comparison, which has the precision to decompose this growth, it comes mostly from non-state firms expanding capacity, in new plants on empty land and in established industrial towns. It shows no sign of having moved from elsewhere in the province; the fence's purchases could account for a sixth to a quarter of it at most (Appendix G.4.2). The design locates this growth; it does not attribute it to the zones.

6 A Model of Land Admission around a Zone

Section 4 shows which firms lost access to land around the zones, but not what their lost entry was worth. We model where firms locate, how the parent zone admits them and where workers seek jobs, to recover that value and to divide the zone's gains and costs among the agents. Because land is allocated by administrative rule at prices that do not clear the market (Tao et al. 2010; Tian et al. 2024), its price must be read from the composition of entry, as the market-design literature reads the price of a seat from who is admitted when tuition does not clear the market (Fu 2014; Azevedo and Leshno 2016).

Three decisions organize the model: where firms produce, which firms the parent zone admits, and where workers seek jobs. The fence's establishments are given; they take land, hire workers, pay taxes and may change the profits of other firms. The model delivers three

results: who gets land, what the fence costs, and who gains and who loses. The second is the core object, which we call the price of the fence: the value that firms turned away to accommodate the fence would have created. Its size depends on how the parent zone rations. If it admits firms in order of value, it turns away only marginal firms and the cost is a Harberger triangle. If it sorts firms into coarse tiers and rations the marginal tier at random, it turns away firms of every size in that tier and the cost is first order.

6.1 Environment

The fence removes industrial land from a parent development zone whose remaining plots are allocated by administrative rule. The prefecture is small in an open economy: national and foreign demand, prices and wages outside it are given. The parent zone allocates its land as a whole: it holds $\bar{K}$ hectares of industrial land outside the fence and decides which firms obtain plots anywhere in it. A firm can produce in the parent zone (E), elsewhere in its prefecture (P), in a neighboring prefecture (N), or not at all. Parent zones are compact: the fence occupies a median of about a quarter of its parent zone and a median fence covers 2.8 square kilometers, so the parent zone lies within a few kilometers of the fence. We therefore observe its admissions as the entry of new establishments within 10 km of the fence; the grid cells of Section 3 are only the units in which we count that entry. We compare the years after a zone opens with the same years without the fence.

6.2 Firms

Firms choose their most valuable feasible location; entry and incumbency are two stages of the same problem.

Profits. A firm produces a differentiated variety under monopolistic competition with elasticity of substitution $\sigma > 1$. With constant-elasticity demand and a constant markup, its operating profit in location ℓ is

$$\pi_{j\ell} = A_\ell \, \varphi_{j\ell}^{\sigma-1}, \tag{3}$$

where $\varphi_{j\ell}$ is its productivity there and A_ℓ the location's profitability, which collects the demand it reaches and its input costs (Appendix T.1). Productivity differs across locations because suppliers, buyers and the entrepreneur's own ties differ. A firm that produces pays a fixed cost F per period and, in the parent zone, needs a plot of d_j hectares, which rises with its size.

Location. Let $V_{j\ell}$ be firm j's lifetime operating profit in location ℓ net of fixed costs, discounted with the survival and growth we measure for new establishments; not producing

is worth zero. A firm chooses

$$\max_{\ell \in \mathcal{L}_j} \left\{ V_{j\ell} - m\,\mathbb{1}[j \text{ moves an established plant}] \right\}, \qquad \mathcal{L}_j = \begin{cases} \{E, P, N, 0\} & \text{if } j \text{ holds a plot,} \\ \{P, N, 0\} & \text{otherwise,} \end{cases} \tag{4}$$

where m is the cost of moving and 0 denotes not producing. A potential entrant applies for a plot when E is its best location; an incumbent in the parent zone already holds one. Both obtain or keep a plot through the same admission rule, at application and at review. A potential entrant refused a plot produces in its best other location or not at all. An incumbent that loses its plot moves if its best other location is worth more than m and closes otherwise (Melitz 2003).

Exports. The same firms choose which products to export. Firm j exports product p if its export profit covers the fixed cost of finding buyers and meeting their standards, $\pi^X_{jp} \geq f_p(N_p)$, which falls with the number N_p of firms in the province that already export the product or a related one, the fence's establishments among them (Krautheim 2012). We write ΔV^X for the resulting gain of the province's exporters.

6.3 The Parent Development Zone

The parent zone allocates scarce plots according to the fiscal and output value it expects from applicants. Local governments in China collect a share of the taxes of the firms they host, and their officials are evaluated on local output (Li and Zhou 2005). The parent zone therefore values an admitted firm by $v_j = (t_{\text{tax}} + \omega)\, r_{jE}$: the tax rate t_{tax} on its revenue r_{jE} plus the weight ω that career incentives place on output. It does not observe productivity. It observes the investment and output per hectare that an applicant commits to and sorts applicants, and plot holders under review, into a high tier H and a low tier L by that signal. Plots are sold at an administered price. The parent zone chooses the probability a_c of admitting a member of tier c to solve

$$\max_{a_H, a_L \in [0,1]} \left\{ G_F + \sum_{c \in \{H,L\}} a_c\, n_c\, \mathbb{E}[v_j \mid c] \right\} \qquad \text{subject to} \qquad \sum_{c \in \{H,L\}} a_c\, n_c\, \mathbb{E}[d_j \mid c] \leq \bar{K}, \tag{5}$$

where n_c is the number of applicants and reviewed holders in tier c and G_F the taxes and credited output of the fence's establishments, which the parent zone receives but does not choose. This is the admissions problem of Fu (2014, equations 12–14) without its tuition stage, with seats measured in hectares. The solution ranks the tiers by expected value per hectare, $\mathbb{E}[v_j \mid c]/\mathbb{E}[d_j \mid c]$; when land binds, it admits the high tier and the share $a_L \equiv e$ of

the low tier that fills the land. The multiplier Λ is the value of a marginal plot to the parent zone, not the value forgone by the firms it turns away. Only the ranking of the tiers matters, so t_{tax} and ω need not be estimated.

6.4 Workers

Workers choose where to seek employment, given wages, amenities, their own tastes and the chance of getting a job. Worker i chooses among the locations of the prefecture and an outside option O, farm work at home or a job in another province:

$$\max_{\ell \in \{E,P,N,O\}} \left\{ q_\ell \, B_\ell w_\ell \, u_{i\ell} + (1 - q_\ell) \, B_O w_O \, u_{iO} \right\}, \tag{6}$$

where w_ℓ is the wage, B_ℓ an amenity, $u_{i\ell}$ an idiosyncratic taste and q_ℓ the probability of obtaining a job, with $q_O = 1$. The labor market closes in one of two ways, and we keep both.

(i) *Flexible wages.* Wages clear the market, $q_\ell = 1$, and (6) is the location choice of Fajgelbaum et al. (2019). A location that is small in its labor market faces a labor supply with elasticity ε, the Fréchet shape of tastes, and the marginal worker is indifferent between a job and her outside option.

(ii) *Rigid wages.* The wage in location ℓ is above the level that clears the market, employment equals labor demand, and workers queue: a worker who chooses ℓ obtains a job with probability $q_\ell = L^d_\ell / L^s_\ell < 1$ (Harris and Todaro 1970). The marginal hire gains $g_\ell = 1 - w^O_\ell / w_\ell$ of the wage, where w^O_ℓ is the wage equivalent of her outside option.

Under the first closure $g_\ell = 0$; under the second $g_\ell > 0$. Section 8 reports the welfare account under both.

6.5 The Fence's Establishments

The fence's establishments are taken as given, but their land, jobs, taxes and spillovers enter the other agents' problems. They are foreign assemblers that import inputs and export output in bond. They employ L_F workers and pay taxes T_F, and the share α_F of their activity would not have taken place elsewhere in the country without the zone. They lower the parent zone's land $\bar{K}$, add L_F to labor demand in the parent zone and add G_F to the parent zone's payoff. They may raise the profitability of every firm in the parent zone by the factor $\exp(\eta)$, as large plants raise the productivity of their neighbors in Greenstone et al. (2010). And they add to N_p for the products they export.

6.6 Equilibrium

An equilibrium consists of firms' locations, workers' choices, admission probabilities and wages consistent with the allocation of land and labor. Given $\bar{K}$, the administered price of land and the fence's establishments, firms solve (4), the parent zone solves (5) and workers solve (6). Firms that obtain or keep plots use the land $\bar{K}$, all of it whenever some applicant is turned away. Each labor market clears, with employment equal to labor demand where jobs are queued. The parent zone's land is thus cleared by Λ rather than by its price.

6.7 Functional Forms

Five assumptions turn the agents' problems into statements about entry and retention. Their full forms are in Appendix T.1.

Profits. Productivity is Pareto across firms, so operating profits are Pareto with tail $\zeta > 1$. Across locations, profits follow the multivariate Pareto distribution of Arkolakis et al. (2018) with correlation $\rho \in [0, 1)$: at $\rho = 0$ a firm does well in at most one location, and as $\rho \to 1$ it does equally well everywhere. The number of potential entrants applying to the parent zone changes with the opening by the factor $\exp(\nu)$, as the attractiveness of the location changes in a fixed pool (Chaney 2008).

Information. The parent zone's signal of firm j is $s_j = \tau z_j + \sqrt{1-\tau^2}\,\xi_j$, where z_j is the normal score of the firm's productivity rank within its industry and year and ξ_j is independent noise, the form Khandelwal et al. (2013, footnote 28) give the allocation of export licenses. A firm is in the high tier if $s_j \geq \bar{s}$. A firm of productivity fifth b therefore obtains a plot with probability

$$P_b = h_b + (1 - h_b)\, e, \tag{7}$$

where h_b, the share of the fifth in the high tier, rises with τ and falls with $\bar{s}$. A share χ of plots is reviewed with the same tiers and the same rule, as export licenses were reallocated on past performance (Khandelwal et al. 2013).

Labor supply. Tastes are Fréchet with shape ε, so with guaranteed employment the location shares take the form of Fajgelbaum et al. (2019).

Spillovers. Agglomeration multiplies the profitability of every firm in the parent zone by $\exp(\eta)$. The fixed cost of exporting a product falls with the number of its exporters, $f_p = \bar{f}_p(1 + N_p)^{-\kappa}$, and export profits are Pareto with the same tail ζ; what matters for welfare is the change $\Delta \ln f$ that the fence induces.

The parameters are thus the signal and the rule $(\bar{s}, \tau, e, \chi)$, the pool change ν, the tail ζ and correlation ρ, the labor supply elasticity ε and wedge g, and the spillovers η and $\Delta \ln f$.

6.8 Three Results

The model delivers three objects: the allocation of land, the price of the fence, and the incidence of the zone's gains and costs. Because a firm's chosen location is independent of the profit it earns there (Arkolakis et al. 2018, Lemma 1), applicants to the parent zone are a representative draw of the pool's productivity ranks (Appendix T.2). The rule then decides which applicants get land while the pool decides how many apply.

Proposition 1 (Who gets land)**.** *Entry of new establishments of productivity fifth b into the parent zone, and the number of its incumbents that keep their plots, change by*

$$\Delta \ln n_b = \nu + \ln P_b, \qquad \Delta \ln n_b^I = \ln \left[1 - \chi \left(1 - P_b\right)\right]. \tag{8}$$

If a share δ_ℓ of the turned-away enters location ℓ instead, the entry of the low fifths in ℓ rises relative to its top fifth by δ_ℓ times the ratio of the turned-away to the low fifths' usual entry there.

The first part is the signature that Section 7 inverts. Coarse tiers produce a step across fifths, not a ramp: fifths in the low tier lose the same share of applicants, the top fifth loses almost none, and the same step appears among incumbents, scaled by the share of plots reviewed. A hard cutoff on productivity is the limit $\tau \to 1$ with $e = 0$; a smooth score is the limit with one tier. The second part is the mirror image of the step: entry that the rule removes near the zone reappears, in the same fifths, wherever the turned-away go, while shocks common to all fifths of a location cancel. It separates moving entry, as the French enterprise zones of Mayer et al. (2017) move establishments within their municipality, from removing it; $\delta_\ell = 0$ when $\rho = 0$.

Proposition 2 (The price of the fence)**.** *Let the fence reduce the parent zone's land from K_0 to K_1, and let $\mathcal{D}$ be the applicants who obtain land at K_0 but not at K_1. The price of the fence is the value they forgo,*

$$\mathcal{C} = \sum_{j \in \mathcal{D}} \left[V_{jE} - \max\left\{0, \max_{\ell \neq E} V_{j\ell}\right\}\right] = \underbrace{\left[V^*(K_0) - V^*(K_1)\right]}_{\textit{scarcity}} + \underbrace{\left[G(K_1) - G(K_0)\right]}_{\textit{rule}}, \tag{9}$$

where $V^*(K)$ is the value admitted when land K is allocated in order of value per hectare and $G(K)$ the value the rule loses relative to that allocation. Random rationing within a tier

makes $\mathcal{C}$ first order in the share of applicants turned away; allocation by value makes it second order.

The price of the fence is the cost of the land the fence took, not of all rationing: it is the change from K_0 to K_1, which is what the step in entry measures. It is a value forgone by firms, not the parent zone's shadow value Λ. The decomposition follows Khandelwal et al. (2013, Section 6), who compare a political allocation of quota licenses with an auction of the same quota and with no quota. The first term is the cost of scarcity, which only more land removes; the second is the change in the cost of the rule, which another allocation of the same land removes. If the parent zone did not ration before the fence, $G(K_0) = 0$ and the second term is the whole cost of the rule at K_1; otherwise it is smaller. The proposition also explains why the price of land did not move. A cutoff and a charge that admit the same firms produce the same entry (Anderson 1985; Khandelwal et al. 2013), and a cutoff that fills a fixed capacity is a market-clearing price (Azevedo and Leshno 2016); under the cutoff the admitted firms keep the rent, so the transaction price of land, which Section 4.1 finds unchanged, need not rise (Appendix T.2).

Proposition 3 (Welfare incidence)**.** *Holding prices outside the prefecture fixed, the zone changes welfare, to first order, by*

$$\Delta W = -\mathcal{C} - \mathcal{I} + \eta\,\Pi_E + \sum_{\ell} g_\ell\, w_\ell\, \Delta L_\ell + \alpha_F\, T_F + \Delta V^X, \tag{10}$$

where $\mathcal{I}$ is the value lost by incumbents that lose their plots and Π_E the operating profit of the firms in the parent zone. Changes in wages are transfers between workers and firms.

Equation (10) is the classic account of a place-based policy in this setting, the general formula of Harberger (1964): each gain is a change in a quantity times the wedge it crosses. The fence's taxes count because its owners are foreign: their profits leave, and the taxes are the part of their surplus that stays; for the prefecture $\alpha_F = 1$. A channel is absent from the account when either its quantity or its wedge is zero, so a zero estimate says which. A flat wage, for instance, says that labor supply was elastic, not that workers were untouched; workers then gain only through the wedge on new jobs, which is zero with guaranteed employment and positive where jobs are queued (Harris and Todaro 1970; Tombe and Zhu 2019). The jobs the zone creates are the fence's, plus the change in incumbents' employment in the parent zone, minus the jobs of turned-away firms that do not produce elsewhere:

$$\Delta L = L_F + \Delta L_E^{\text{inc}} - \sum_{j \in \mathcal{D}} \mathbb{1}\Big[\max_{\ell \neq E} V_{j\ell} < 0\Big]\, l_j. \tag{11}$$

Table 4 lists what the three results need. They do not need the size of the pool, the level of the fixed cost, the parent zone's valuation or the land itself. Two inputs cannot be estimated from outcomes around the zones, the labor wedge g and the share α_F; Section 8 reports the account across their range.

Table 4: Inputs of the three results

Input	Role	Data	Status
Rule $\bar{s}$, τ, e, χ; pool ν	who gets land	entry and retention within 10 km by fifth	estimated
Lifetime value of entrants	price of the fence	new domestic establishments, 1999–2004 cohorts	measured
Tail ζ	triangle or rectangle; exporters	upper tail of new exporters' exports	estimated
Re-entry δ_ℓ	where the turned-away go	low-minus-top entry by location	bounded
Agglomeration η	gain within 10 km	sales of incumbents that remain	estimated
Labor supply ε	wages or jobs	employment and wage per worker	bounded below
Fence's jobs L_F, taxes T_F	gains of the fence	survey and customs records of fenced firms	measured
Export cost $\Delta \ln f$	exporters' gain	related export entry 40–100 km away	estimated
Wedge g; new activity α_F	value of jobs; national account	literature; not identified	range

7 Estimation and the Channels

The data pin down the cost of the fence's claim on land and say, channel by channel, which of the gains are present. We estimate the admission rule from the entry and retention of establishments within 10 km, then take each channel of Proposition 3 in turn. For every channel we keep three things apart: the mechanism, stated in Section 6; the estimator and how much it can detect; and the estimate. A channel can be absent because the quantity does not move or because the move crosses no wedge, and the model says which reading each zero supports.

7.1 Comparison and Moments

All moments come from one comparison. Treated cells are the 0.1-degree cells within 150 km of a zone, grouped into distance rings. Comparison cells lie at the same distances from special customs zones that opened after 2007, the end of our survey data, so they share the coastal and administrative settings that drew zones without having one in our years. For an outcome y_{ct} of cell c in year t, we estimate by Poisson pseudo-maximum likelihood

$$\mathbb{E}[y_{ct}] = \exp\Big(\alpha_c + \delta_{r(c)t} + \sum_h \sum_k \beta_{hk}\, \mathbb{1}\{c \in h,\ t - t_{z(c)} = k\}\Big), \tag{12}$$

where α_c are cell effects and $\delta_{r(c)t}$ are effects for each ring and year, shared by treated and comparison cells. The index h runs over groups of cells: rings, and within rings the host prefecture, the neighboring prefectures of the same province and other provinces. Event years k include leads -4 to -2. A moment is the mean of β_{h0}, β_{h1} and β_{h2} minus half of $\beta_{h,-2}$; the alternative subtracts the mean of the leads. Standard errors are clustered by prefecture, and the covariance of all moments comes from stacking the prefecture-level influence functions of the separate regressions.

We treat entry within 10 km as admission to the parent zone, which the compactness of parent zones supports (Section 6.1); unrationed entry within 10 km but outside the parent zone would dilute the step and make the price of the fence too small. The model counterpart of each moment is the same statistic: the log change of the same outcome in the same tiers, computed from the model's entry and retention with the same productivity fifths. Entry of fifth b within 10 km changes by (8), retention of incumbents by the second part of (8), and re-entry by Proposition 1. Productivity fifths are ranked within four-digit industry and year by revenue productivity, for new domestic establishments in their first year and for incumbents in 1999–2000. Table 5 reports the ten fitted moments. Before the openings, about 216 new domestic establishments entered within 10 km of the zones each year and 762 entered 10–40 km away.

7.2 The Admission Rule

We fit the ten moments by minimum distance,

$$\hat{\theta} = \arg\min_{\theta}\ J(\theta) = \big(\hat{m} - m(\theta)\big)'\, \hat{\Sigma}^{-1} \big(\hat{m} - m(\theta)\big), \tag{13}$$

where $\hat{m}$ stacks the moments of Table 5, $m(\theta)$ is the model's counterpart and $\hat{\Sigma}$ their estimated covariance. With the efficient weight, J is distributed χ^2 with degrees of freedom equal to the

Table 5: Entry and retention within 10 km by productivity fifth

	Fifth 1	Fifth 2	Fifth 3	Fifth 4	Fifth 5
Entry of new domestic establishments	−0.65	−0.22	−0.61	−0.17	0.23
	(0.21)	(0.21)	(0.30)	(0.25)	(0.20)
Retention of 1999–2000 incumbents	−0.14	−0.07	−0.09	−0.04	0.03
	(0.08)	(0.05)	(0.05)	(0.05)	(0.05)

Notes: Log changes from equation (12): mean of event years 0–2 minus half of event year −2, for 0.1-degree cells within 10 km of the zones relative to cells at the same distance from special customs zones opened after 2007, with cell effects and ring-by-year effects. Fifths of revenue productivity within four-digit industry and year. Standard errors clustered by prefecture in parentheses.

moments minus the parameters. We fit four families of rules. Coarse tiers follow (T3) with the high-tier cutoff $\bar{s}$, the low-tier admission e, the pool change ν, the review share and the correlation τ, which we profile over a grid; the pool either grows alike in every fifth or only in the high tier. A smooth noisy score is the single-tier limit with a cutoff. The last two rank applicants by first-year sales per hectare and by value per hectare over their first four years.

The rule's form is identified only up to a set. Following the profile criterion of Chernozhukov et al. (2007), we keep every rule whose J lies within 3.84 of the smallest J among all rules; 23 rules on the parameter grid pass. The best coarse-tier rule fits with $J = 11.1$ on 5 degrees of freedom ($p = 0.05$). It places 34 percent of applicants in the high tier, admits the share 0.54 of the low tier and reviews the share 0.15 of plots; the pool rises by 0.11. The smooth score and ranking by first-year sales also pass. Ranking by value over the first four years fails ($J \geq 20.3$): the parent zone did not allocate land by what establishments turned out to be worth. On the alternative baseline that subtracts the mean of the leads, ranking by first-year sales fails as well. Figure 5 compares the fitted moments with the data: the low fifths fall together, the fourth fifth falls less and the top fifth rises, and incumbents follow the same pattern at a smaller scale. Figure 6 shows the fit and the implied price of the fence for every rule on the grid.

The step is not the trace of sorting. Had the opening drawn relatively more productive firms toward the zone, as complementarity between productivity and place would (Gaubert 2018), entry would tilt toward the top fifths as a ramp. The data neither require such a tilt nor exclude it, and Appendix T.4 shows that it would lower the price of the fence by about half.

The loss of the excluded entrants is stable across the set, while how much of it a better rule could recover is not. In the best coarse-tier rule the excluded lose 3.5 billion yuan of operating profit a year, 22 percent of the value of all applicants; across the set the loss ranges from 2.0 to 5.6 billion yuan (2.5 to 5.3 on the alternative baseline). This range reflects which

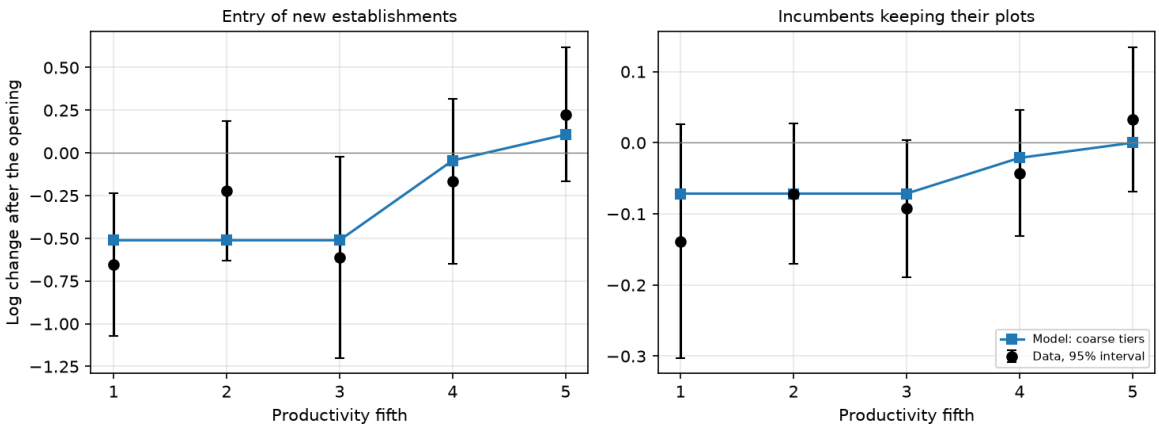


Figure 5: Entry and retention within 10 km by productivity fifth: data and model

Notes: Dots are the moments of Table 5 with 95 percent intervals. Squares are the best coarse-tier rule.

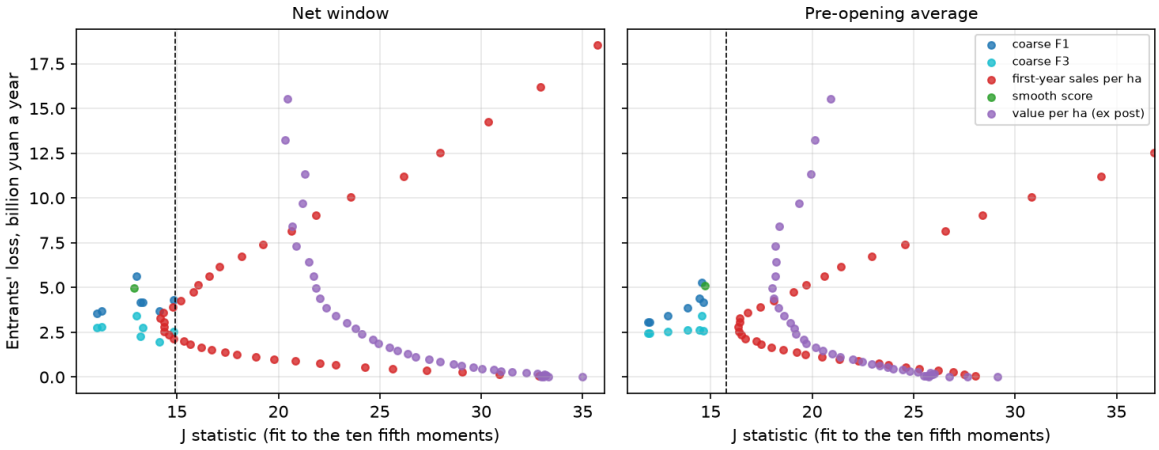


Figure 6: Fit and the price of the fence across rules

Notes: Each point is one rule on the grid of Section 7.2. The horizontal axis is the minimum-distance statistic J for the ten moments; rules left of the dashed line, $J \leq J_{\min} + 3.84$, are in the identified set. The vertical axis is the price of the fence in billion yuan a year. Left: net window; right: pre-opening average.

rule the parent zone used; sampling error within each rule adds to it. About 70 applicants a year are turned away across the zones. Section 8 asks how much of this loss another rule for the same land would recover, and who bears it.

7.3 The Channels

Table 6 lists each channel with its mechanism, its estimator, the estimate and the condition under which the data reject the channel's presence or absence. The rows that follow the rule answer four questions: where the excluded went, whether establishments near the fence gained, what workers gained, and what the fence's establishments brought.

Table 6: Channels: identification, estimator and estimate

Channel (parameter)	Mechanism	Moment and estimator	Estimate	Rejected if
Panel A. Cost				
Admission rule $(\bar{s}, e, \chi, \nu)$	applicants in the low tier turned away	ten moments of Table 5; minimum distance	loss 3.5 bn yuan a year; set 2.0–5.6	entry falls alike in every fifth
Panel B. Where the excluded went				
Re-entry, own prefecture 10–40 km (δ_P)	turned-away enter next best area	entry of fifths 1–3 minus fifth 5, Proposition 1	0.03 (0.16); share 0.15, upper 1.72 (no power)	low-minus-top rises
Re-entry, neighboring prefectures 10–40 km (δ_N)	same	same	−0.35 (0.21); share upper 0.07	same
Re-entry, other provinces 10–150 km	same	same	0.01 (0.10); share upper 0.76	same
Panel C. Establishments near the fence				
Agglomeration (η)	profits within 10 km rise	sales per remaining incumbent within 10 km	0.08 (0.09); 4.6 bn yuan, [−5.3, 14.6]	sales rise
Panel D. Workers				
Labor supply (ε)	employment versus wage	employment within 10 km; wage per worker	0.15 (0.06); −0.035 (0.052); $\varepsilon \geq 2.3$	wage rises with employment
Incumbents' hiring	incumbents expand	employment per remaining incumbent	0.11 (0.03)	—
Panel E. The fence's establishments				
Fence's jobs (L_F)	fenced firms hire locally	fenced firms in the survey, 2006	0.19 million matched; 1.25 million scaled	—
Fence's taxes (T_F)	fiscal wedge	taxes over sales, fenced firms versus same industry, year and ownership	1.41 versus 3.09 percent; 6.1 bn yuan a year	—
Panel F. Exporters in the province (customs comparison of products, not the later-treated design)				
Fixed cost $(\Delta \ln f)$	fenced exports lower entry costs of related products	related export entry and new exporters' size, 40–100 km	−0.54 (0.16); net profits +1.02 percent of base exports	related entry no higher

Notes: Panels A–D use equation (12) on the later-treated comparison with prefecture-clustered standard errors in parentheses. Re-entry shares are the low-minus-top contrast times the area's pre-opening entry of fifths 1–3, divided by the applicants of those fifths turned away within 10 km a year (63); "upper" is the 95 percent upper bound. The bound on ε divides the employment change by the upper bound of the wage change. Scaled fence jobs divide by the matched share of fenced exports. Panel F: customs records, standard errors clustered by zone. —: measured, not tested.

The excluded did not enter in neighboring prefectures; within the host prefecture the data cannot tell. If the applicants turned away within 10 km had entered elsewhere, the low fifths would have risen, relative to the top fifth, wherever they went (Proposition 1). In the neighboring prefectures 10–40 km from the zones, the pre-opening entry of the low fifths, 62 a year, is about as large as the number turned away, so the test has power there. The contrast is −0.35 (0.21), and the upper bound of its confidence interval implies that at most 0.07 of the turned-away entered there. In other provinces the bound is 0.76. Within the host prefecture the contrast is 0.03 (0.16), but entry there is five times the number turned away, so the test cannot exclude any feasible share (upper bound 1.72). The share of displaced incumbents that relocate, 0.35 percent, is consistent with a high cost of setting up elsewhere, but it concerns establishments that already hold plots and is not a bound for applicants. Across a prefecture boundary, then, the rule removed entry rather than moving it, unlike the French enterprise zones of Mayer et al. (2017), which divert establishments from the rest of their municipality. This fits the evidence that entrepreneurs start firms where they already live and work (Michelacci and Silva 2007): an applicant turned away by its own development zone faces another jurisdiction's rationed land, not an open market. Within the host prefecture, re-entry remains open, and Section 8 reports the loss when half or all of the turned-away re-enter there and keep their value.

The agglomeration gain is the least precise channel. An agglomeration shift would raise the sales of every incumbent within 10 km that keeps its plot. Sales per remaining incumbent change by 0.08 (0.09), 0.08 (0.11) in the top fifth, which the review does not reach, against 0.01 (0.04) 10–40 km away. The top fifth's entry, which the same shift would raise, changes by 0.23 (0.20). The point estimates are positive and none is statistically distinguishable from zero. Revenue per worker of remaining incumbents changes by −0.04 (0.09). With constant returns and an elastic labor supply, a shift in productivity or demand raises output and employment in proportion, so this is what the shift predicts, and it does not separate the shift from zero. The precision matters for welfare, because the base is large. Incumbents within 10 km sold 238 billion yuan a year before the openings, so the point estimate is worth 4.6 billion yuan of operating profit a year, with a confidence interval from −5.3 to 14.6. The customs rule suggests why a local gain could be small: fenced establishments buy and sell abroad in bond, so the shared suppliers and buyers through which large plants raise their neighbors' productivity in Greenstone et al. (2010) are largely absent. The gains from the fence's know-how that we can detect appear 30–100 km away, in the related products of Panel F.

Employment rose without a detectable change in wages. Employment within 10 km rose by 0.15 (0.06) while the wage per worker changed by −0.035 (0.052). By the workers' problem (6), labor supply to the area was elastic: dividing the employment change by the upper confidence bound of the wage change gives ε of about 2.3 or more, a rough bound rather than a joint confidence set. The transfer from employers to workers was therefore close to zero. The wage bill within 10 km, 30.9 billion yuan a year, times the change in the log wage is −1.1 billion yuan, with a confidence interval from −4.3 to 2.1. Incumbents that kept their plots hired 0.11 (0.03) more workers each, with no detectable change in their sales. Employment also rose in the rest of the host prefecture, 0.08 (0.05), and in neighboring prefectures, 0.19 (0.10). The rise is thus not specific to the land around the fence. The leads also show that employment within 10 km was falling relative to the comparison before the openings (mean lead −0.11, standard error 0.04; on the alternative baseline the change is 0.19, standard error 0.08). We therefore do not attribute the regional change in employment to the zone. We count the terms of equation (11) instead. Total employment of incumbents within 10 km changes by 0.07 (0.06), about 73 thousand jobs on a base of 1.05 million. Fenced firms matched to the survey employed 0.19 million workers in 2006, or 1.25 million scaled to all fenced exports. A new domestic establishment employed 141 workers in its first year, so three cohorts of turned-away entrants would have employed about 29 thousand, 2.3 percent of the fence's jobs. The fence and the rule thus move employment in opposite directions, and the fence's addition is some forty times the rule's subtraction. What the fence's jobs are worth depends on the labor wedge, which Section 8 varies.

The fence's establishments paid low taxes on large sales. Fenced firms paid taxes of 1.41 percent of sales, against 3.09 percent for plants of the same industry, year and ownership outside fences. On the 54 billion US dollars that fenced firms exported in 2006, that rate yields 6.1 billion yuan a year. Whether these taxes are a gain to the country depends on α_F, the share of the fence's activity that would not have located elsewhere in China, which the data around the zones cannot identify.

8 Welfare and Counterfactuals

This section asks three questions in turn. How large is the price of the fence? How much of it would a different rule for the same land recover, and who bears it? And do the zone's gains cover it? The first two questions use only the estimated rule and the measured value of new establishments, so their answers are model predictions with the precision of Section 7. The third adds the channels of Table 6 and two inputs that the data around the zones cannot supply. Value is an establishment's operating profit over its first four years, discounted at 5

percent a year; for each year's cohort of applicants to the parent zones, all applicants together would create 15.8 billion yuan. We hold the pool of applicants, wages and the fence's own activity fixed and, except in Section 8.4, the land.

8.1 The Price of the Fence

The rule turned away firms worth several billion yuan a year, and it did so by turning away firms of every size in the low tier. Under the estimated rule, the parent zones admit 71 percent of applicants and turn away 29 percent, who would have created 22.3 percent of the pool's value: 3.5 billion yuan a year. Across the rules the data accept, the price ranges from 2.0 to 5.6 billion yuan. These values are operating profits, gross of the fixed costs the firms would have paid, which we do not observe. With Pareto profits, the marginal entrant earns 7.4 percent of the average entrant's operating profit, which is the fixed cost. Charging every turned-away firm that amount against four-year value, which overstates fixed costs because value includes growth, lowers the price to 20.2 percent of the pool's value, so the price net of fixed costs is at least 3.2 billion yuan.

The value turned away depends on who is turned away, and Table 7 brackets it. A rule that admits applicants in order of value per hectare turns away the least valuable and loses 3.7 percent of the pool's value; this is the Harberger triangle, small because the marginal applicant is worth little more than its costs. A random allocation of the same land turns away a representative slice and loses 29.4 percent; this is the rectangle. The estimated rule loses 22.3 percent, 76 percent of the way to the random benchmark. The reason is the coarse signal. The parent zone admits a high tier it is sure of, about the top 34 percent of applicants by productivity, and treats the rest as interchangeable. Within the low tier it turns away applicants of every size, so the cost is first order in the land it rations. Random rationing within the low tier is the rule of Fu (2014), and the step in entry cannot test it. If the parent zone partly ranked the low tier by value, the price would lie between the estimated rule and the government ranking of Table 7, so the estimate is an upper bound given the tiers.

The firms turned away were small at entry but would have grown. They hold 29 percent of entrants' first-year employment and 19 percent of their first-year sales, but 22.3 percent of their four-year value. The less productive half of new establishments survives to age three about as often as the more productive half, 69 percent, and grows faster, by 0.37 against 0.24 log points a year. A signal recorded at entry thus misses the establishments whose measured productivity is low in their first year and whose sales would catch up. The values are those of comparable entrants nationwide over four years, a conservative choice: establishments admitted near the zones survive longer than the national cohort, and small entrants keep

Table 7: The price of the fence under alternative rules for the same land

Rule (applicants ranked by)	Applicants turned away, percent of all: Establishments	First year: Sales	First year: Employment	Four-year value	Billion yuan a year
Panel A. Estimated rule and alternatives					
Estimated rule: tiers, low tier at random	29	19	29	22.3	3.5
Random	29	29	29	29.4	4.7
Bids on own first-year sales per hectare	31	6	22	10.7	1.7
Auction on application-time information	32	11	18	11.5	
Government ranking on application-time information	33	11	16	11.0	
Oracle: four-year value per hectare	32	9	20	3.7	0.6
Panel B. Rule estimated against the pre-opening average					
Estimated rule	25	17	25	19.9	
Bids on own first-year sales per hectare	26	5	20	9.1	
Oracle	27	8	17	2.7	

Notes: Applicants are new domestic establishments of the 1999–2004 cohorts nationwide, weighted to the mix of productivity fifths that entered within 10 km of the zones before the openings. The estimated rule admits each fifth with the probability of equation (7) at the estimates of Section 7.2; the land it uses, with plot area proportional to employment to the power 0.24, is the capacity that the other rules fill in their own order within two-digit industry and year. Four-year value is discounted sales at ages zero to three, 5 percent a year, zero after an establishment leaves the survey; operating profit is the share $1/\sigma$ of sales, $\sigma = 4$. The rows differ in what the allocator knows. The oracle knows each applicant's four-year value and is a benchmark, not a policy. Bids on own first-year sales assume that applicants know their first-year sales and bid them per hectare. The auction and the government ranking use only what the parent zone sees at application (employment, capital, assets, leverage, industry and year): in the auction, applicants bid their predicted first-year sales; the government ranks them by predicted four-year value. Billion yuan: operating profit of each year's cohort.

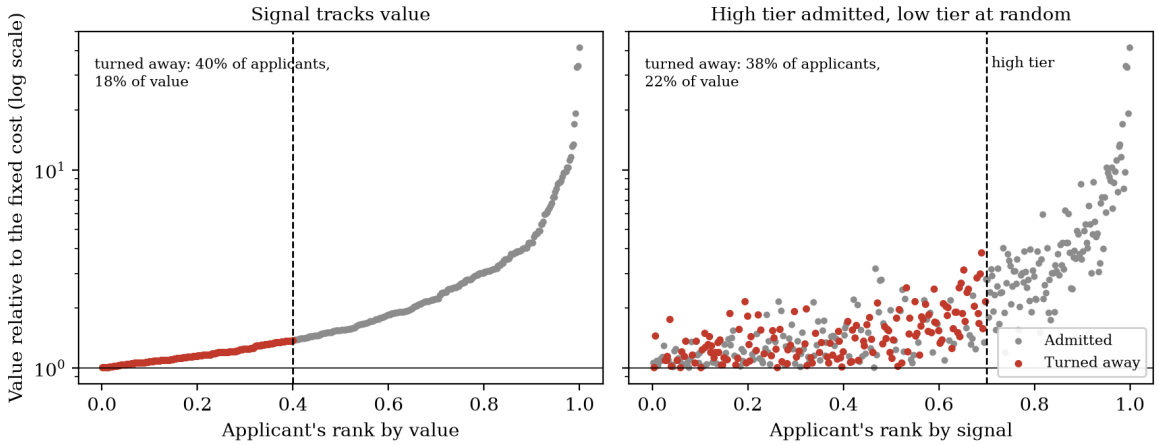


Figure 7: Applicants turned away when admission tracks value and when a low tier is rationed at random

Notes: Illustration with simulated applicants, not estimates. Each dot is an applicant; the vertical axis is its value relative to the fixed cost, Pareto with shape 1.5. Both rules admit 60 percent of applicants. Left: applicants are admitted in order of value. Right: the top 30 percent by a signal with correlation 0.9 with log value form the high tier and are admitted; the rest are admitted at random.

growing after age three, both of which would raise the price.

8.2 Scarcity, Information and the Rule

Most of the price lies in how the remaining land was allocated, not in how much land the fence took. Proposition 2 splits the price into the cost of scarcity, the value an allocation by value would lose when land falls from K_0 to K_1, and the change in the cost of the rule. The cost of scarcity is 3.7 percent of the pool's value, or 0.6 billion yuan a year; only more land removes it. The cost of the rule at K_1 is the remaining 18.6 percentage points, or 2.9 billion yuan, 83 percent of the price. It equals the change in the cost of the rule if the parent zones did not ration before the openings and exceeds it otherwise, so 83 percent is an upper bound on the rule's share. Whether they rationed before is a limitation the data cannot resolve: we observe admissions only relative to the years before the fence. The price itself is the change from K_0 to K_1 and does not rest on that assumption.

Information explains part of the rule's cost, and the rule the rest. No rule at application can foresee which entrants will grow. An allocation by the value predicted from what the parent zone could see at application, the applicant's employment, capital, assets, leverage, industry and year, turns away 11.0 percent of the pool's value. The 7.3 points between this feasible allocation and the efficient one are the value of foresight. The 11.3 points between the estimated rule and the feasible allocation are the cost of the rule given the information available.

A price would recover about half of the price of the fence, and how much depends on what bidders know. If the parent zone sold the same land to the applicants that bid most per hectare, and applicants bid their own first-year sales, it would turn away 10.7 percent of the pool's value, saving 1.8 billion yuan a year, 52 percent of the price. An auction in which bidders know only what the parent zone sees at application does about as well, 11.5 percent, and so does a government that ranks applicants by the value predicted from the same information, 11.0 percent. Only the oracle, which no allocator could implement, reaches the cost of scarcity. This counterfactual depends on which rule the parent zones used. If they already ranked applicants by sales, a rule the data do not reject on the main baseline, there is nothing to recover, and across the rules the data accept the recovery runs from 0 to 61 percent. Against the pre-opening average, where ranking by sales is rejected, the recovery is 54 percent. For comparison, Khandelwal et al. (2013) attribute 14 to 18 percent of the productivity gain from removing China's textile export quotas to the end of their political allocation.

8.3 Who Bears the Cost

Absent re-entry, the rule placed the whole price on firms that never entered. Table 8 divides the cost among the agents of the model under the estimated rule and under a price for the same land. Under the rule, the admitted firms pay the administered price and keep the scarcity rent of the land, as holders of quota licenses kept the rent of the quota (Khandelwal et al. 2013). The parent zone gains nothing in rent, and workers, whose wages did not move, neither gain nor lose through wages. The whole price falls on the applicants turned away, who appear in no data on the zones. Under a price, the admitted firms would pay up to 16.9 percent of the pool's value, 2.7 billion yuan a year, to the parent zone, and the loss would shrink toward the cost of scarcity. The rent is computed with oracle bids, the highest losing value per hectare, so it is an upper bound on what a feasible auction would collect.

Table 8: Incidence of the price of the fence

	Estimated rule	Price for the same land
Percent of the four-year value of all applicants		
Applicants turned away	−22.3	−3.7
Admitted firms	0	−16.9
Parent development zone	0	16.9
Workers, through wages	0	0
Total: efficiency cost	−22.3	−3.7
Total, billion yuan a year	−3.5	−0.6

Notes: Model predictions at the estimated rule, relative to entry without the fence's claim on land. The price column admits applicants in order of four-year value per hectare and charges the highest losing bid per hectare. Workers' entries use the wage estimate of Table 6.

8.4 More Land

Returning the fence's land to the parent zone would remove the whole price, 3.5 billion yuan a year, and transfer nothing, since the shadow value of land would fall to zero. Reallocating the land that remained would remove 83 percent of the price without returning any land. Re-entry in neighboring prefectures is tightly bounded (Section 7.3); re-entry elsewhere in the host prefecture would offset part of either gain. The policy margin with the larger return is therefore the rule by which the parent zone allocates land, not the amount of land it keeps.

8.5 The Gains and the Full Account

Did the gains from concentrating foreign assembly behind the fence cover its price? Table 9 evaluates Proposition 3 for all zones together. The price of the fence and the plots lost by incumbents are estimated. The fence's taxes are measured. The agglomeration gain is estimated but imprecise, and the exporters' gain comes from a different comparison and has no yuan base; both are shown and kept out of the totals. The account holds prices outside the prefecture, the pool of applicants and the fence's own activity fixed, a partial-equilibrium simplification that suits zones small relative to national markets. Two inputs are ranges, not estimates: the labor wedge g, reported at zero, a quarter and a half of the wage, and the share α_F of the fence's activity that is new to the country, zero in the national column.

The account reads in three layers. For the area around the zones, the fence's taxes, 6.1 billion yuan a year, are about as large as the price of the fence and the plots lost by incumbents together. The account is therefore close to zero when jobs carry no wedge: 1.7 billion yuan under the best rule, between -0.4 and 3.3 across the rules. It turns positive once jobs carry a wedge, 6.7 billion yuan at $g = 0.25$, because the fence employed about 1.25 million workers, some forty times the jobs of the firms turned away. For local governments, which keep about 38 percent of the fence's taxes, the account at $g = 0$ is -2.0 billion yuan and turns positive once the wedge exceeds 0.10 of the wage. For the country, the fence's taxes and jobs count only to the extent that its activity is new: with none of it new, the account is -4.4 billion yuan, and it breaks even when the new share exceeds 0.72 at $g = 0$ or 0.38 at $g = 0.25$.

Two channels could move these signs, and the data cannot rule either out. The agglomeration gain to incumbents in the parent zones is 4.6 billion yuan a year at the point estimate, as large as the fence's taxes, with a confidence interval from -5.3 to 14.6. Re-entry within the host prefecture would shrink the price: if half of the turned-away entered elsewhere in their prefecture and kept their value, it would fall to 1.8 billion yuan. Both are reported as ranges because the estimates that bear on them are imprecise, not because the channels are absent. The exporters' gain beyond 10 km, 1.02 percent of related exporters' base exports, is conditional on the fixed cost of exporting being the channel.

9 Conclusion

China's export processing zones concentrated foreign assembly behind customs fences, and the places next to the fences did not grow. The firms there hired more workers at unchanged wages, industrial land around later zones of the same program did not become dearer, and

Table 9: The account of the zones, billion yuan a year

Channel	Area	Local governments	Country ($\alpha_F = 0$)	Range or scenario
Panel A. Channels				
Price of the fence, $-\mathcal{C}$	−3.5	−3.5	−3.5	rules: [−5.6, −2.0]; half re-enter in the prefecture: −1.8
Plots lost by incumbents, $-\mathcal{I}$	−0.8	−0.8	−0.8	upper bound
Fence's taxes, $\alpha_F T_F$	6.1	2.3	0	local share 38 percent
Jobs at $g = 0.25$: fence	4.8	4.8	0	matched firms only: 0.7
incumbents in the parent zone	0.3	0.3	0.3	employment 0.07 (0.06)
turned-away entrants	−0.1	−0.1	−0.1	three cohorts
Agglomeration, $\eta\,\Pi_E$ (not in totals)	4.6	4.6		[−5.3, 14.6]
Exporters, ΔV^X (not in totals)	+	+	+	1.02 percent of base exports
Panel B. Totals				
$g = 0$	1.7	−2.0	−4.4	area across rules: [−0.4, 3.3]
$g = 0.25$	6.7	2.9	−4.2	area across rules: [4.6, 8.2]
$g = 0.5$	11.7	7.9	−4.0	area across rules: [9.6, 13.2]
Panel C. Break-even values				
Share α_F new to the country at which the country breaks even:				
scaled fence jobs	0.72 ($g = 0$), 0.38 ($g = 0.25$), 0.26 ($g = 0.5$)			
matched fence jobs only	0.72 ($g = 0$), 0.61 ($g = 0.25$), 0.53 ($g = 0.5$)			
Wedge g at which local governments break even: 0.10 (scaled fence jobs), 0.56 (matched only)				
Transfer, not in the totals: wages, $\sum wL\Delta \ln w$	−1.1			[−4.3, 2.1]

Notes: All zones together. The account combines the values of entrant cohorts with the fence's activity in 2006; the labor wedge and the new-activity share are scenarios, the ranges over rules are not confidence intervals, and the agglomeration and exporters' gains are kept out of the totals. Price of the fence: best coarse-tier rule, no re-entry; the range over rules is the identified set of Section 7.2, not a sampling interval. Incumbents: value of plots lost under the best rule, no relocation. Taxes: 1.41 percent of 2006 fenced exports at 8.0 yuan per US dollar; the local share applies the tax-sharing rules of the period (a quarter of value-added tax, two fifths of enterprise income tax, all other taxes) to the fenced firms' tax mix. Jobs: the fence's 1.25 million jobs (0.19 million matched to the survey) at the pre-opening wage per worker within 10 km, 15,300 yuan. Agglomeration: change in sales per remaining incumbent times incumbents' operating profit within 10 km, 60 billion yuan.

the parent development zones admitted fewer new domestic establishments, in a step that left the most productive entrants untouched and turned away the rest in similar proportions. Growth appeared instead 40 to 100 km away, in the zones' provinces.

Read through a model of how the parent zone allocates its land, the step in entry is a price: the value of the entry turned away to make room for the fence, 3.5 billion yuan a year. The price is high because the rule was coarse. A parent zone that admitted a high tier for sure and rationed the rest at random turned away firms of every size, so the loss came close to that of a random allocation, while an allocation by value would have lost a small fraction of it. The cost fell on firms that never entered and therefore appear in no data on the zones, and the scarcity rent stayed with the firms that were admitted.

For the policy question, three conclusions follow. First, the local account of the zones is close to balance: the fence's taxes roughly offset the price of the land it took, and the zones' local gain depends on whether the jobs they created paid more than the workers' alternatives. Second, the national account depends on how much of the fenced activity was new to China, which outcomes around the zones cannot reveal. Third, and independent of both, the larger margin was the rule rather than the land: selling the remaining land by bids per hectare would have recovered about half of the price under the best-fitting rule, and between 0 and 61 percent across the rules the data accept, without returning any land.

The analysis has limits that suggest further work. The price of the fence is identified from the change in entry, but how much of it is due to the rule rather than to scarcity depends on whether the parent zones rationed before the fences arrived. The agglomeration gain to firms near the fence is imprecise. And the growth 40 to 100 km away is located, not attributed. Records of individual land applications, which would show who applied and who was refused, would resolve the first; the second and third need variation in where the zones were placed that our comparison does not provide.

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Appendix

Guide to the appendix

The appendix has five parts. Part I documents the setting and the data. Part II reports the design of the main text in detail and checks it. Part III reports the main results on a second comparison, never-exposed cells matched on region, port access and 1998 size, the finer decompositions that need its precision, and supporting evidence on relocation, plant size and factor prices. Part IV documents the evidence on learning, and Part V the model. Each row below names the statement in the main text that the appendix supports.

	Contents	Supports
Part I. Setting and data		
A	Policy and institutions: approval batches, opening dates, admission into the fence, siting inside development zones, customs rules, zone types, the fence's industry mix	Section 2: what the fence separates and how land is allocated
B	Data construction: the firm survey, customs records, the zone directory, input–output links, relocation records, corrections	Section 3: outcomes and definitions
C	Which zones enter which analysis	Sections 3–8
D	Geocoding and sample composition	Section 3: cell totals
E	Symbols	All sections
Part II. The later-treated design		
F	The later zones and comparison cells; anticipation; estimates by opening wave; leaving out one province; sensitivity to pre-opening trends	Section 4: no growth within 10 km, the step in entry, growth 40–100 km away
Part III. Alternative comparisons and supporting evidence		
G	Design, results and mechanisms on matched never-exposed cells: output within 10 km by wave, who enters, land, the hinterland, supplier industries, how the hinterland grew	Sections 3, 4, 5: the facts on a second comparison and their decompositions
I	Supplementary exhibits: the export series, outcomes within 40 km, timing, binary designs, related products, the hinterland 100–150 km away	Sections 2–5
H	Inference with few treated places, matched specification	Appendix G
J	Estimation details of the matched comparison	Appendix G
K, L	Cell and year effects only; specification checks	Appendix G
M, O, P, Q	The hinterland: robustness, relocation, plant size, factor prices	Section 5.5
Part IV. Learning		
S	Design of the learning tests, checks, timing, distance, persistence and size	Section 5.3: the compressed account
R	Related products and copied markets: all specifications (see also Appendix I)	Section 5.3: the main estimates
Part V. The model		
T	Functional forms, proofs, sorting by	Sections 6–8

Part I. Setting and Data

A Policy and institutions

This appendix documents the policy behind Section 2: which zones were approved and when, how a zone's opening is dated, who is admitted to a fence, where a fence is placed, which customs rules govern it and how they changed, how the EPZ compares with other zone types, what the 2005 extension of carry-over did, and which level of government decides.

A.1 Origins and approvals

Export processing zones (EPZs) are a customs special supervision zone created by the State Council to concentrate processing-trade activity under unified customs supervision and to curb smuggling that scattered processing arrangements had made hard to police.[1] They are the oldest and most common form of special economic zone: most of the roughly 5,400 zones operating in more than 140 economies in 2019 offer duty-free processing for export behind a customs fence (UNCTAD 2019; Farole 2011). By 2000 processing trade, imported inputs assembled for re-export, made up more than half of China's exports but was scattered across thousands of bonded factories, each separately inspected and bonded.

The State Council approved the first wave of 15 zones on April 27, 2000: Beijing Tianzhu, Tianjin, Dalian, Hunchun, Yantai, Weihai, Kunshan, Suzhou Industrial Park, Songjiang, Hangzhou, Xiamen, Shenzhen, Guangzhou, Wuhan, and Chengdu. Further approvals followed in 2001–2003 (24 zones) and 2005–2006 (19 zones), and one zone each was approved in 2010 (Hefei) and 2011 (Jinggangshan), for 60 directory entries in all. A zone's planned area was capped at 3 km^2; zones that grew beyond it received a separately approved second area, which is why several zone names carry "and Area B." One entry, Jinqiao, duplicates another zone under the name of an older development zone: its directory entry dates to a 1990 development-zone designation rather than to a customs-registered EPZ, and the zone itself appears separately as the Jinqiao South Area. The analysis therefore uses 59 zones, of which 52 had opened by 2007 (Appendix Table C1 lists them).

A.2 From approval to customs operation

Approval by the State Council and the start of customs supervision are distinct dates. A prefecture applied through its province and the State Council approved. The host prefecture

[1] General Office of the State Council (2002); see Kee and Tang (2016) and Koopman et al. (2012) for the broader processing-trade regime that EPZs sit inside.

then built the fence, a customs office dedicated to the zone with round-the-clock supervision, and the infrastructure inside; customs and other central agencies inspected the zone jointly before it began operating. Shenzhen, for example, was approved in the first wave but its zone was accepted into customs operation only on March 31, 2001, and Tianjin's on June 29, 2001. Each zone received its own customs district code, so a firm legally inside a fence is identified by its customs registration code rather than by its address (Appendix B.3).

The paper dates every opening by customs operation. A zone's opening year is the year of its official customs acceptance, counted from the following year when acceptance falls in the second half of a year; an official acceptance date is recorded for 36 of the 59 zones. For the remaining 23 zones the opening year is the approval year plus two, the median lag between approval and acceptance among zones with a recorded date. The median zone began operating 2 years after approval (Figure A1), and approval itself set off construction and investment promotion, so approval and operation are not interchangeable timing conventions. Appendix Table C1 reports for each zone which kind of date it has.

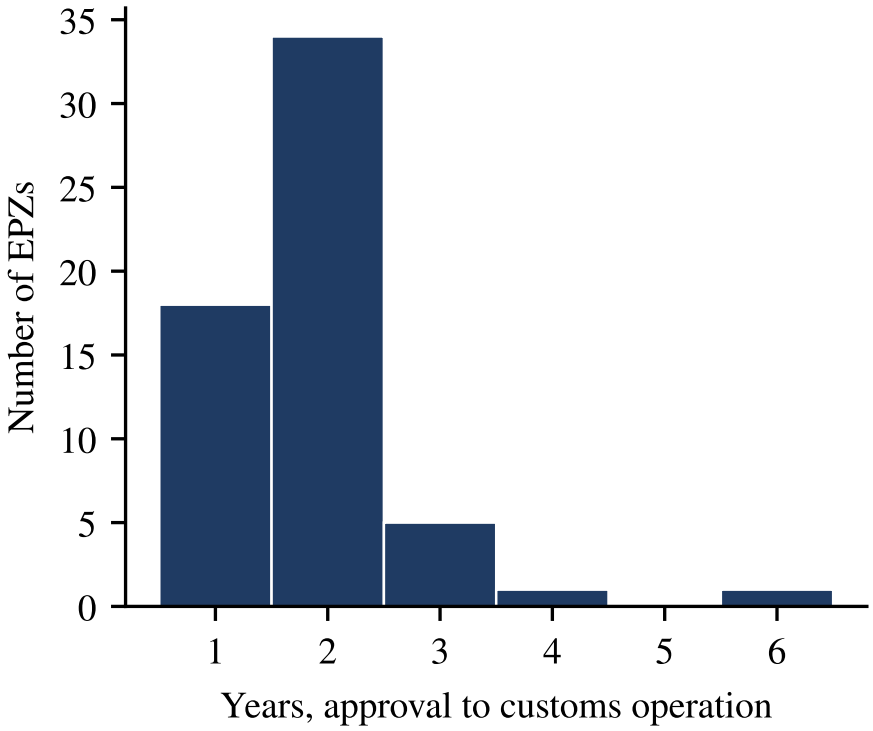


Figure A1: Years from State Council approval to customs operation

Notes: 59 EPZs; operation is the official customs acceptance year where available (36 zones), otherwise approval plus two years, so the lag is two years by construction for the zones without a recorded acceptance date. Median 2 years.

A.3 Investment promotion, firm admission, and who is inside the fence

Once customs accepted the zone, the host prefecture's zone administration recruited firms into the fence; investment promotion, like land and infrastructure, is the host prefecture's own responsibility (Appendix A.10). The founding customs order allowed three kinds of firms inside: export processors, warehousing firms serving them, and transport firms licensed to move goods in and out of the zone. The resulting activity is concentrated and foreign: among firms with customs records inside an EPZ, foreign-invested firms account for 99 percent of fenced exports in 2000–2006, and processing with imports, in which the zone firm itself owns the imported inputs and the exported output, as opposed to pure assembly on a foreign principal's account, accounts for 100 percent of fenced processing exports.

Admission rules broadened over time. The founding customs order restricted zone firms to export processing only, unchanged through a 2011 revision. Successive entry guidelines in 2012, 2018, and 2021 progressively widened the menu of admissible activity to include bonded logistics, research, testing and repair, entrepôt trade, domestic sales, and processing on commission for firms outside the fence, classifying admissible investments by the combination of input sources and output destinations that fits the tax regime; the 2018 guideline, for instance, lists investments using only domestic inputs as admissible. None of the guidelines conditions entry on local sourcing or on linkages with local firms.

If the EPZ merely relocated the processing firms a development zone would host anyway, the firms inside should look like processing exporters outside it. Table A1 compares fenced exporters with processing-trade and ordinary-trade exporters in the same customs districts. By firm count, 75 percent of fenced exporters are foreign-invested, close to the 81 percent among processing exporters outside and far above the 28 percent among ordinary-trade exporters; weighted by exports, fenced firms are almost entirely foreign-invested. What sets the fence apart is organization, not ownership: processing with imports is 100 percent of fenced processing exports against 76 percent outside. Fenced firms are not selected on productivity: their mean within-industry TFP percentile in their first four operating years is 0.481, against 0.503 for processing exporters outside. And they leave little locally. The wage bill of the fenced firms the firm survey records is 1.1 percent of the median zone's fenced exports, a lower bound because the survey records only some of the fenced exporters; relative to those firms' own exports, wages are 6 percent in the median zone. The domestic value added ratio (DVAR) of fenced exports is in Table 1. The industry mix of the fence depends on the weighting: electronics is 87 percent of fenced exports, 21 percent in the median zone, and 55 percent of fenced firms on a broad definition (Appendix A.8).

Table A1: Who is inside the fence

	Fenced exporters	Outside the fence, processing trade	Outside the fence, ordinary trade
Foreign-invested (% of firms)	75	81	28
Main export in HS chapter 84, 85, or 90 (% of firms)	55	31	26
Processing-with-imports share of processing exports, 2000–2006 (%)	100	76	–
Median export value (US$10,000)	41.6	38.6	14.9
Within-industry-year TFP percentile, own first four years (mean)	0.481	0.503	0.533
Wage bill / exports, first four years (%)	1.1	7.2	18.8

Notes: Customs data, 2000–2011. Fenced exporters: firms whose registration code places them inside an EPZ. The two comparison groups are exporters in the same customs districts that are in no special customs zone, split by whether processing is at least half of their exports. Each firm enters once, in its first year in the data. The first two rows count firms; the processing-with-imports row is export-weighted and uses firms first observed in 2000–2006, the years in which customs separate the two processing regimes. TFP percentile: mean within-industry-year percentile of fenced firms matched to the firm survey, production function estimated by the control-function method. Wage bill over exports: for fenced firms, the wage bill of the firms matched to the firm survey over all of the zone's fenced exports, median across zones, which is a lower bound on the fence's wage share because unmatched firms' exports enter the denominator but not their wages; for the comparison groups, over the firms' own exports. "–": not available for the comparison group.

A.4 Siting inside existing development zones

The founding customs regulation allowed an EPZ only inside an existing development zone approved by the State Council, one EPZ per development zone, with a planned area of at most 3 km^2. In practice EPZs were placed inside national economic and technological development zones or high-technology zones, with Qingdao and Beihai among the few sited outside one. Of the 60 directory entries, 44 sit inside a parent development zone, 13 adjoin one, and 2 stand alone; the sixtieth is the duplicate entry of Appendix A.1.

Land, investment promotion, and infrastructure for the EPZ are the host prefecture's responsibility, so an EPZ occupies development-zone land and admission capacity that the host prefecture could otherwise have allocated to other firms: plots, entry approvals, and administrative attention come out of the same allocation. This is the institutional basis for treating land next to a fence as an allocation made by the host prefecture (Section 6). Because the customs fence is built inside a pre-existing development zone rather than on a bare site, distinguishing a newly established EPZ from the mere conversion, renaming, or later revocation of an existing special-zone designation matters for dating an opening (Appendix B.4).

A.5 Customs rules and later reforms

Rules at the founding. Inside the fence, firms imported materials and equipment free of duty and exported without tax, but the founding regulation confined them to export processing: ordinary trade, entrepôt trade, and retail were prohibited, and output could not be sold on

the domestic market; the prohibition survived the 2011 revision. Deep-processing carry-over, the sale of fenced components to processing firms outside the fence, was piloted only in the Kunshan and Songjiang zones in 2002 and extended to all zones by General Administration of Customs Order No. 126, effective May 1, 2005, the only reform inside the firm-survey window that identifiably opened a channel between zone and non-zone firms. A firm subcontracting a stage of production to a supplier outside the fence had to post a bond equal to the tariffs and import value-added tax that would otherwise have been due on the subcontracted materials. Domestic materials bought by zone firms qualified for export rebates only under a 2008 rule, and only for listed products. The 2007 deposit requirement on processing trade in restricted products, which applied to processing firms elsewhere, did not apply inside customs special supervision zones. The test of the 2005 extension is in Appendix A.9.

Later reforms. The customs regime was not static. Between 2002 and 2019 the central government issued a sequence of rules that progressively lowered the cost of transactions between firms inside a zone and firms outside it (Table A2). Two features of the sequence matter for empirical work. First, the reforms that most directly open the fence—integration into comprehensive bonded zones, value-added-tax general-taxpayer status for zone firms, and selective tariffs on domestic sales—date from 2015 or later. Second, only two reforms change the rules within the period covered by firm-level customs records with registration codes (2000–2016): the 2005 national extension of carry-over, and the conversion of individual zones into comprehensive bonded zones, which occurred zone by zone rather than all at once. The comprehensive bonded zones that absorbed the EPZs numbered 168 in 2024 and handled about a fifth of China's foreign trade (General Administration of Customs of China 2025).

No central document diagnoses weak local linkages of export processing zones as such: the recurring official diagnosis of zones through 2011–2012 concerns land use, operating returns, and the proliferation of zone types, and local linkage enters policy documents as an objective only from 2011–2012. Official statements in the same period also assert that processing trade had already "driven the development of supporting industries," citing a rise in domestic purchases and value added from 32 to 44 percent of processing exports between 2002 and 2012; that ratio is measured by place of production and includes foreign-owned suppliers operating in China, so it is the domestic value added concept of Kee and Tang (2016) and Koopman et al. (2012), not a measure of local firms' capabilities.

The official *Catalog of China's Development Zones* lists every zone with its approval date and tracks the conversion into comprehensive bonded zones: the 2006 edition lists 58 export processing zones; the 2018 edition lists 135 special customs zones, in which zones that had been converted appear under their comprehensive-bonded-zone name and approval date, and 27 export processing zones appear unchanged. Matching the two editions by city

Table A2: Rules governing transactions across the fence of export processing zones, 2000–2020

Year	Document	Content
2000	Customs General Administration Order 81 (revised 2003, 2011)	Zone firms may only conduct export processing; ordinary trade, entrepôt trade, and retail are prohibited. The prohibition is unchanged in the 2011 revision.
2002	State Council General Office letter [2002] 64	Pilot of deep-processing carry-over (bonded sales of intermediate goods from zone firms to processing firms outside the fence) in the Kunshan and Songjiang zones.
2005	Customs General Administration Order 126	Carry-over extended to all export processing zones from May 1, 2005.
2008	Ministry of Finance, Customs, and Tax Administration notice [2008] 10	Listed domestic inputs purchased by zone firms regain export rebates when they enter the zone.
2011	Ministry of Commerce and five ministries, [2011] 448	Processing trade described as of "low growth quality and returns"; zones instructed to attract "large firms with strong capacity to drive local supporting industries."
2012	State Council [2012] 58	First stand-alone task to "cultivate supporting capacity outside the zones" and to "bring qualified firms into processing-trade production and supply chains"; zones with low land use and poor performance to be rectified.
2015	State Council General Office [2015] 66	Integration of special customs zones into comprehensive bonded zones; zone firms allowed to take processing orders from firms outside the fence.
2016	Tax Administration, Finance, Customs announcement 2016 no. 65 (extended 2018, 2019)	Value-added-tax general-taxpayer status for zone firms, starting with seven zones: purchases from domestic suppliers outside the fence become creditable.
2016	Finance and Customs [2016] 40; announcement 2020 no. 20	Selective tariffs on domestic sales of zone output, extended to all comprehensive bonded zones in 2020.
2019	State Council [2019] 3	Twenty-one measures, including processing on commission for firms outside the fence.
2020	Customs and five ministries, performance evaluation rules	The only quantitative linkage indicator ("radiation and service": second-line trade with the rest of China) receives 10 percent of the score.

Notes: Document titles are translated; the 2009 functional expansion of export processing zones (bonded logistics, research and development, testing, and repair) is documented indirectly in the 2011 and 2012 texts and its originating document number is omitted.

and name gives the conversion date of each zone; zones whose successor cannot be identified unambiguously are treated as unmatched. By 2024 no zone of the export-processing type remained in official counts.

A.6 Five zone types compared

Table A3 sets the EPZ beside the four other zone types a Chinese prefecture could host, and Table A4 adds each type's approval authority and main preferences. Three points qualify them. First, national economic and technological development zones and high-technology zones are labels for bundles of preferences whose content varied across places and years; the rules we can document from primary legal sources are income-tax rates, 15 percent for productive foreign-invested firms inside a national economic and technological development zone (1991–2007) and for certified high-technology enterprises inside a national high-technology zone (1991–2006), and each applies to a firm inside the zone's legal boundary that meets the stated conditions. These zones are not customs special supervision zones, so no customs rule separates their firms from the rest of the economy; their firms sell at home and buy from and subcontract to local firms without customs restriction. Second, the customs special supervision zones form a sequence: bonded zones from 1987, EPZs from 2000, bonded logistics parks and bonded ports in the mid-2000s, and comprehensive bonded zones from the Suzhou Industrial Park pilot of December 2006. A 2012 State Council guideline made the comprehensive bonded zone the only type for new special customs zones, and the 2015 integration plan converted the existing EPZs zone by zone (Figure A2). Third, the comprehensive bonded zone reversed the EPZ's defining restrictions in stages (Table A2) and added a performance evaluation in 2020 under which a zone graded lowest for two consecutive years and ranked among the bottom five nationally starts an exit procedure; the only quantitative linkage indicator in that evaluation measures goods moving between the zone and the rest of China and carries a tenth of the score.

Among the types whose rules we can document, the EPZ is the one that both confined its firms to export processing and restricted their dealings with firms outside. The comprehensive bonded zones that absorbed the EPZs reopened those dealings step by step: processing orders from firms outside, creditable purchases from domestic suppliers, and domestic sales at selective tariffs. Neither the EPZ rules nor the later admission guidelines condition entry on local sourcing.

Table A3: Five zone types a Chinese prefecture could host

Zone type (first year)	Customs line	Sales to the domestic market	Transactions with firms outside	Siting and admission
Economic and technological development zone (1984)	None: not a customs special supervision zone	Unrestricted (no customs line)	Unrestricted (no customs line)	—
High-technology development zone (1988)	None: not a customs special supervision zone	Unrestricted (no customs line)	Unrestricted (no customs line)	—
Bonded zone (1987)	Yes: bonded warehousing, processing, and trade	—	—	—
Export processing zone (2000)	Yes: dedicated customs office, round-the-clock supervision; export processing only	No: general trade and retail prohibited; selective-tariff pilots in a few zones from 2016	Restricted: carry-over sales to outside processors from 2005 (two pilots in 2002); outward processing only against a bond; rebates on listed domestic inputs from 2008	Inside a State-Council-approved development zone, one per zone, up to 3 km^2; export processors and the warehousing and transport firms serving them
Comprehensive bonded zone (2006)	Yes: bonded processing, logistics, research, testing, and repair	Yes, paying duties on entry; selective tariffs in all zones from 2020	Processing orders from outside firms (2015, 2019); creditable purchases from domestic suppliers (2016 pilot)	The only type for new zones since 2012; EPZs converted from 2015; projects screened on input sources and output destinations

Notes: National-level zones, all approved by the State Council (development and high-technology zones also exist at the provincial level). First year: earliest approval year of a zone of that type in the official Catalog of China's Development Zones; for comprehensive bonded zones, the Suzhou Industrial Park pilot (December 2006). "Unrestricted (no customs line)": development and high-technology zones are not customs special supervision zones, so no customs rule limits these transactions. "—": no rule we could document from primary sources. No documented admission rule conditions entry on local sourcing. Approval and main preferences: Table A4; later rules: Table A2.

Table A4: Approval and main preferences of the five zone types

Zone type (first year)	Approval	Main preferences for firms inside
Economic and technological development zone (1984)	State Council	Tax, land, and approval preferences varying by place and year; e.g., 15% income tax for productive foreign-invested firms (1991–2007)
High-technology development zone (1988)	State Council	Same kind of bundle; e.g., 15% income tax for certified high-technology firms (1991–2006)
Bonded zone (1987)	State Council	Bonded warehousing, processing, and trade behind a customs fence
Export processing zone (2000)	State Council, on application through the province	Duty-free imports of materials and equipment; tax-free exports; export processing only
Comprehensive bonded zone (2006)	State Council, zone by zone	Bonded processing, logistics, research, testing, and repair; general-taxpayer VAT status piloted from 2016

Notes: National-level zones; development and high-technology zones also exist at the provincial level, approved by provinces. First year: earliest approval year of a zone of that type in the official Catalog of China's Development Zones; for comprehensive bonded zones, the Suzhou Industrial Park pilot (December 2006). Bonded zones and EPZs were exempt from the 2007 deposit on processing trade in restricted products.

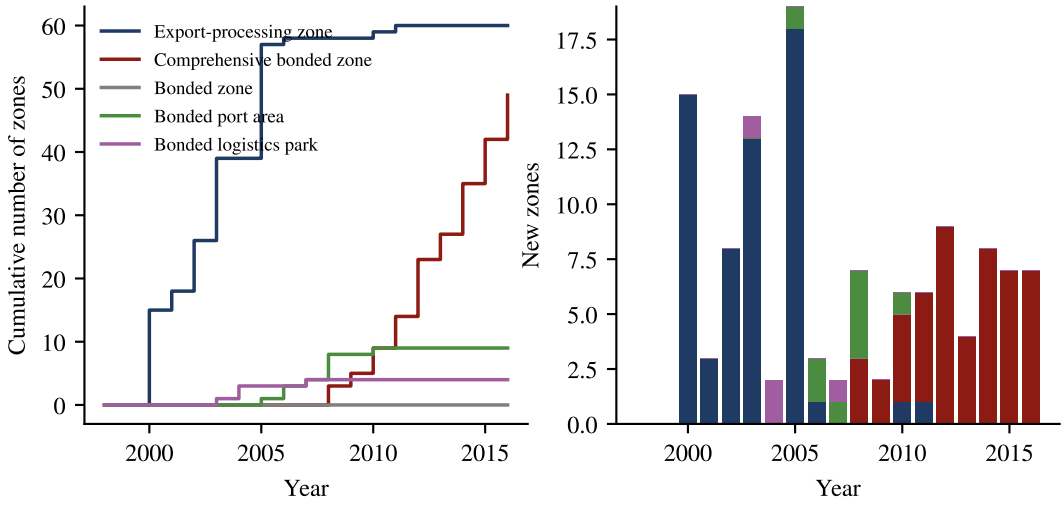


Figure A2: Special customs supervision zones by type and year

Notes: Zone directory. EPZs are identified by name. Left: cumulative number of zones; right: new zones per year. Comprehensive bonded zones absorbed most EPZs after 2015.

A.7 Where China's exports were registered

The fifth digit of a customs registration code records the kind of zone in which a firm is registered. Pooling all exports over 2000–2011, firms registered in no designated zone shipped 69.8 percent of China's exports. Firms in national economic and technological development zones shipped 9.4 percent, special economic zones 6.1 percent, EPZs 5.8 percent, national high-technology zones 4.1 percent, bonded zones 3.4 percent, comprehensive bonded zones and bonded ports 0.5 percent, and bonded logistics parks 0.2 percent; the remaining 0.7 percent carry no zone code. All customs special supervision zones together account for 10.0 percent. The EPZs were thus one instrument for concentrating processing trade, not the place where most of it happened: the fences shipped a small minority of national exports throughout (Figure I1 and Figure A3).

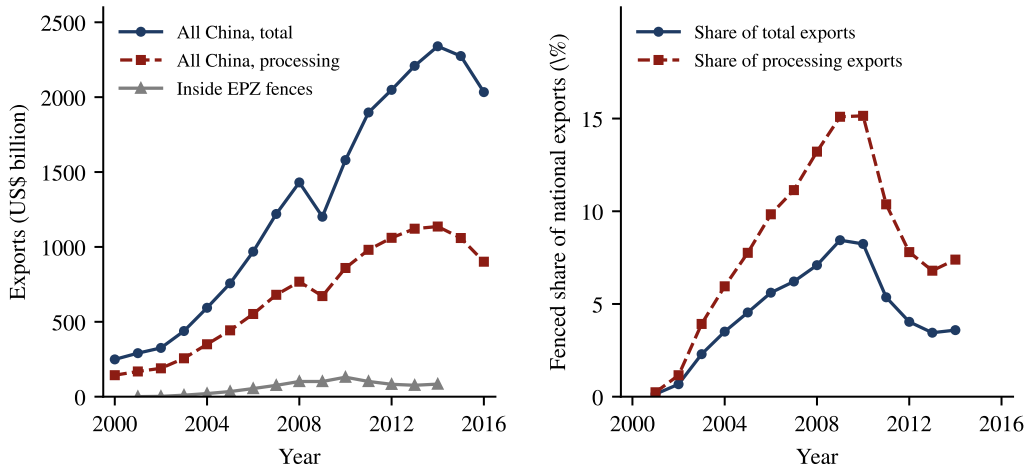


Figure A3: China's exports and exports from inside EPZ fences, 2000–2016

Notes: Fenced exports are those of firms whose customs identifier marks an EPZ in the year of the record (2001–2014). The fall after 2010 is mostly recoding: as zones were converted into comprehensive bonded zones, their firms re-registered under that zone type's identifier (between 2010 and 2011, 130 to 102 billion US dollars under the EPZ identifier against 4.6 to 53 billion under the comprehensive-bonded-zone identifier). Processing exports are total exports minus ordinary trade in every year.

A.8 The fence's industry mix: three measures

The industry composition of fenced firms is used twice in the paper: to describe which industries they are in, and to define their suppliers in the input–output analysis (Appendix B.5). Products are mapped from six-digit HS codes to Chinese industries and then to the 42 sectors of the 2002 input–output table; a manual rule at the four-digit HS level covers the few codes the concordance misses. Electronics is the sector "communication equipment, computers, and other electronic equipment," which contains computers and their parts (HS 8471 and 8473)

and the telecommunications, audio-visual, electronic-component, and semiconductor headings of chapter 85 (HS 8517–8529, 8534, 8540–8542); general and special-purpose machinery, office machinery, and electrical machinery are separate sectors.

Three measures answer different questions. (i) By export value: weighted by exports and pooled over all fenced firms in 2000–2006, electronics is 87 percent of fenced exports. (ii) Zone by zone: over each zone's first three operating years, among the 37 zones whose fenced exports can be assigned to a named zone, the median zone's electronics share is 21 percent, 14 zones have more than half their fenced exports in electronics, and 9 export none. (iii) By number of firms: the share of fenced firms whose main export is in HS chapters 84, 85, or 90, a broad definition that includes general machinery and instruments, is 55 percent (Table A1). The export-weighted share is high because most of the largest fences, the hubs, are electronics fences; the median zone is not an electronics enclave, and a statement about the fence as a whole describes the hubs. None of these product-based measures is affected by the 2003 revision of the Chinese industry classification (Appendix B.7).

A.9 The 2005 extension of carry-over

Were the zones enclaves because of the rules, or because of what local firms could do with the fence's products? The rules changed once inside the firm-survey window. Deep-processing carry-over, the transfer of fenced output to processing firms outside the fence for further processing and re-export, not general domestic-market access for the fence's products, was permitted from 2002 in two pilot zones, Kunshan and Songjiang, and extended to all EPZs in May 2005. The reform opened the forward linkage from the fence to downstream users. It did not change the terms on which local suppliers sold into the fence: domestic inputs bought by zone firms regained export rebates only under a 2008 rule, and only for listed products. Downstream user industries—general and special equipment, transport equipment, electrical machinery, and instruments, which each use at least 0.02 units of the fence's products per unit of output (Appendix B.5)—should have gained relative to other industries within 20 km of zones already in operation if local processors could use the fence's components; upstream industries, on which the reform did not act, serve as a within-location comparison. Table 1 shows the other side of the reform: the domestic-sales share of the fenced firms did not move after 2005.

Table A5 shows that relaxing the carry-over restriction generated no detectable short-run expansion of eligible downstream industries over the 2005–2007 post-reform window available in the firm survey: for employment the estimates rule out increases above 10 percent (output: upper bound 23 percent). Relative to other industries, output of downstream users within 20

Table A5: The 2005 extension of carry-over transfers

	(1) Extension, 2005	(2) Extension, excl. 2005	(3) Placebo: bonded zones	(4) Pilot zones, 2003
Panel A. Downstream user industries				
Output	−0.067	−0.083	0.095	0.029
	(0.140)	(0.150)	(0.138)	(0.155)
Employment	−0.026	−0.048	0.127	0.227
	(0.061)	(0.076)	(0.106)	(0.172)
Export delivery value	−0.220	−0.257	−0.220	−0.245
	(0.116)	(0.122)	(0.082)	(0.195)
Panel B. Upstream supplier industries				
Output	−0.073	−0.096	−0.007	−0.046
	(0.108)	(0.120)	(0.145)	(0.106)
Employment	−0.076	−0.065	0.056	−0.018
	(0.085)	(0.093)	(0.086)	(0.135)
Export delivery value	0.088	0.053	−0.103	0.133
	(0.130)	(0.147)	(0.136)	(0.157)
Panel C. Electronics				
Output	−0.105	−0.138	0.098	1.123
	(0.161)	(0.172)	(0.136)	(0.155)
Employment	0.171	0.155	0.283	0.527
	(0.151)	(0.166)	(0.148)	(0.250)
Export delivery value	−0.076	−0.133	0.062	1.048
	(0.147)	(0.154)	(0.111)	(0.159)
Treated grid cells	267	267	21	24

Notes: Coefficients on $T_c \, \text{Post}_t \, \mathbb{1}[k]$ for each industry group k in a Poisson triple difference with cell × group, cell × year, and group × year fixed effects, 1999–2007. Treated cells lie within 20 km of an EPZ operating by 2004, excluding the two pilot zones; comparison cells are those around special customs zones that open later, with no special customs zone within 40 km before 2008 and one by 2013. Column (2) drops the transition year 2005. Column (3) treats cells near 1990s bonded zones, to which the 2005 rule did not apply. Column (4) treats cells near the two pilot zones with 2003 as the first post year. Downstream users: industries with direct requirements of the fence's products of at least 0.02 per unit of output. Export delivery value is reported in the firm survey. Standard errors clustered by prefecture in parentheses.

km of operating zones changes by −0.067 (0.140) and employment by −0.026 (0.061). Their export delivery value falls (−0.220, s.e. 0.116), but by the same amount near bonded zones, where the reform did not apply (−0.220, s.e. 0.082); the decline reflects a common trend in how processing exporters report, not the reform. The event study in Figure A4 shows no break in 2005.

The pilot zones show larger point estimates for downstream employment (0.227, s.e. 0.172) and a large relative expansion of electronics after 2003. Two zones in one of the densest electronics clusters in China cannot separate the pilot rule from the scale of foreign investment that arrived there at the same time, and we do not read these estimates as the effect of

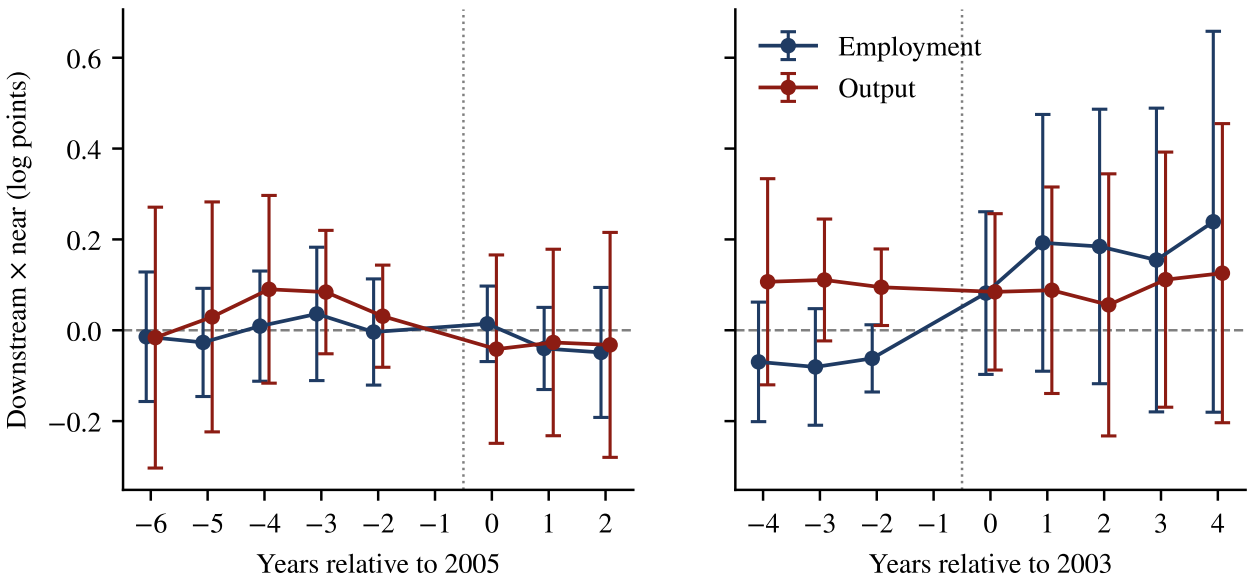


Figure A4: Downstream industries around the carry-over reform

Notes: Coefficients on treated cell × year × downstream industry from the triple difference in Table A5. Left: all operating zones except the pilots, reference year 2004. Right: Kunshan and Songjiang, reference year 2002. Bars are 95 percent confidence intervals clustered by prefecture.

carry-over.

Before 2005, fenced firms outside the two pilot zones could not transfer components to processing firms elsewhere under bond; once they could, the users did not expand. Regulatory permission was therefore not sufficient to generate forward linkages in 2005–2007: local processing firms that could use the fence's components did not expand once they were allowed to buy them. This does not mean that regulation never matters for these linkages; it shows that removing this particular restriction was not, on its own, enough. Later reforms that equalized tax treatment inside and outside the fence, and the conversion of EPZs into comprehensive bonded zones after 2015, fall outside the firm-survey window (Appendix A.5).

A.10 The prefecture as the decision unit

In China's administrative hierarchy, the prefecture-level city is the effective decision-making unit for the policy instruments relevant to a zone: land allocation, investment promotion, most industrial policy, and the fiscal revenue that a zone generates, and the unit along which local officials are evaluated. An export processing zone is proposed and operated by the local government, ordinarily the prefecture, and approved by the State Council; once approved, the zone's land, infrastructure, and investment promotion are the host prefecture's responsibility, and the zone's tax revenue accrues to the host prefecture. Municipalities directly under the central government (Beijing, Tianjin, Shanghai, Chongqing) are a special case: their districts

and counties do not hold the same administrative powers as an ordinary prefecture-level city, so the relocation records treat each municipality as a single prefecture and a single province (Appendix B.6).

A firm's legal identity is a registration in a specific county-level jurisdiction, and it pays tax where it is registered. Moving a firm's registration across a prefecture boundary requires changing or re-filing its tax registration; moving it across a provincial boundary ordinarily requires deregistering the firm and registering it anew, which can break the cross-year identifier that links a firm's records over time (Appendix B.6). By contrast, the channels through which a place-based policy transmits an economic shock across a jurisdictional boundary, commuting, input purchases, and output sales, do not require any change in registration. Migrant labor inflows are a further source of asymmetry: they make local labor supply highly elastic at the scale of an EPZ's labor demand, so a zone's effect on nearby wages need not resemble its effect on nearby firm entry or on the allocation of development-zone land.

B Data construction

This appendix describes how the data of Section 3 are constructed and lists the data traps that the paper handles.

B.1 The annual survey of industrial firms

Coverage. The firm-level data are China's Annual Survey of Industrial Firms, 1998–2007. The survey covers all state-owned firms and all non-state firms with sales above 5 million RMB (Brandt et al. 2012); the paper applies no cut on employment or on any other measure of size. Firms are followed across years using the data provider's cross-year firm identifier, which links a firm's records across survey waves under a stable number.

The 2004 census year. The 2004 wave of the survey was assembled from the economic census and contains two kinds of new records: census passers-by, firms that appear in 2004 only, and established firms registered for the first time that remain in the survey afterwards, the census additions. The paper keeps every 2004 record. Firms first recorded in 2004 but founded earlier, the passers-by among them, are not counted as new establishments, because they were founded before the year in which they first appear (below). Appendix Table M2, panel B, removes all firms first recorded in 2004 and founded before 2003 in every year, and Section 4.4 reports that the hinterland's response does not change.

Entry and exit. A firm is a new establishment in year t if it appears in the survey for the first time in t *and* its self-reported founding year is no earlier than $t-1$; the second condition

excludes firms that are new to the survey only because they passed the size threshold or were brought in by the census, rather than because they were newly established. A firm exits in year t if it is present in $t-1$ and absent in t, and it is assigned to its location in $t-1$. Exit means leaving the survey, which is not the same as closing: a non-state firm leaves when its sales fall below the threshold, and a firm also disappears from the records when it ceases operating or when its identifier changes. Exits and early entries are not reliable on a file that keeps only firms with exact coordinates before 2004 (Appendix B.7).

Value added and output. The 2004 wave does not report value added. To keep one definition in every year, value added is measured as gross output minus intermediate inputs, both reported in every year including 2004, and set to zero where it would be negative; it is 23 to 24 percent of output in every year from 1998 to 2007. All money values are nominal and are aggregated over the firms of a cell before estimation. In the 2010 wave gross output is stored as zero for every firm rather than as missing; the firm panel used here ends in 2007 and no analysis uses output from 2010.

Industry codes. Analyses that compare industries across years use a single two-digit industry code for each firm that does not change over time, in the 2003 revision of the national industry classification (GB/T 4754-2002). The classification changed in 2003: several manufacturing codes in the tail of the schedule shifted by one digit, so that a firm coded 40 under the pre-2003 classification ("electrical machinery") corresponds to code 39 under the 2002 classification, while code 40 under the 2002 classification is instead "electronics and telecommunications equipment." The full correspondence used is $\{40\to39,\ 41\to40,\ 42\to41,\ 43\to42\}$. Firms with any post-2003 record are assigned the mode of their post-2003 codes; firms observed only before 2003 are assigned the mode of their pre-2003 codes after applying this correspondence. Grouping firms by their *current-year* reported code instead, without this harmonization, misclassifies pre-2003 electrical-machinery firms as electronics, because 94.5 percent of firms reporting the old code 40 in a pre-2003 year report code 39 from 2003 onward. The electronics share used as a baseline covariate and the input–output industry groups use the harmonized code. Measures defined within an industry and year, such as productivity relative to the industry-year mean and size fifths within industry and year, use the code the firm reports in that year, which the revision does not affect because each year is coded under a single classification; the product classes of the hinterland analysis are assigned from four-digit industry names.

B.2 Locating firms and zones

Firms. A firm is located by its exact coordinates where its address can be geocoded and otherwise by the centroid of its town or county; the share of firm-years located by exact coordinates rises from 33 percent in 1998 to 90 percent in 2007 (Appendix Table D1). The file of the survey that keeps only firms with exact coordinates is a survivor sample before 2004, and the paper therefore uses all surveyed firms, with centroid locations for the rest (Appendix B.7).

Zones. A zone's location is constructed, in order of preference, as: (i) the median coordinates of its legally registered member firms under the customs fifth-digit code, provided at least 5 member firms are matched to coordinates and their median distance to that median point does not exceed 15 km; (ii) failing that, an empirical centroid computed from the locations of firms independently attributed to the zone, provided it is based on at least 10 firms with a median spread of at most 5 km; (iii) failing that, a zone-directory "recommended" coordinate; and (iv) failing all of the above, the zone directory's raw coordinate. This fallback hierarchy privileges coordinates revealed by firms that are legally, rather than merely nominally, inside the fence.

Administrative units. Prefecture-level boundaries are taken from GADM version 4.1, level 2, simplified with a tolerance of approximately 200 meters. Each polygon is assigned to a prefecture by taking the modal administrative code among the firms whose coordinates fall inside it, rather than trusting the polygon dataset's own labels; the firm-level administrative code agrees with the polygon it falls inside for 99.6 percent of firms.

B.3 Customs data

Coverage. The customs data record transactions at the firm × product × destination × year level for 2000–2016. Not every field covers every year. The trade-regime field, which separates processing with imports from pure assembly, is reported only through 2006, so analyses that use the regime are restricted to 2000–2006, and after 2006 processing exports are measured as exports other than ordinary trade (trap 9 below). From 2012 the records no longer carry the postcode that locates newly registered firms. The analyses of exporters' product entry (Section 5.3) and of intermediaries use 2000–2007, the years that overlap the firm survey.

Identifying firms inside a fence. Each customs registration code carries a fifth digit that identifies the customs special-supervision status of the firm's registered location: 1, special economic zone; 2, economic and technological development zone; 3, high-technology zone; 4,

bonded zone; 5, export processing zone; 6, comprehensive bonded zone; 7, bonded logistics park. This fifth digit is a legal fact about where a firm is registered, not an inference from its address or postal code, and is the basis on which a firm is classified as legally located inside an EPZ.

Processing with imports and pure assembly. Customs records distinguish two organizational forms of processing trade. Under *processing with imports*, the exporting firm itself purchases the imported inputs and owns both the inputs and the resulting output. Under *pure assembly*, a foreign principal owns the imported inputs and the output throughout, and the domestic firm performs only the processing step under contract. This distinction is a legal and contractual fact recorded in the trade regime field, not a measure of production technology: two firms performing identical physical transformations can be recorded under either regime depending on who holds title to the goods.

Domestic value added ratio. For a set of fenced firms, the domestic value-added ratio (DVAR) of their exports, as in Kee and Tang (2016), is one minus the ratio of processing imports to processing exports,

$$\text{DVAR} \;=\; 1 - \frac{\text{processing imports}}{\text{processing exports}}.$$

Because processing exports also embed domestic wages and profits that are not themselves evidence of local intermediate-input sourcing, DVAR is an upper bound on the share of a fence's exports actually sourced from local suppliers, and customs records do not identify carry-over transfers to outside firms, so it bounds local intermediate sourcing from above rather than measuring it directly. It can be computed for an individual fence or aggregated across fences. Aggregated over the firms assigned to a named EPZ, using their processing imports and exports in each calendar year, it rose from 0.27 in 2003 to 0.48 in 2006, consistent with the economy-wide rise in the DVAR of Chinese processing exports documented by Kee and Tang (2016). This yearly series is not the measure of Section 2.1 and Table 1, which is computed from all fenced firms over each zone's first four operating years and reported as the median zone (0.30), the export-weighted average (0.34), and the most generous variant (0.57). The two differ in aggregation, calendar years pooled across zones against each zone's own first four years, and in definition, the firms assigned to a named zone against all fenced firms, so their levels cannot be compared directly.

B.4 The zone directory

The zone directory records, for each customs special supervision zone, an approval year, an operating (customs-acceptance) date, and a geographic location, together with the zone's

type: export processing zone, comprehensive bonded zone, bonded port, bonded logistics park, or (plain) bonded zone. EPZs are identified by name and checked against the approval documents, not by the directory's type field (Appendix B.7). How an EPZ's opening year is set is described in Appendix A.2; the approval years of other zones are the directory's own recorded start years. The non-EPZ zones are not the object of study; the later special customs zones enter the land test of Figure Q1.

A change in a location's customs special-supervision status is not always a newly established zone. A site can instead be *converted* from one special-zone type to another, *retained* under a new administrative category, or *renamed*, without a first customs-supervision event actually taking place there; a site's special-zone status can also later be *revoked*. Because these status changes do not represent a first opening, the land test uses only zones opened on new sites, and conversions, retentions, renamings, and revocations of existing designations are not counted as openings.

B.5 Input–output linkage measures

The fence's industry mix. Let w_k be the share of sector k of the 2002 42-sector national input–output table in the exports of firms legally registered inside an EPZ, pooled across all zones over each zone's first three operating years. Fenced firms are identified by the fifth digit of the customs registration code (Appendix B.3), and their HS exports are mapped to input–output sectors. The national mix is used, rather than each zone's own mix, so that industry groups are fixed before any zone opens and do not depend on what a particular zone later attracted. Electronics is the fence's own industry and its share depends on the weighting: 87 percent of fenced exports in 2000–2006, 21 percent in the median zone, and 55 percent of fenced firms on a broad product definition (Appendix A.8).

Upstream and downstream intensity. With A the 42×42 matrix of direct requirements from the 2002 table,

$$U_j = (A\,w)_j, \qquad D_j = (w'A)_j. \tag{B1}$$

U_j is the direct input from sector j required per unit of fence output: how much fenced firms need what j sells (backward linkage). D_j is the input of fence products required per unit of j's output: how much j needs what fenced firms sell (forward linkage). Both are mapped from input–output sectors to two-digit industries of the 2003 national classification, and industries are assigned to firms by the time-invariant harmonized code of Appendix B.1.

Industry groups. Electronics and telecommunications equipment (two-digit code 40) is the fence's own industry. It is treated as a separate group and excluded before any quantile of U_j

or D_j is taken. The groups are:

- *Suppliers*: industries in the top quartile of U_j (excluding electronics). These are the supplier industries of Section 5 (Appendix J). Where the 2005 reform is analyzed, industries classified as downstream users are removed from this group.
- *Downstream users*: industries with $D_j \geq 0.02$ (general equipment 35, special equipment 36, transport equipment 37, electrical machinery 39, and instruments 41). This group is used only in the analysis of the 2005 reform (Appendix A.9), which acted on forward linkages.
- *Electronics*: code 40.
- *Other*: all remaining manufacturing industries, the comparison group in that analysis.

B.6 Firm relocation records

The firm relocation records cover county-to-county moves of above-scale manufacturing firms for 1999–2007, with 9,102 candidate moves and 3,982 confirmed moves. A *candidate* relocation is a change in a firm's recorded county-level administrative code between consecutive appearances in the firm-level data, that is, an origin county-year record and a destination county-year record linked by the same firm identifier with a different county code. A *confirmed* relocation is a candidate that has been independently verified as an actual physical move of the firm's operations, as opposed to a change in recorded location arising from administrative boundary reclassification or a data-entry correction. Municipalities directly under the central government are treated as a single prefecture and a single province for the purpose of classifying a relocation's origin and destination (Appendix A.10).

The records have a known limitation. Because a cross-province move ordinarily requires a firm to deregister in its origin province and register anew in its destination province, a firm's cross-year identifier can fail to survive a cross-province relocation. The records therefore capture within-province moves more completely than cross-province moves: a firm that relocates across a provincial boundary and re-registers under a new identifier will generally not appear in them as a relocation at all, but as an exit at the origin and an unrelated entry at the destination. Appendix O uses the records.

B.7 Data traps and how they are handled

Chinese firm and customs microdata contain structural breaks and coding conventions that can manufacture results. This section lists those that matter for the paper. For each we

state the trap, how it could mislead, how it is handled, and what it would do to a cell-level outcome if left unhandled.

1. The 2003 revision of industry codes. Trap. The national industry classification was revised in 2003, and several two-digit manufacturing codes shifted by one: code 40 means electrical machinery before 2003 and electronics and telecommunications equipment from 2003 on. *Risk.* Grouping firms by the code they report in each year relabels pre-2003 electrical-machinery firms as electronics, producing a jump in "electronics" near the zones exactly when the fence, itself mostly electronics, arrives. *Handling.* Every firm gets one harmonized two-digit code that does not change over time (Appendix B.1), used wherever industries are compared across years, including the electronics share used as a baseline covariate and the input–output groups. *What it would do.* Left uncorrected, cell-level electronics output would jump in 2003 in the cells that hold the earliest zones, and the supplier groups defined by industry would change composition at the same time.

2. The 2004 economic census. Trap. The 2004 survey wave was assembled from the economic census and contains two kinds of new records: census passers-by that appear in 2004 only, and established firms registered for the first time that stay in the survey afterwards. *Risk.* Counting both as entrants inflates entry in 2004; dropping the whole year removes legitimate firms and shortens every event window that crosses it. *Handling.* Every 2004 record is kept, but firms founded before the year they first appear are not counted as new establishments, and a robustness check removes all firms first recorded in 2004 and founded before 2003 (Appendix B.1). Because the 2004 wave does not report value added, value added is measured as output minus intermediate inputs in every year. *What it would do.* Counting census additions as entrants would add spurious new establishments in 2004 to every cell that holds firms the census brought in; using reported value added would leave a hole in 2004 that the year effects would absorb only if it were common to all cells.

3. Output recorded as zero in 2010. Trap. In the 2010 wave of the firm survey, gross output is stored as zero for every firm rather than as missing, while employment is recorded normally. *Risk.* A Poisson regression treats these zeros as real, so any design that reaches 2010 would record a collapse of output in that year. *Handling.* The firm panel ends in 2007 and no analysis uses output from 2010. *What it would do.* Nothing on the reported estimates; it limits how far the firm panel can be extended.

4. A survivor sample in the geocoded file. Trap. The file of the survey that keeps only firms with exact coordinates is a survivor sample before 2004: most firms in the file in 1998–2002 reappear in 2004 or 2007, far more than in the full survey. *Risk.* Exits before 2004 are almost invisible in the file and appear mainly in the later years, and early entry is affected by the

same selection, so exits and early entry measured on it are unreliable. *Handling.* The paper uses all surveyed firms and locates those without exact coordinates at the centroid of their town or county (Appendix Table D1); exit is reported as a record leaving the survey, not as a closure. *What it would do.* On the survivor file, cell-level counts of exiting firms would be too low before 2004 and would rise afterward for mechanical reasons, whatever happens near a zone.

5. Where firms and zones are. Trap. Firm locations come from geocoded addresses where these are available and from town or county centroids otherwise, and the official coordinates of several national zones, including the largest EPZ by exports, lie far from where their member firms actually are. In customs records a minority of firms, mostly small and short-lived, report an address and a name that point to different provinces. *Risk.* A misplaced zone puts cells in the wrong distance group, and a firm placed at a centroid is assigned to the cell that contains the centroid, which can differ from the cell of its plant. *Handling.* A zone is located from the median coordinates of its legally registered member firms where enough of them are geocoded, and from the directory only as a last resort (Appendix B.2). Whether a firm is inside the fence is read from its registration code, not from its location. Customs firms whose address and name conflict are flagged, not dropped; they carry a small share of exports. *What it would do.* Cells near the boundary of a distance group can be assigned to the wrong group; the innermost group, 10 km wide, is the most exposed.

6. Zone types and dates in the zone directory. Trap. The official directory lists 26 of the 60 EPZs, including 8 of the 15 in the first wave, under bonded zones, and for one zone, Caohejing, it records 1992, the year of its parent high-technology park; the EPZ was approved in March 2003 and began operating in March 2004. *Risk.* Splitting EPZs by the directory's type field compares two arbitrary groups of genuine EPZs and invites a spurious "true versus mislabeled" interpretation; a parent-zone year would date an opening more than a decade too early. *Handling.* EPZs are identified by name and checked against the approval documents; openings are dated as in Appendix A.2, and Caohejing's approval year is corrected in every table. *What it would do.* Dating Caohejing by the directory year would place its opening a decade before it began operating, and a split by the type field would sort genuine EPZs into arbitrary groups.

7. Three measures of the fence's electronics share. Trap. "Share of the fence in electronics" can be computed by export value or by firm count, nationally or zone by zone, and with a narrow or broad product definition. *Risk.* The export-weighted national share, 87 percent, describes the hubs, not a typical zone; read as a typical zone, it would suggest that the fence is an electronics enclave everywhere, and that nearby electronics firms are the natural suppliers

or rivals of every fence. *Handling.* We report three measures: 87 percent by export value, 21 percent as the median zone's share (with 14 of 37 measurable zones mainly electronics), and 55 percent by number of firms on a broad definition (Appendix A.8, Table A1). The product classification is by HS code and is therefore unaffected by trap 1. *What it would do.* A statement about the electronics fence would be read as a statement about every zone; the input–output groups use the national mix, fixed before any zone opens, and aggregate facts are best reported zone by zone and without the hubs.

8. Two scopes for the domestic value-added ratio. Trap. The ratio $1 - M/X$ for fenced firms depends on which flows and which firms enter it. The yearly series of Appendix B.3 uses processing imports and processing exports of firms assigned to a named EPZ: 0.27 in 2003, rising to 0.48 in 2006. Counting all imports and exports of every firm registered inside an EPZ, including duty-free equipment and other imports outside the processing regimes, gives 0.20 in 2003; the two firm sets export almost the same amount, so the gap comes from the imports counted. Adding firms of the comprehensive bonded zones and bonded ports that succeeded the EPZs changes nothing before 2007, when they barely traded, but lowers the ratio afterwards: in 2011 it is 0.41 with them and 0.46 without. *Risk.* Comparing ratios across studies, or across the yearly series and the zone-by-zone measure of Table 1, without noting the scope attributes a difference of definition to a difference in the fence. *Handling.* Every ratio is reported with its scope; the paper's measure uses processing imports and exports only and is treated as an upper bound on local sourcing, since it also includes wages and profits paid inside the fence and customs records do not show carry-over sales. *What it would do.* The processing-only ratio is the higher of the two scopes, so any other scope would leave even less locally.

9. Identifying fenced exports year by year. Trap. After 2006 customs no longer separate the two processing regimes, and as zones were converted into comprehensive bonded zones, fenced firms re-registered under new codes. *Risk.* A series built from the regime columns stops in 2006, and a series built from a fixed list of fenced firms shows a spurious decline after 2010; either suggests that the zones faded. *Handling.* Fenced firms are identified each year from that year's registration code, and processing exports are exports other than ordinary trade in every year. The resulting series reproduces the official EPZ export total for 2012 closely; it stops in 2014, before the conversion wave recodes most fenced firms (Figure I1). *What it would do.* Descriptive only; the series plays no part in the cell-level estimates.

C Which zones enter which analysis

Appendix Table C1 lists the 59 EPZs in order of approval with their approval and opening years. The opening year is the year of the zone's official customs acceptance, counted from the following year when acceptance falls in the second half of a year, for the 36 zones with a recorded acceptance date, and the approval year plus two for the other 23 (Appendix A.2). The approval years follow the State Council documents; the zone directory's year for Caohejing (1992) belongs to its parent zone and is corrected to 2003. The five hubs are the zones with the largest fenced exports. The last column marks the zones whose fenced exports can be assigned to a named zone in the customs records.

The analyses of the main text use different sets of zones. By 2007, 52 of the 59 zones had opened (Section 2), and the seven that opened later (2008, 2012, and 2013) have no post-opening year in the firm survey. The domestic value added ratio (DVAR) of fenced exports is computed for 32 zones, from the customs records of the fenced firms over each zone's first four operating years (Table 1). The analysis of new products among non-fenced exporters uses exporters with known locations near 36 zones, with standard errors clustered by the nearest of the 36 (Section 5.3). The land test uses the 49 special customs zones that opened in 2008–2016 on new sites, with no earlier export processing zone, comprehensive bonded zone, bonded port or bonded logistics park within 10 km (Figure Q1): 39 comprehensive bonded zones, 4 bonded ports and 6 of the 7 EPZs of Table C1 that opened after 2007.

Table C1: The 59 EPZs: approval, opening, and customs matching

Zone	City	Approval year	Opening year	Opening date	Hub	Fenced exports matched
Approved on April 27, 2000 (first wave, 15 zones)						
Beijing Tianzhu EPZ	Beijing	2000	2001	Official		Y
Sichuan Chengdu EPZ (incl. West Area)	Chengdu	2000	2002	Imputed		Y
Liaoning Dalian EPZ	Dalian	2000	2001	Official		Y
Guangzhou EPZ	Guangzhou	2000	2002	Imputed		Y
Zhejiang Hangzhou EPZ	Hangzhou	2000	2001	Official	Y	Y
Shanghai Songjiang EPZ (incl. Area B)	Shanghai	2000	2001	Official	Y	Y
Guangdong Shenzhen EPZ	Shenzhen	2000	2001	Official		Y
Jiangsu Kunshan EPZ	Suzhou	2000	2001	Official	Y	Y
Jiangsu Suzhou Industrial Park EPZ (incl. Area B)	Suzhou	2000	2001	Official	Y	Y

Continued on the next page

Zone	City	Approval year	Opening year	Opening date	Hub	Fenced exports matched
Tianjin EPZ	Tianjin	2000	2001	Official		N
Shandong Weihai EPZ	Weihai	2000	2001	Official		Y
Hubei Wuhan EPZ	Wuhan	2000	2001	Official		Y
Fujian Xiamen EPZ	Xiamen	2000	2002	Official		Y
Jilin Hunchun EPZ	Yanbian	2000	2002	Imputed		Y
Shandong Yantai EPZ (incl. Area B)	Yantai	2000	2002	Imputed		Y
Approved in 2001–2003 (23 zones)						
Chongqing EPZ	Chongqing	2001	2003	Official		N
Shanghai Jinqiao EPZ (South Area)	Shanghai	2001	2003	Imputed		N
Inner Mongolia Hohhot EPZ	Hohhot	2002	2008	Official		N
Jiangsu Nantong EPZ	Nantong	2002	2004	Imputed		Y
Zhejiang Ningbo EPZ	Ningbo	2002	2004	Official		Y
Hebei Qinhuangdao EPZ	Qinhuangdao	2002	2004	Imputed		Y
Anhui Wuhu EPZ	Wuhu	2002	2003	Official		Y
Jiangsu Wuxi EPZ	Wuxi	2002	2003	Official	Y	Y
Shaanxi Xi'an EPZ	Xi'an	2002	2004	Official		Y
Henan Zhengzhou EPZ	Zhengzhou	2002	2004	Official		Y
Guangxi Beihai EPZ	Beihai	2003	2005	Official		Y
Zhejiang Jiaxing EPZ	Jiaxing	2003	2005	Imputed		N
Jinan EPZ	Jinan	2003	2006	Official		Y
Jiangsu Lianyungang EPZ	Lianyungang	2003	2004	Official		Y
Jiangsu Nanjing EPZ (incl. South Area)	Nanjing	2003	2005	Imputed		Y
Shandong Qingdao EPZ	Qingdao	2003	2004	Official		Y
Shanghai Caohejing EPZ	Shanghai	2003	2004	Official		Y
Shanghai Qingpu EPZ	Shanghai	2003	2004	Official		Y
Shenyang EPZ	Shenyang	2003	2007	Official		N
Jiangsu Suzhou New & Hi-Tech Zone EPZ	Suzhou	2003	2005	Imputed		Y
Xinjiang Urumqi EPZ	Urumqi	2003	2005	Imputed		Y
Weifang EPZ	Weifang	2003	2006	Official		N
Jiangsu Zhenjiang EPZ	Zhenjiang	2003	2004	Official		Y
Approved in 2005–2006 (19 zones)						
Jiangsu Changzhou EPZ	Changzhou	2005	2007	Imputed		N
Hunan Chenzhou EPZ	Chenzhou	2005	2008	Official		N

Continued on the next page

Zone	City	Approval year	Opening year	Opening date	Hub	Fenced exports matched
Fujian Fuqing EPZ	Fuzhou	2005	2007	Official		N
Fujian Fuzhou EPZ	Fuzhou	2005	2007	Official		N
Guangdong Nansha EPZ	Guangzhou	2005	2007	Imputed		N
Guangdong Huizhou EPZ	Huizhou	2005	2007	Imputed		N
Jiangxi Jiujiang EPZ	Jiujiang	2005	2006	Official		Y
Yunnan Kunming EPZ	Kunming	2005	2008	Official		N
Hebei Langfang EPZ	Langfang	2005	2007	Imputed		N
Sichuan Mianyang EPZ	Mianyang	2005	2007	Imputed		N
Zhejiang Cixi EPZ	Ningbo	2005	2007	Imputed		N
Fujian Quanzhou EPZ	Quanzhou	2005	2007	Official		N
Shanghai Jiading EPZ	Shanghai	2005	2007	Imputed		N
Liaoning Shenyang (Zhangshi) EPZ	Shenyang	2005	2007	Imputed		N
Jiangsu Changshu EPZ	Suzhou	2005	2007	Imputed		N
Jiangsu Wujiang EPZ	Suzhou	2005	2007	Official		N
Jiangsu Wuzhong EPZ	Suzhou	2005	2008	Official		N
Jiangsu Yangzhou EPZ	Yangzhou	2005	2007	Imputed		N
Shandong Qingdao West Coast EPZ	Qingdao	2006	2008	Official		N
Approved in 2010–2011 (2 zones)						
Anhui Hefei EPZ	Hefei	2010	2012	Imputed		N
Jiangxi Jinggangshan EPZ	Ji'an	2011	2013	Imputed		N

Notes: Opening year: official customs acceptance (acceptance in the second half of a year counts from the next year), otherwise approval year plus two. Opening date: "Official" marks the former and "Imputed" the latter. Y: yes; N: no. Hub: the five zones with the largest fenced exports. Fenced exports matched: the zone's fenced exports can be assigned to a named zone in the customs records. The directory's sixtieth entry, Jinqiao, duplicates another zone under the name of an older development zone and is not listed; its South Area is a separate zone.

D Geocoding and Sample Composition

Table G1 in Section 3 compares cells near future zones with all comparison cells and with matched comparison cells before the openings.

E Notation

Table D1: How firms are located: share of manufacturing firm-years by location source

	1998	1999	2000	2001	2002	2003	2004	2005	2006	2007
Exact coordinates	0.33	0.38	0.44	0.57	0.63	0.73	0.79	0.89	0.89	0.90
Town centroid	0.36	0.34	0.32	0.24	0.20	0.12	0.08	0.03	0.03	0.02
County centroid	0.31	0.29	0.25	0.19	0.17	0.16	0.13	0.08	0.08	0.07

Firms outside the fence. Fixing every firm at its most precise location in all years leaves the paths within 10 km unchanged (Section 4.1).

Table E1: Symbols used in the main text

Symbol	Meaning	Section
Panel A. Design		
c, t	0.1-degree cell; year	3
g_c	year in which the first EPZ within 150 km of cell c opens	3
$r(c)$	distance ring of cell c from that zone	3
k	event year, $t - g_c$	3
α_c, $\delta_{r,t}$	cell effects; ring-by-year effects shared with rings around later zones	3
β_{rk}	change in the log of a cell total in ring r at event year k	3
b	productivity fifth within four-digit industry and year	4
Panel B. Model		
E, P, N	the parent zone, the rest of the prefecture, neighboring prefectures	6
$\pi_{j\ell}$, A_ℓ, $\varphi_{j\ell}$	operating profit of firm j in location ℓ; profitability of the location; productivity	6
$V_{j\ell}$, F, m	lifetime value net of fixed costs; fixed cost per period; cost of moving	6
$\bar{K}$, d_j	industrial land of the parent zone outside the fence; plot of firm j	6
v_j, t_{tax}, ω	parent zone's value of an admitted firm; tax rate; weight on output	6
H, L, a_c, e	high and low tier; admission probability of tier c; admission probability of the low tier	6
Λ	value of a marginal plot to the parent zone	6
s_j, τ, $\bar{s}$	signal; its correlation with the productivity rank; high-tier cutoff	6
h_b, P_b	share of fifth b in the high tier; probability that a firm of fifth b obtains a plot	6
ν, χ, ζ, ρ	change in the pool of applicants; share of plots reviewed; Pareto tail of profits; correlation of profits across locations	6
δ_ℓ	share of the turned-away that enter location ℓ	6
w_ℓ, B_ℓ, $u_{i\ell}$, q_ℓ	wage; amenity; idiosyncratic taste; probability of obtaining a job	6
ε, g_ℓ	elasticity of labor supply (Fréchet shape); labor wedge	6
L_F, T_F, G_F, α_F	the fence's employment; its taxes; its receipts credited to the parent zone; share of its activity new to the country	6
η, $\Delta \ln f$	agglomeration shift; change in the fixed cost of exporting related products	6
$\mathcal{C}$, $\mathcal{I}$	price of the fence; value lost by incumbents that lose plots	6
$V^*(K)$, $G(K)$	value admitted when land K is allocated by value per hectare; value the rule loses relative to it	6

Part II. The Later-Treated Design

F The Later-Treated Design and Its Robustness

This appendix describes the comparison of the main text and checks the four facts on which the paper rests: no growth of output within 10 km, the rise of employment there, the step in entry across productivity fifths, and the growth of output 40 to 100 km away. Table F1 collects the checks.

F.1 The Comparison

The comparison cells lie within 150 km of the 69 special customs zones that opened after the survey ends in 2007: 55 comprehensive bonded zones, 7 export processing zones, 6 bonded ports and one bonded logistics park. Their surroundings form the same distance rings as the surroundings of the zones in our sample, and 5,312 cells lie within 150 km of a later zone without lying within 150 km of an earlier one. Equation (1) gives each ring its own year effects, shared by treated and comparison cells, so each coefficient compares a ring around a zone that has opened with the same ring around a zone that will open only later. Outcomes in cells within 150 km of neither are set to missing and do not identify the coefficients.

F.2 Checks

Anticipation. Firms and local governments may have acted on an approval before the customs acceptance that dates the opening. We therefore treat the year before the opening as the first year of exposure, which shifts every event year by one and moves the reference year back. Output within 10 km and 40 to 100 km away is unchanged. The step in entry halves, to −0.36 (0.14), and remains distinct from zero; the employment rise within 10 km falls to 0.09 (0.06).

Heterogeneous effects across waves. With staggered openings, a pooled event study can weight the effects of different cohorts in ways that do not average them (Sun and Abraham 2021). Our comparison consists only of cells around zones not yet open in our years, so no treated cell serves as a comparison for another. We nevertheless estimate the effects separately for the zones opened in 2001–02, 2003–05 and 2006–07, each against the same later-treated comparison, and average them with weights equal to the number of treated cells. The average is 0.04 (0.04) within 10 km and 0.10 (0.03) 40 to 100 km away, close to the pooled estimates.

Leaving out one province. Zones cluster in a few coastal provinces. Dropping the treated cells of one province at a time moves output within 10 km between 0.02 and 0.07, output 40 to 100 km away between 0.08 and 0.15, and the step in entry between −0.80 and −0.49: no single province drives the facts.

Sensitivity to pre-opening trends. Following Rambachan and Roth (2023), we ask how far the trend after the opening may deviate from parallel trends, relative to the largest deviation before it, before the 95 percent interval of the estimate includes zero. For output 40 to 100 km away this breakdown value is 7.9: the post-opening deviation would have to be several times the largest pre-opening one. For the step in entry it is 0.7, and for employment within 10 km 0.2. The step therefore survives a deviation of about two thirds of the largest pre-opening one, which is why the main text reports the model on the alternative baseline that subtracts the mean of the leads; the employment rise does not survive any, which is why the main text does not attribute it to the zones.

Table F1: Robustness of the main facts on the later-treated comparison

	Baseline	Year before treated	By wave, weighted	Leave one province out (min, max)	Breakdown $\bar{M}$
Output, 0–10 km	+0.04 (0.05)	+0.07 (0.05)	+0.04 (0.04)	[+0.02, +0.07]	0.0
Employment, 0–10 km	+0.15 (0.06)	+0.09 (0.06)	—	[+0.12, +0.19]	0.2
Entry step, 0–10 km	−0.72 (0.20)	−0.36 (0.14)	—	[−0.80, −0.49]	0.7
Output, 40–100 km	+0.11 (0.04)	+0.10 (0.04)	+0.10 (0.03)	[+0.08, +0.15]	7.9
domestic firms	+0.09 (0.04)	+0.09 (0.04)	—	—	1.5
foreign-invested firms	+0.10 (0.07)	+0.07 (0.09)	—	—	0.0

Notes: Baseline: equation (1), mean of event years 0–2 minus half of event year −2, prefecture-clustered standard errors. Year before treated: event years shifted by one. By wave: the effect estimated separately for zones opened in 2001–02, 2003–05 and 2006–07 and averaged with treated-cell weights. Leave one province out: minimum and maximum over the provinces with a zone. Breakdown $\bar{M}$: the largest ratio of post-opening to pre-opening deviations from parallel trends at which the 95 percent interval excludes zero (Rambachan and Roth 2023). Entry step: mean of the log changes of fifths 1–3 minus the change of fifth 5, new domestic establishments.

What is not checked here. Spatial correlation beyond prefectures is handled by the inference of Appendix H, which treats the few treated places directly. A dose response in the size of the fence relative to its parent zone has no power: the fence's share of its parent zone is recorded for 34 zones only.

Part III. Alternative Comparisons and Supporting Evidence

G The Matched Comparison

The main text compares the cells around each zone with cells at the same distance from special customs zones opened after 2007. This appendix reports the design that compares them instead with never-exposed cells matched on region, port access and 1998 size, and the results on that comparison. The matched comparison is more precise. The step in entry across productivity fifths, the flat wage and the growth 40 to 100 km away appear on both; output within 10 km rises more on the matched comparison, mostly around the zones that opened in 2001–02 into places already growing.

G.1 Design of the Matched Comparison

We estimate how each outcome in a cell responds to the first EPZ opening within 150 km, separately by distance from the zone and by years since the opening. Let g_c be the year in which the first EPZ within 150 km of cell c opens, and let $r(c)$ be the cell's distance group relative to that zone: 0–10, 10–20 and 20–40 km; 40–100 and 100–150 km inside the zone's province; and 40–150 km in another province (Figure 2). For each outcome y_{ct} we estimate

$$y_{ct} = \exp\left(\alpha_c + \lambda_{q(c),t} + \boldsymbol{x}_c'\boldsymbol{\mu}_t + \sum_r \sum_{e=0}^{3} \beta_{re}\, \mathbb{1}\left[r(c) = r,\ t - g_c = e\right] + \chi_{ct}\right)\eta_{ct} \tag{G1}$$

by Poisson pseudo-maximum likelihood (Santos Silva and Tenreyro 2006), which requires only that η_{ct} have mean one conditional on the regressors.

The indicator marks a cell of group r in year e after the opening, and α_c are cell effects. The term χ_{ct} collects the exposure terms for the years four and later and for the opening of a second, closer zone: zones cluster, many cells lie near more than one, and a later zone gets its own terms rather than being attributed to the first (Section G.3.1 shows what counting zones reveals). The year effects $\lambda_{q(c),t}$ are specific to the cell's matched stratum $q(c)$, defined by macro-region, terciles of distance to the nearest port, and bins of the cell's number of surveyed firms and output in 1998 (Appendix J); $\boldsymbol{x}_c$ are baseline firm density, output, electronics share and port distance, with year-specific coefficients $\boldsymbol{\mu}_t$. We report $\bar{\beta}_r = \frac{1}{4}\sum_{e=0}^{3} \beta_{re}$, the mean of the four post-opening coefficients, without subtracting the leads, and, where levels matter, the level change $\sum y_{ct}(1 - e^{-\beta_{re}})$ over the exposed cell-years. Coefficients are changes in logs throughout: a coefficient of 0.10 is a rise of about 10 percent. Standard errors are clustered by prefecture (329 clusters).

Every cell enters in every year. The comparison consists of cells never within 150 km of a zone and, for each cell, the years before its first opening; each distance group has its own exposure terms, so no cell serves as a comparison after an opening. A binary design, which marks only the cells close to a zone as exposed, differs in this respect. If cells 40 to 100 km from a zone respond, as they do, that design puts responding cells into the comparison and biases the estimate for cells close to the zone (Butts 2023); Appendix Table L2 shows by how much.

G.2 Matched Comparison Cells

Before the openings, the cells around future zones were industrial places and most comparison cells were not (Table G1). Within 10 km of a future zone, 92 percent of cells had manufacturing above the survey threshold in 1998, against 28 percent of never-exposed cells, and log output was 6.2 against 1.1; the mean standardized difference over baseline characteristics is 1.21. Within matched strata it falls to 0.21, and the remaining differences in baseline size enter the equation through $\boldsymbol{x}_c'\boldsymbol{\mu}_t$. One difference remains that the strata do not remove: cells around future zones lie closer to their provincial capital than their matched comparisons, by about half a standard deviation within 10 km and three quarters of one 40 to 100 km away. Section G.3.5 therefore also compares hinterland cells only with cells at the same distance from the province's central cities.

Employment growth in 1999–2000 is similar under either comparison; levels are not, and in the 2000s cells with different baseline manufacturing grew at different rates. Comparing an industrial core with the national average of mostly rural cells would attribute that difference to the zone. The strata therefore use bins of baseline firm counts and output rather than terciles, because 79 percent of cells had at most one surveyed firm in 1998 and terciles separated little more than cells with two or more firms from the rest.[2]

To trace the timing, we replace the post-opening terms with one coefficient for every year from four years before to three years after the opening (event-study coefficients), with the year before the opening as reference, and test that the three leads are jointly zero.

G.3 Results on the Matched Comparison

This section reports where the openings changed manufacturing outside the fence, and two results organize it. Within 10 km, the zones changed who entered more than how much

[2]We adopted the matched-strata comparison after estimating the design with cell and year effects alone, which showed how different the cells around zones were. Appendix K reports the results with cell and year effects alone in full, Appendix Table L1 adds the controls one at a time, and Appendix Table J2 shows how each step moves the estimate of entry within 10 km of the zone.

Table G1: Cells near future zones and comparison cells before the openings

		Comparison cells		Standardized difference	
	Cells near zones	All	Matched	All	Matched
A. Within 10 km of a future zone					
Any manufacturing in 1998	0.92	0.28	0.92	+1.74	+0.02
Firms in 1998 (log)	2.26	0.19	1.98	+1.76	+0.24
Output in 1998 (log)	6.22	1.10	5.56	+2.25	+0.29
Electronics share of output	0.07	0.00	0.01	+0.63	+0.51
Employment 1999–2000 (log)	8.12	2.10	7.72	+2.05	+0.14
Employment growth 1999–2000	0.11	0.14	0.09	-0.03	+0.02
Distance to port (km)	369	615	389	-0.47	-0.04
Distance to provincial capital (km)	127	246	201	-0.73	-0.45
Mean absolute standardized difference				1.21	0.21
B. 40–100 km from a future zone, in its province					
Any manufacturing in 1998	0.49	0.28	0.49	+0.45	+0.01
Firms in 1998 (log)	0.60	0.19	0.56	+0.50	+0.05
Output in 1998 (log)	2.29	1.10	2.20	+0.52	+0.04
Electronics share of output	0.01	0.00	0.01	+0.15	+0.07
Employment 1999–2000 (log)	3.70	2.10	3.61	+0.48	+0.03
Employment growth 1999–2000	0.22	0.14	0.16	+0.08	+0.05
Distance to port (km)	361	615	407	-0.58	-0.11
Distance to provincial capital (km)	124	246	230	-0.86	-0.75
Mean absolute standardized difference				0.45	0.14

Cells within 10 km (panel A) and 40–100 km in the zone's province (panel B) of the first zone to open, against the cells never within 150 km of a zone open by 2007: all of them, and reweighted to the exposed cells' distribution over matched strata. Firms: log of the number of surveyed manufacturing firms in 1998 (at least one); output and employment: logs of one plus the value. Standardized difference: difference in means over the pooled standard deviation.

was produced: production rose mainly where it was already rising, and entry of the least productive domestic establishments fell for about two years, while land around later zones was neither sold in larger amounts nor dearer (Sections G.3.1–G.3.4). Forty to 100 km away inside the province, output rose with no movement before the openings (Sections G.3.5 and G.3.6). Section 5 asks why the zone did not work as an engine next to its fence, and Sections 6 and 8 price the change in entry.

G.3.1 Within 10 km of the zone

The first place to look for an engine is next to the fence: if a zone made the places around it more productive, incumbents within a few kilometers would expand and new establishments would arrive. Manufacturing within 10 km of a zone grows after the first opening, but the growth is concentrated around zones that opened into places that were already growing.

Where the comparison has no detectable pre-opening trend, production shows no detectable change over four years and the entry of new domestic establishments falls; across all zones, the least productive entrants are deterred for about two years. We estimate equation (G1) for every outcome of manufacturing outside the fence within 10 km, first for all zones and then by wave of openings (Section G.3.2); we then trace the number and productivity of new domestic establishments (Section G.3.3), and the area and price of industrial land sold near later zones and the effect of further openings (Section G.3.4).

Table G2 reports the effect of the first opening on every outcome of manufacturing outside the fence within 10 km, for all zones and for each wave of openings. Across all zones, output rises by 0.15 (0.03) and value added by 0.21 (0.05) over the first four years, and employment, the wage bill, fixed capital, profits and taxes rise with them, export deliveries less precisely. Between 10 and 40 km the estimates are small and almost all are not statistically significant (Figure G1; Appendix Table I1). The rise is not an artifact of fenced firms that the customs match missed and that therefore appear outside the fence: output rises as much 5 to 10 km from the zone as within 5 km, and the estimate for foreign-invested firms is larger 5 to 10 km from the zone than within 5 km.[3]

Production rises within 10 km and again 40 to 100 km away inside the zone's province; at the other distances out to 150 km the estimates are smaller and mostly not statistically significant (Figure G1). Within 10 km the figure also shows the estimate for the zones of 2003–05, whose surroundings had no detectable trend before the opening: it is smaller than the pooled estimate and not distinguishable from zero, for output and for value added. Section G.3.5 returns to the rise 40 to 100 km away.

G.3.2 Production by wave of openings

The pooled rise within 10 km mixes zones that opened into different local economies (Table G2, columns 2–4; Figure G2). Each cell belongs to the wave of the first zone opened within 150 km of it, so the waves are waves of first openings: a zone opened in 2003–05 near an earlier zone counts as a second exposure of that zone's cells, not as part of the second wave. Pooled over all zones, output and value added show no detectable difference from the comparison before the opening (joint p-values of the leads 0.20 and 0.17), while employment was already rising (joint $p = 0.001$; Appendix Figure I2). The flat pooled leads conceal different waves. The first wave, the fifteen pilot zones approved in 2000 and opened in 2001–02 in the leading industrial cities, arrived where employment and output were already rising (joint p-values of the leads 0.000 and 0.02); around these zones output rises by 0.16 and employment by

[3]The estimate for foreign-invested firms is +0.18 (0.11) 5 to 10 km from the zone and +0.04 (0.10) within 5 km (Appendix Table M2); neither is distinguishable from zero on its own.

Table G2: Manufacturing outside the fence within 10 km of the zone, all zones and by wave of openings

	All zones	Zones opened in		
		2001–02	2003–05	2006–07
A. Production				
Output	+0.15	+0.16	+0.08	+0.16
	(0.03)	(0.04)	(0.06)	(0.03)
Value added	+0.21	+0.23	+0.16	+0.13
	(0.05)	(0.05)	(0.12)	(0.06)
Fixed capital	+0.11	+0.10	+0.07	+0.25
	(0.04)	(0.04)	(0.05)	(0.10)
Profits	+0.18	+0.21	+0.20	+0.08
	(0.08)	(0.10)	(0.15)	(0.07)
Taxes	+0.16	+0.16	+0.12	+0.17
	(0.05)	(0.07)	(0.08)	(0.06)
B. Labor				
Employment	+0.18	+0.29	+0.04	−0.01
	(0.05)	(0.06)	(0.05)	(0.07)
Wage bill	+0.19	+0.20	+0.10	+0.25
	(0.04)	(0.04)	(0.06)	(0.05)
C. Firms and exports				
Number of firms	+0.06	+0.15	−0.18	+0.14
	(0.05)	(0.05)	(0.07)	(0.08)
New domestic establishments	−0.08	+0.08	−0.41	−0.01
	(0.11)	(0.21)	(0.11)	(0.18)
Domestic exits	+0.13	+0.25	−0.06	+0.18
	(0.11)	(0.16)	(0.12)	(0.13)
Export deliveries	+0.15	+0.16	+0.23	+0.01
	(0.08)	(0.09)	(0.23)	(0.09)
Exporting firms	+0.11	+0.14	+0.04	−0.08
	(0.05)	(0.06)	(0.10)	(0.09)
Pre-opening test, output (p)	0.20	0.02	0.11	0.00
Pre-opening test, employment (p)	0.00	0.00	0.25	0.03
Cells within 10 km	89	40	29	20

Equation (G1): mean of the coefficients for the first four years after the opening, in changes in logs; columns 2–4 give the cells within 10 km of zones opened in each period their own exposure terms, where a cell's wave is that of the first zone opened within 150 km of it; for zones opened in 2006–07 the mean covers the available post-opening years, at most two. Standard errors clustered by prefecture (329 clusters) in parentheses. All manufacturing firms outside the fence; new establishments and exits are domestic. Pre-opening tests: joint p-values of the leads at −4, −3, −2 (reference −1). 19,149 cells, 1998–2007.

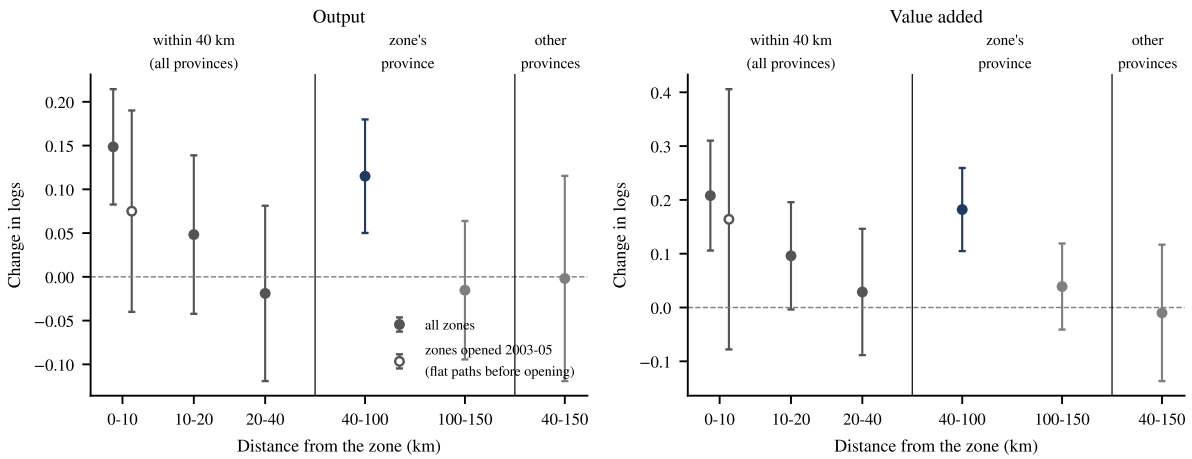


Figure G1: Effect of the first opening within 150 km on output and value added of manufacturing outside the fence, by distance group: mean of the first four years after the opening. Filled markers: all zones. Hollow marker at 0–10 km: zones opened in 2003–05, the wave without a detectable trend before the opening (Table G2). PPML estimates of equation (G1), with cell effects, matched-stratum-by-year effects and baseline characteristics by year; 95 percent confidence intervals, standard errors clustered by prefecture.

0.29 over the four years, continuing the earlier path. The third wave, opened in 2006–07, has at most two post-opening years, and its leads are jointly different from zero (p-values 0.000 and 0.026). The second wave, opened in 2003–05 in nine prefectures, opened where output and employment show no detectable pre-opening trend (p-values 0.11 and 0.25), so the comparison around these zones does not rely on extrapolating earlier growth. Around them output shows no detectable change over the four years, +0.08 (0.06), and the number of firms falls.[4] The output coefficient of the fourth year is 0.17 (0.08), so a later rise cannot be excluded.

G.3.3 Entry and who enters

Around the second wave, new domestic establishments within 10 km fall by 0.41 (0.11) over the first four years relative to comparison cells, 33 percent of the entry that would otherwise have occurred (Figure G2b). Entry before the opening is noisier than output: the joint p-value of its leads is 0.09, mostly because of a high value four years before the opening. Measured against its own path before the opening, the fall is −0.46 (0.10). With only nine treated prefectures, the wild cluster bootstrap gives a p-value of 0.023; the fall holds without any one of the nine and around zones with an official opening date (Appendix J), and its

[4]Value added changes by +0.16 (0.12) and employment by +0.04 (0.05); neither is distinguishable from zero.

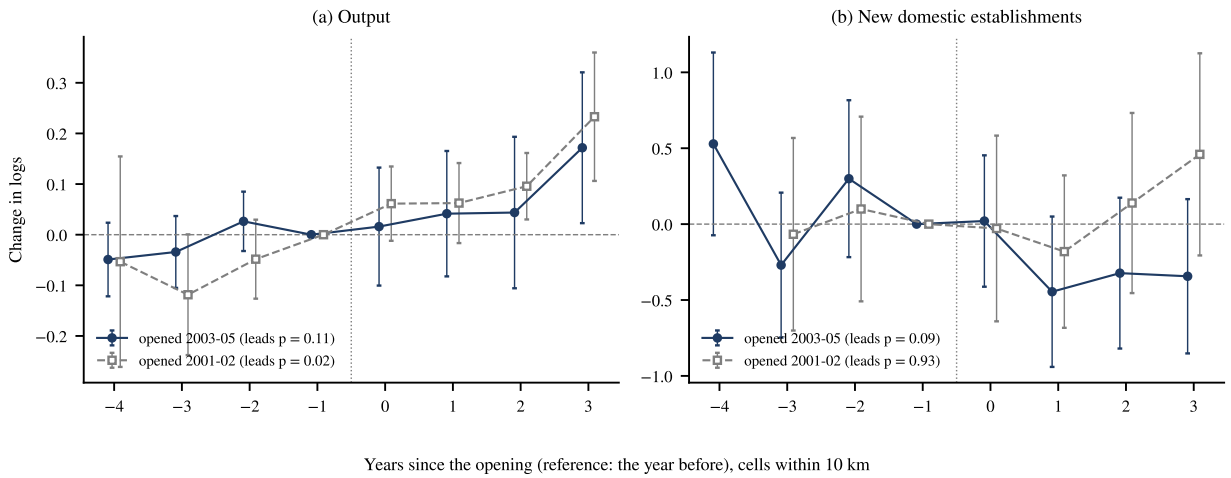


Figure G2: Output and new domestic establishments within 10 km, by wave of openings: zones opened in 2003–05 (no detectable trend before the opening) and in 2001–02 (placed where activity was already rising). PPML estimates of equation (G1), with cell effects, matched-stratum-by-year effects and baseline characteristics by year, one term per year from four years before to three years after the opening and separate terms for each wave; reference: the year before the opening; 95 percent confidence intervals, standard errors clustered by prefecture; joint p-values of the leads in the legends.

t-statistic exceeds that of every placebo zone built from never-exposed cells.[5] The estimates for the first and third waves are imprecise and show no detectable change in entry. Across the full set of outcomes in Table G2, output, value added, fixed capital, the wage bill and taxes rise around the first and third waves; around the second, no outcome rises at the 5 percent level and the number of firms falls.

Across all zones, the number of new domestic establishments within 10 km shows no detectable change, −0.08 (0.11), but their composition does (Figure G3); the fall around the second wave does not show up in this pooled estimate.[6] Pooling all waves and ranking entrants within the national distribution by revenue productivity relative to their industry and year, entry of the least productive fifth falls by 0.42 (0.17) relative to comparison cells over the first four years. The event path shows when (Figure G3a): relative to the year before the opening, its coefficients average −0.23 in the three years before that and −0.51 after the opening, with a trough of −1.03 in the year after the opening year and −0.21 by the fourth year. For the most productive fifth the four-year estimate is positive, 0.26 (0.14), but not significant at the 5 percent level, and its path shows no rise at the opening: its coefficients average 0.38 before and 0.45 after. The zone deters the least productive entrants for about

[5]The placebo zones were specified in advance; comparing their t-statistics rather than their coefficients was added after the first placebo run, because placebo cells are sparser and their estimates noisier (Appendix H).

[6]Domestic exits change by 0.13 (0.11) and the output of domestic firms present in 2000 by +0.01 (0.08); neither is distinguishable from zero.

two years; it does not detectably draw in more productive ones.

Entry differs more by productivity than by capital, though the difference between the two gradients is not statistically significant (Figure G3b). When entrants are cross-classified by halves of productivity and of capital at entry, within their industry and year, the productivity gradient, the average difference in the effect between entrants above and below the median, is 0.43 (0.14) and the capital gradient is 0.14 (0.17). Admission rules that set a minimum investment per unit of land would screen on capital; instead, the point estimates are negative for low-productivity entrants at either level of capital and positive for high-productivity entrants at either level.

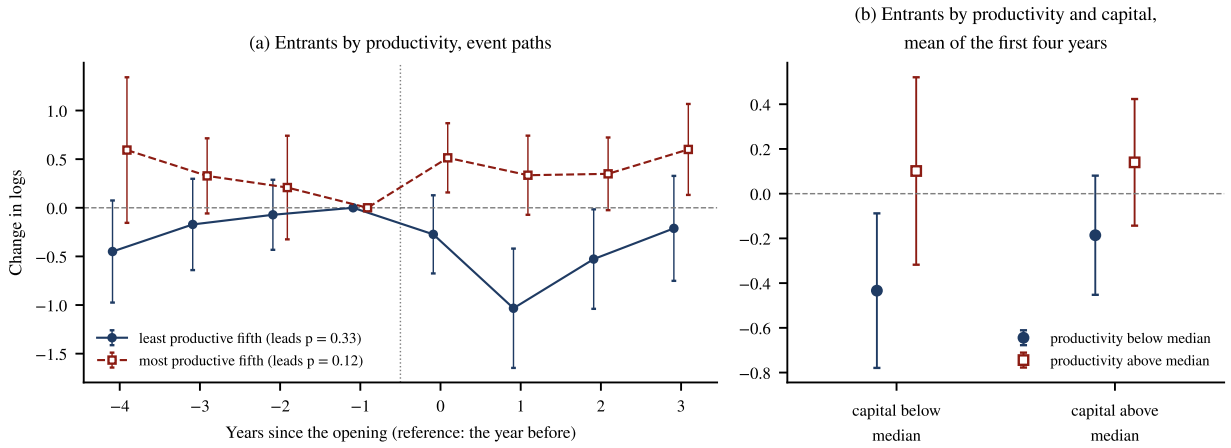


Figure G3: Who enters within 10 km of the zone, all zones. (a) Entrants in the least and the most productive fifth of the national distribution of new domestic establishments (revenue productivity relative to the mean of the entrant's industry and year), one term per year; reference: the year before the opening; joint p-values of the leads in the legend. (b) Entrants by halves of productivity and of fixed capital at entry, each relative to the median entrant of the same two-digit industry and year: mean of the first four years after the opening. PPML estimates of equation (G1), with cell effects, matched-stratum-by-year effects and baseline characteristics by year; 95 percent confidence intervals, standard errors clustered by prefecture.

G.3.4 Land and several zones

Industrial land next to an EPZ belongs to the construction-land quota of its parent development zone, which local governments allocate establishment by establishment at administered prices (Section 2). If demand for sites near a new zone were cleared by price, land there would become dearer or be sold in larger amounts. Land transactions are recorded from 2007, so we test this on the 49 special customs supervision zones, mainly comprehensive bonded zones (the zone type into which EPZs were later converted), that opened on new sites in 2008–2016 (Appendix Figure Q1).

Three to 10 km from these zones, the area of industrial land sold changes by −0.19 (0.16) and its price per hectare by −0.02 (0.06), relative to land 40 to 100 km away in the same prefecture; the upper end of the 95 percent confidence interval for price is a rise of 9 percent. Neither quantities nor prices show a detectable change. This is what admission rather than price clearing demand for sites near a zone would produce, but also what no shift in demand would produce; it is informative only together with the finding that, across all zones, the composition of entry changes without a detectable change in its number. Land around these later zones was sold under the auction rules of 2007, so the test is indirect evidence on allocation around the 2001–07 zones, not a measurement of it.

Zones cluster, and 739 cells receive a second zone within 40 km by 2007. Each additional zone within 40 km lowers new domestic establishments by −0.36 (0.06), while output shows no detectable change (−0.02 (0.04)); entry around the additional zone shows no detectable difference from the comparison before it opens (joint p-value of the leads 0.80). Where zones accumulate, domestic entry is lower with each additional one. A binary design, which marks whether a zone has opened within 40 km of a cell and treats every other cell as a comparison, finds no detectable change in entry or output within 40 km (Appendix Table I2), because it puts the growing places 40 to 100 km away into the comparison and averages places near one zone with places near several.

G.3.5 Growth 40 to 100 km away

The growth that accompanied the zones appears farther away than an engine of local linkages would put it. Manufacturing grew 40 to 100 km from the zones, inside their provinces, a ring we call the provincial hinterland. Within the province the evidence is consistent with new activity rather than relocation, and most of the growth did not run through the purchases of fenced firms. The estimates are the effect of an opening together with whatever else changed in the province when it opened, which the design cannot separate from the zone. Between 40 and 100 km from the zone, inside its province, manufacturing grows: averaged over the first four years after the opening, output rises by 0.11 (0.03) and value added by 0.18 (0.04) in logs, and employment, fixed capital, profits and taxes rise as well (Figure G1; Appendix Table M1).[7] Before the opening, output and value added there do not differ detectably from the comparison cells (joint p-values of the leads 0.89 and 0.47); after it they rise year by year, output by 0.07, 0.11, 0.12 and 0.16 and value added by 0.11, 0.17, 0.20 and 0.25 (Figure G4). At the same distance in other provinces, output and value added show no rise after the opening; their output leads are jointly different from zero at the 10 percent level ($p = 0.07$),

[7] Employment rises by 0.08 (0.04), fixed capital by 0.15 (0.06), profits by 0.16 (0.06) and taxes by 0.19 (0.06), all in logs. Estimates in this section are in logs: 0.11 is a rise of 11 log points.

because of the earliest year.

The response does not extend beyond this ring. Output changes by −0.02 (0.04) 100 to 150 km away and by −0.04 (0.03) in the rest of the province beyond 150 km, neither detectably different from zero.

The hinterland's point estimates also exceed those 10 to 20 km away and those at the same distance in other provinces, although most of these differences are less than twice their standard errors.[8] The difference between the response in the hinterland and the response within 10 km is −0.03 (0.04), not statistically distinguishable from zero.

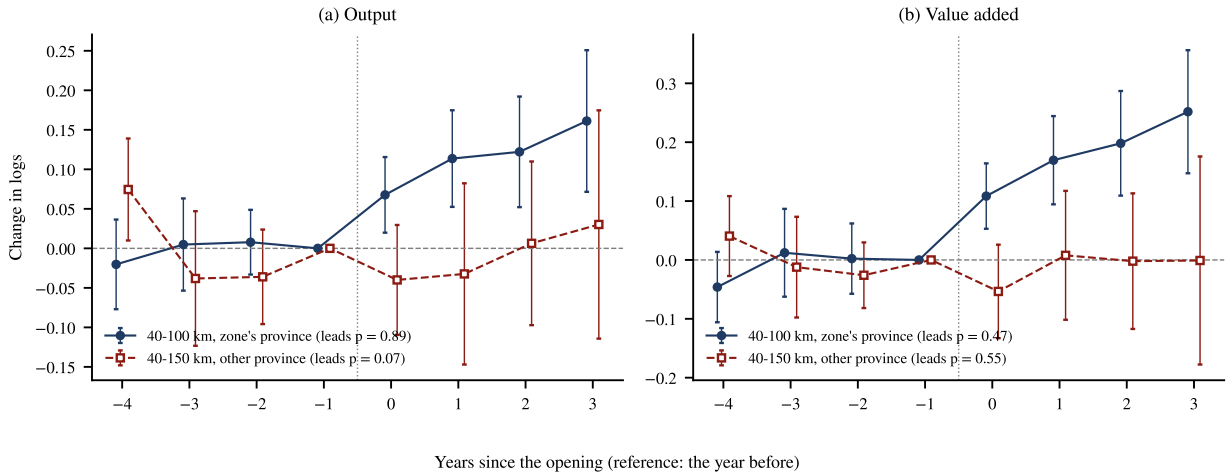


Figure G4: Output and value added 40–100 km from the zone inside its province, against 40–150 km from it in other provinces, by years since the first opening. PPML estimates of equation (G1), with cell effects, matched-stratum-by-year effects and baseline characteristics by year, one term per year; reference: the year before the opening; 95 percent confidence intervals, standard errors clustered by prefecture; joint p-values of the leads in the legends. The path 100–150 km away inside the province is in Appendix Figure I5.

The choice of comparison decides whether places 100 to 150 km away appear to grow; it changes the size of the hinterland's response but not its sign. With a single year effect for all of China, places 100 to 150 km from a zone also appear to grow (employment +0.15), and the apparent gain disappears once they are compared within their macro-region, which is consistent with its reflecting the difference between coastal and inland China. The response 40 to 100 km away stays between 0.09 and 0.20 for output under every comparison in the upper block of Appendix Table L1. It falls to 0.06 (0.04) only when the comparison is restricted to 241 cells around zones that open after 2007 and geography is interacted with year; those cells are the cores of places that were about to receive zones.

[8]The differences from the response 10 to 20 km away are 0.07 (0.05) for output and 0.09 (0.06) for value added; the differences from the response at the same distance in other provinces are 0.12 (0.07) and 0.19 (0.07).

G.3.6 What the hinterland is, and who grows there

The hinterland comprises the 2,709 cells 40 to 100 km from a zone inside its province, beyond the zone's commuting zone but within its province and trucking range. Two thirds of them are in neighboring prefectures and one third in the outer counties of the host prefecture; half had no manufacturing above the survey threshold in 1998.

Table G3: The provincial hinterland: where the growth is and which firms account for it

A. Where: cells 40–100 km away in the zone's province, by pre-opening trait	Output	Value added	Employment	Firms
Jurisdiction				
Host prefecture, outer counties	+0.10 (0.06)	+0.19 (0.07)	+0.09 (0.04)	+0.03 (0.06)
Neighbouring prefectures	+0.12 (0.04)	+0.18 (0.04)	+0.08 (0.04)	+0.06 (0.03)
Land in 1998				
No manufacturing (greenfield)	+0.23 (0.12)	+0.24 (0.11)	+0.21 (0.11)	+0.23 (0.09)
Low industrial density	−0.04 (0.06)	−0.00 (0.06)	+0.02 (0.05)	−0.06 (0.05)
High industrial density	+0.13 (0.04)	+0.20 (0.04)	+0.08 (0.04)	+0.06 (0.04)
Distance to own prefecture's core				
Near (median 23 km)	+0.16 (0.04)	+0.24 (0.05)	+0.07 (0.04)	+0.05 (0.04)
Far (median 76 km)	+0.14 (0.06)	+0.19 (0.05)	+0.12 (0.05)	+0.07 (0.06)

B. Who: domestic firms 40–100 km away, by status at first record	Output	Employment	Firms
Not exporting at first record	+0.13 (0.04)	+0.09 (0.04)	+0.06 (0.04)
Exporting at first record	+0.06 (0.06)	+0.04 (0.05)	+0.02 (0.05)
New establishment (founded ≤1 year before)	+0.12 (0.06)	+0.10 (0.05)	+0.10 (0.04)
Founded 10 or more years before	+0.14 (0.07)	+0.11 (0.04)	+0.04 (0.04)

Equation (G1); cells 40–100 km from the zone of the first opening, inside its province, split by traits fixed before the opening (panel A), or domestic firms split by export status and age at their first record (panel B). Mean of the first four years after the opening; standard errors clustered by prefecture. Firms by who controls their paid-in capital: Appendix Table N2.

The growth is not one city's (Table G3). Its size is similar in neighboring prefectures and in the outer counties of the host prefecture (+0.12 and +0.10 for output, respectively), and it is present both near and far from the economic core of the cell's own prefecture. It takes two forms. In cells with no manufacturing above the survey threshold in 1998, firms enter: the number of firms rises by 0.23 (0.09) and the number of new domestic establishments by 0.30 (0.11). In cells of high industrial density in 1998, activity expands: value added rises by 0.20 (0.04).

Non-state firms account for more of the growth than of the output. We classify firms by who controls their paid-in capital rather than by registration type, which counts corporatized state firms as limited-liability companies. The point estimates for output are 0.10 in foreign-

controlled firms, 0.19 in privately controlled firms and 0.13 in firms controlled by other corporate (legal-person) shareholders, against 0.08 in state-controlled and 0.04 in collective firms (Appendix Table N2). Non-state firms held about half of the hinterland's output before the openings and account for about two thirds of its growth. Among non-state groups, foreign firms founded before 1995 contribute the largest share, 19 percent of the growth from 16 percent of the output.

The point estimates for output are positive in almost every product class, from 0.06 to 0.23, with no class carrying more than 22 percent of the gain and general and special-purpose machinery flat (Appendix Table N4). Electronics, 87 percent of fenced exports by value but 21 percent in the median zone (Appendix A), is measured poorly in the hinterland: its output there lies well below the comparison both before and after the opening. The foreign electronics firms that resemble fenced firms are too few to show whether they moved: their output changes by −0.21 (0.62).

The growth is also not concentrated in the development zones listed in the national Catalog of Development Zones. Output of firms inside listed zones changes by −0.10 (0.15), with no detectable growth. Firms around those zones, outside their cores, account for 71 percent of the growth. Firms that report a zone address absent from the catalog, most likely county and township industrial parks, have a larger output response, +0.23 (0.36), but one too imprecise to be distinguished from the rest.

G.4 Mechanisms on the Matched Comparison

G.4.1 Goods: the response depends on the industrial base

If purchases are the only link in goods between fenced firms and the economy around them, they should matter only where the zone has suppliers to buy from. We divide zones into terciles by the 1998–2000 output, in the cells around them, of the industries that supply fenced firms' inputs, and estimate the effect within 20 km for each tercile (Appendix Figure I3). The output of supplier industries is higher around zones with a thick supplier base than around zones with a thin one, by 0.22 (0.09).[9] The output of unrelated industries shows no detectable difference between the two, 0.02 (0.11).

This is the pattern that purchases from existing suppliers would produce: where a supplier base exists, fenced firms can buy from it, and where it does not, which is the typical case, they would have little to buy locally and would import their inputs. The survey does not record transactions, so the evidence is the pattern of output rather than the purchases themselves.

[9]The point estimates are a fall of 0.13 (0.08) around zones with a thin base and a rise of 0.09 (0.07) around zones with a thick one; neither is distinguishable from zero on its own.

The contrast is absent before the openings, but a thick supplier base is also a thick industrial base, and splitting zones by the output of unrelated industries gives a similar contrast (Appendix Table J5). The data therefore show which industries respond, but not whether it is the supplier base or the industrial base as a whole that conditions the response.

G.4.2 How the hinterland grew

Within firms, the growth is an expansion of capacity with no detectable change in profit rates. Among the firms present in 2000, relative to the years before the opening, sales rise by 0.12, fixed capital by 0.20, and employment and intermediate inputs by 0.09 and 0.09, with profits in step with sales (Appendix Table N5). The profit rate, the subsidy rate, the tax rate, the interest rate, leverage and the export share each change by less than one standard error of the numerator's response, and the decline in the share of new products is smaller than two standard errors (Appendix Table N5, panel B). We find no detectable change in subsidy or interest rates and no shift into new products or exports; the firms sold more of what they already made.

The output of non-state firms begins to rise in the year of the opening and builds up over four years. Entry of new private firms behaves differently. It shows a rising path before the opening (coefficients of -0.28, -0.19 and -0.10 four to two years before, relative to the year before; joint $p = 0.15$) and no further rise after it (Appendix Table N3).

The additional output appears to have been sold in the broader domestic market. Output of household goods rises by $+0.17$ against $+0.09$ for goods bought by firms, and the estimates for necessities are similar to those for goods with high income elasticities, which are imprecisely estimated. Export value changes by $+0.03$ against $+0.11$ for output as a whole. The data show no sign of local buyers: retail sales, wages and savings in the hinterland's own cities show no detectable change, and neither does their registered population (Appendix Table N6). Nor were fenced firms the buyers, as the calculation of their purchases below shows.

New activity or relocated activity? Growth in the hinterland could be new, or it could be activity that would otherwise have located within 40 km of the zone or elsewhere in the province. Relocation leaves the province's manufacturing unchanged and new activity does not, and the two make different predictions (Table G4).

Relocation within the province would lower activity elsewhere in it. The places of the zone's province more than 150 km away show no detectable decline when the zone opens: output changes by -0.04 (0.03) and value added by -0.01 (0.04). Output within 10 km of the zone does not fall either.

For the province as a whole, adding the level effects of every distance group with their

Table G4: What the hinterland's growth is not: relocation or the purchases of fenced firms

Test	Estimate	Prediction
Panel A. Rest of the zone's province, more than 150 km away (log points)		
Output	−0.04 (0.03)	negative
Value added	−0.01 (0.04)	negative
Employment	−0.03 (0.04)	negative
Panel B. Whole province: value added gained per RMB of fenced exports		
Sum over all distance groups	+2.6 (1.3)	about zero
Panel C. Industry fingerprint (28 two-digit industries)		
Correlation of entry effects within 40 km and 40–100 km away	+0.28 [-0.10, +0.59]	negative
Panel D. Purchases of fenced firms (RMB per RMB of fenced exports)		
Value added gained 40–100 km away, zone's province	2.1 [1.3, 2.9]	at most 0.34–0.57

Panels A–C test relocation; the last column gives what relocation predicts. Relocation would also lower output within 10 km, where it rises instead (Table 2). Panel A: cells of the zone's province more than 150 km away, timed by the province's first opening; comparison: provinces without an EPZ by 2007. Panel B: sum of the level effects of every distance group of the province with their joint covariance, divided by fenced exports; each cell belongs to the group of its first zone. Panel C: correlation across 28 two-digit industries of the effects on new domestic establishments within 40 km and 40–100 km away; 95 percent interval from Fisher's transformation. Panel D: value added gained 40–100 km away per RMB of fenced exports, against the DVAR of fenced exports (Kee and Tang 2016; 0.34 export-weighted, 0.57 in the most generous variant in the median zone), an upper bound on what the purchases of fenced firms could add. Standard errors in parentheses; 95 percent intervals in square brackets.

joint covariance gives a rise in value added of 2.6 RMB (1.3) per RMB of fenced exports, of which 1.8 (0.7) comes from cells 40 to 150 km away inside the province (Appendix Table M2).

The industries do not line up as relocation would require. Across 28 two-digit industries, the correlation between the effect on new establishments within 40 km and 40 to 100 km away is +0.28 (95 percent interval −0.10 to 0.59), whereas relocation of entrants from places within 40 km of the zone to the hinterland implies a negative correlation. The interval includes zero but excludes a negative correlation stronger than −0.10; the positive point estimate is consistent with industry-wide conditions common to both places.

Dropping the census additions changes the hinterland's response little (output +0.10, value added +0.17). Within the province, the evidence is consistent with the hinterland's growth being new activity rather than relocated activity; whether some of it would otherwise have gone to other provinces the design cannot show.

Not the purchases of fenced firms. Value added gained 40 to 100 km away is 2.1 RMB (1.3 to 2.9) per RMB of fenced exports; it is 1.8 for all cells 40 to 150 km away inside the province and 2.6 for the province, in the regression that adds up all the province's distance groups (Table G4). The DVAR of fenced exports, domestic value added per RMB of exports, is 0.34 weighted by exports and 0.57 in its most generous variant, a median-zone value (Table G4, panel E); it is an upper bound on the domestic purchases of fenced firms because it also

contains their wages, profits and indirect domestic content. Even if every RMB of that value added were bought in the hinterland and were additional, it would account for about a sixth of the point estimate of the gain at the weighted value and a little more than a quarter at the most generous one. Against the lower end of the gain's interval, even the most generous variant would account for less than half. Spending by the workers and owners of fenced firms is not bounded by this calculation, but local wages per unit of fenced exports, measured for survey-matched fenced firms, are about 6 percent. Most of the hinterland's growth therefore did not run through the supply chain of fenced firms.

What the design identifies. The design identifies where the growth occurs and in which firms. The hinterland's growth is measured relative to places of the same kind elsewhere, and it begins at the opening and builds after it. Its point estimates are larger inside the zone's province than at the same distance outside it, and it is absent beyond 100 km. The evidence is consistent with its not being relocation from the zone's neighborhood or from the rest of the province.

Hinterland cells lie in larger, denser cities, closer to ports and megacities, than their matched comparisons, but controlling for that leaves most of the response intact (Appendix Table N1). Adding each cell's baseline industry mix interacted with year leaves the output response at 0.10 to 0.11 (0.04), net of the leads. Adding the size, administrative rank and urban core of its city keeps 74 to 85 percent of the baseline response, and replacing the matched-stratum year effects with province × year effects on top of these keeps 87 percent, with an output response of 0.10 (0.04).

Comparing only cells within the same province and year at the same distance from the provincial capital, the nearest megacity and the city core reduces the response substantially: to 0.06 (0.04) for output, 57 percent of the baseline response, and to 0.12 (0.05) for value added, 69 percent. Part of what distinguishes the hinterland is its position around the province's central cities.

What the design cannot separate is the zone from anything else that changed in the province when it opened. We do not observe such changes, so we do not attribute the hinterland's growth to the zone. The estimates are the effect of an opening together with whatever accompanied it, and the evidence on channels is consistent with fenced firms' own transactions not having carried it: the fence's purchases are too small to account for the added output, and the industry pattern of entry shows no sign that entrants deterred near the fence moved to the hinterland.

H Inference with Few Treated Places on the Matched Comparison

This appendix uses the matched comparison of Appendix G, with its window (event years 0–3) and all pre-opening years as the reference; it checks inference for that specification, not for the later-treated estimand of the main text. The cells within 10 km of the zones number 89, 29 of them around the zones opened in 2003–05, so we check the main estimates in two further ways (Table H1).

Placebo zones. We assign the opening years of the real cells to never-exposed cells that resemble them in 1998 and re-estimate the main specification with these placebo zones added. The placebo estimates are widely dispersed. For the four estimates within 10 km that we check under both matches, the randomization p-values range from 0.24 to 0.66 under the pre-specified match and from 0.15 to 0.71 under the stricter one; across all estimates in the table they range more widely. The dispersion has a clear source: even under the stricter match, placebo cells are sparser than the cells within 10 km of the zones, with a quarter or less of their median output in 1998, and their estimates are correspondingly noisier, with a spread close to their own standard errors. Comparing t-statistics rather than coefficients corrects for this. Of the four estimates checked against the stricter placebo zones, the two that are significant in the matched specification, the fall in new domestic establishments around the 2003–05 zones and the rise in output within 10 km of all zones, are more extreme than every placebo zone under that comparison. The studentized comparison was added after the first placebo run; we report both.

Pre-trends. Following Rambachan and Roth (2023), we ask how large a departure from parallel trends after the opening would overturn each estimate, measured relative to the largest departure before it. Allowing half that departure, the robust interval for output within 10 km still excludes zero, while the intervals for entry and for the hinterland include it; their pre-opening paths are noisier. These checks qualify the strength of the estimates within 10 km and in the hinterland without changing their sign.

I Supplementary Exhibits

This appendix collects exhibits that support statements in the main text and in Appendix efapp:matched, most of them on the matched comparison: the program's export series, every outcome at every distance within 40 km, the pooled timing within 10 km, the comparison with a binary design and with the number of zones nearby, the supplier-base

Table H1: Inference with placebo zones and sensitivity to pre-trends

	Estimate (s.e.)	Randomization p			Robust interval, $\bar{M} = 0.5$
		Pre-specified	Stricter match	Studentized	
2003–05 wave, 0–10 km					
Output	0.08 (0.06)	0.66	0.54	0.350	[-0.12, 0.27]
Value added	0.17 (0.13)	0.41	–	–	[-0.11, 0.54]
Employment	0.04 (0.06)	0.80	–	–	[-0.15, 0.20]
Establishments	-0.17 (0.08)	0.33	–	–	[-0.36, 0.07]
New domestic establishments	-0.36 (0.12)	0.24	0.15	0.000	[-2.13, 1.69]
All waves, 0–10 km					
Output	0.15 (0.03)	0.30	0.25	0.000	[0.00, 0.35]
Value added	0.21 (0.05)	0.20	–	–	[-0.01, 0.50]
Employment	0.18 (0.05)	0.08	–	–	[-0.04, 0.33]
Establishments	0.06 (0.05)	0.54	–	–	[-0.11, 0.29]
New domestic establishments	-0.08 (0.11)	0.66	0.71	0.562	[-1.08, 0.65]
Same province, 40–100 km					
Output	0.11 (0.03)	–	–	–	[-0.01, 0.24]
Value added	0.18 (0.04)	–	–	–	[-0.00, 0.38]
Employment	0.08 (0.04)	–	–	–	[-0.03, 0.20]
Establishments	0.05 (0.03)	–	–	–	[-0.05, 0.14]
New domestic establishments	0.10 (0.09)	–	–	–	[-0.35, 0.49]

Notes: Estimates are the paper's main specification (post-opening years 0–3 against all pre-opening years), re-estimated on the survey cell aggregates used for the placebo runs, with each statistic's own exposure terms; they differ slightly from the main text (for example, -0.36 against -0.41 for new domestic establishments around the 2003–05 zones). Randomization p: share of placebo estimates at least as large in absolute value, from never-exposed cells assigned the opening years of the real cells; the pre-specified match uses 1998 output within region, the stricter match also requires the same matching stratum. Placebo cells remain sparser than near-field cells, so their estimates are noisier; the studentized version, added after the first placebo run, compares t-statistics. Robust intervals follow Rambachan and Roth (2023), allowing post-opening departures from parallel trends up to half the largest pre-opening departure.

split, the full set of results on related products, and the hinterland path 100–150 km away.

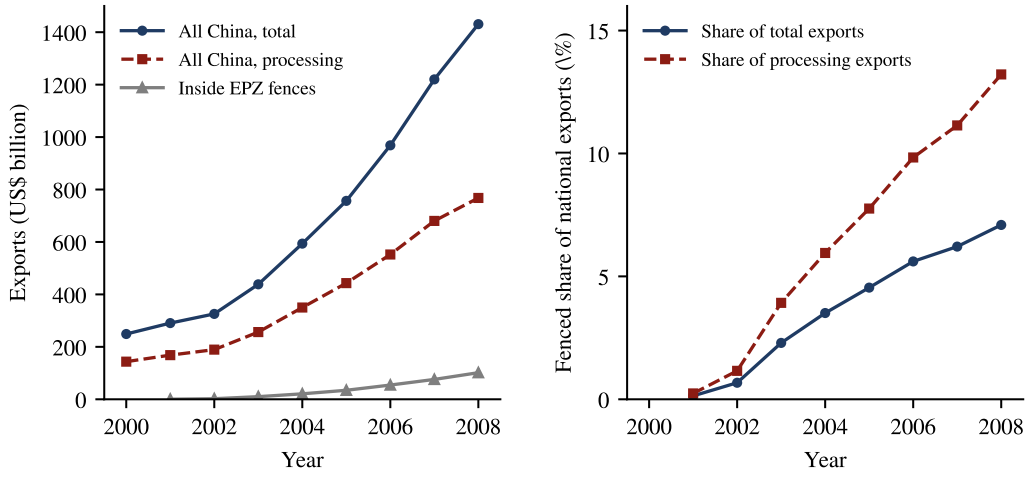


Figure I1: China's exports and exports of fenced firms (customs identifiers that mark an EPZ), 2000–2008. The figure stops in 2008, the end of the period this paper studies. From 2010 most zones were converted into comprehensive bonded zones and their firms re-registered under that zone type's identifier, so a series built on the EPZ identifier falls mechanically. Between 2010 and 2011, exports under the EPZ identifier fell from 130 to 102 billion US dollars, while exports under the comprehensive-bonded-zone identifier rose from 4.6 to 53 billion. The series to 2014 is in Appendix Figure A3.



Figure I2: Output, employment and new domestic establishments of manufacturing outside the fence within 10 km and 10–20 km of the zone, by years since the first opening, all zones. Equation (1) with one term per year from four years before to three years after the opening; reference: the year before the opening; standard errors clustered by prefecture; 95 percent confidence intervals. Joint p-values of the leads in Appendix Table L3.

Table I1: Manufacturing outside the fence after the first EPZ opening, by distance within 40 km

	0–10 km	10–20 km	20–40 km
A. Production			
Output	+0.15	+0.05	−0.02
	(0.03)	(0.05)	(0.05)
Value added	+0.21	+0.10	+0.03
	(0.05)	(0.05)	(0.06)
Fixed capital	+0.11	+0.05	−0.08
	(0.04)	(0.05)	(0.08)
Profits	+0.18	+0.04	+0.05
	(0.08)	(0.07)	(0.08)
Taxes	+0.16	+0.04	+0.04
	(0.05)	(0.06)	(0.07)
B. Labor			
Employment	+0.18	+0.02	+0.03
	(0.05)	(0.05)	(0.05)
Wage bill	+0.19	+0.02	−0.00
	(0.04)	(0.05)	(0.06)
C. Firms and exports			
Number of firms	+0.06	+0.05	−0.00
	(0.05)	(0.05)	(0.05)
New domestic establishments	−0.08	−0.05	−0.09
	(0.11)	(0.11)	(0.12)
Domestic exits	+0.13	−0.09	−0.19
	(0.11)	(0.09)	(0.09)
Export deliveries	+0.15	+0.06	+0.00
	(0.08)	(0.07)	(0.07)
Exporting firms	+0.11	+0.06	+0.01
	(0.05)	(0.06)	(0.07)
Cells	89	188	610

Equation (1): mean of the coefficients for the first four years after the opening, in changes in logs; standard errors clustered by prefecture (329 clusters) in parentheses. All manufacturing firms outside the fence; new establishments and exits are domestic. 19,149 cells, 1998–2007. The first column is the first column of Table 2.

Table I2: Binary and distance-group designs, with the number of zones nearby

	Binary design		Distance groups (this paper)	
	(1)	(2)	(3)	(4)
New domestic establishments				
Opening within 40 km (binary) or 10 km	−0.08 (0.08)	−0.04 (0.08)	−0.08 (0.11)	−0.03 (0.12)
40–100 km, zone's province	–	–	+0.10 (0.09)	+0.08 (0.08)
Each additional zone within 40 km		−0.40 (0.06)		−0.36 (0.06)
Output				
Opening within 40 km (binary) or 10 km	−0.00 (0.04)	−0.00 (0.04)	+0.15 (0.03)	+0.15 (0.03)
40–100 km, zone's province	–	–	+0.11 (0.03)	+0.11 (0.03)
Each additional zone within 40 km		−0.03 (0.04)		−0.02 (0.04)
Employment				
Opening within 40 km (binary) or 10 km	+0.03 (0.04)	+0.03 (0.04)	+0.18 (0.05)	+0.18 (0.05)
40–100 km, zone's province	–	–	+0.08 (0.04)	+0.08 (0.04)
Each additional zone within 40 km		−0.04 (0.04)		−0.02 (0.05)
New domestic establishments around the opening of an additional zone within 40 km				
Years relative to the opening	-4: +0.09 (0.16), -3: +0.12 (0.13), -2: +0.01 (0.10)			
	0: −0.07 (0.09), 1: −0.15 (0.15), 2: −0.24 (0.15), 3: −0.29 (0.23)			
Joint test of the leads (p); mean after	0.80; −0.19 (0.13)			

Columns (1)–(2): a binary design, one set of exposure terms for cells within 40 km of the first zone to open and every other cell in the comparison, no other exposure terms. Columns (3)–(4): equation (1), every distance group with its own exposure terms. Columns (2) and (4) add the number of export processing zones open within 40 km beyond the first. Mean of the first four years after the opening; Poisson regression with cell effects, year effects by matched stratum and baseline characteristics × year; standard errors clustered by prefecture. Bottom panel: cells that receive an additional zone within 40 km, one term per year relative to its opening (reference: the year before), within the design of column (3).

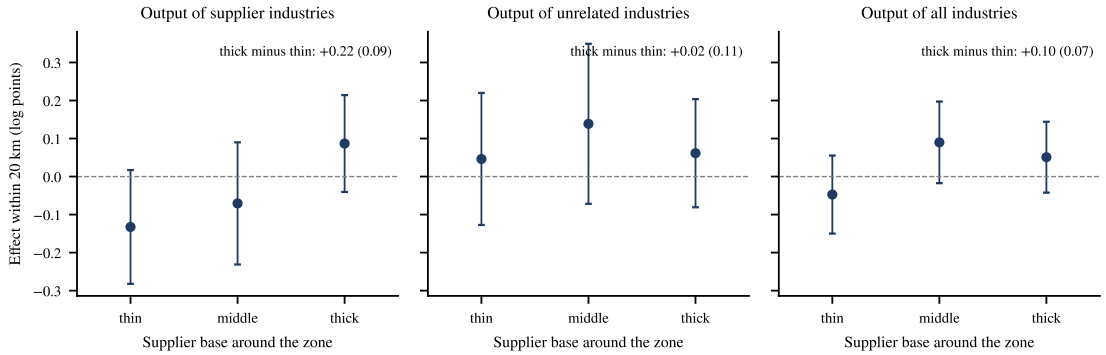


Figure I3: Effect within 20 km of the first opening by the zone's supplier base (terciles of the 1998–2000 output of the industries that supply fenced firms' inputs, in the cells around the zone): output of supplier industries, of unrelated industries and of all industries. Equation (1); standard errors clustered by prefecture; 95 percent confidence intervals.

Table I3: Entry of non-fenced exporters into fenced firms' markets and into related products: all results

Panel A. Markets: entry into a product–destination pair a fenced firm within 20 km exported (percentage points; baseline 6.0)				
	Without next year	Previous year	Next year	
Fenced exports of the pair	+2.9 (0.8)	+0.9 (1.1)	+3.5 (0.9)	
Panel B. Related products: entry into a new product, by its proximity to the fence's portfolio within 100 km (percent of baseline)				
	0.05–0.10	0.10–0.20	≥0.20	Same product
Fence portfolio, previous year	+17 (7)	+51 (20)	+95 (45)	+134 (39)
Fence portfolio, next year	−3 (3)	−4 (6)	+19 (15)	+36 (14)
Difference (percent of baseline)			+76 (36)	+98 (38)
Wild cluster bootstrap p-value			0.077	0.011
Fences' new products of the next year[b]	−1 (5)	+29 (12)	+62 (25)	+7 (26)
Panel C. Where: same product, by the fence's province (percent of baseline)				
	Previous year	Next year	Difference	p-value
Fences in own province	+141 (48)	+54 (16)	+87 (41)	0.04
Fences in other provinces	+27 (32)	+24 (32)	+3 (30)	0.91
Own minus other province			+84 (50)	0.10
Panel D. Entry, scale and persistence: same product, entry years 2003–05				
	Estimate			
Entries still exported the next year (percent of baseline)	+197 (43)			
Entries not exported the next year (percent of baseline)	+143 (47)			
Among entries: exported the next year (percentage points)	+3.7 (2.3)			
Among entries: log export value in the first year (log points)	+0.23 (0.10)			
Panel E. Checks under a different specification (see notes; percent of baseline)				
	0.05–0.10	0.10–0.20	≥0.20	Same product
Zones without EPZ status	−6 (2)	−14 (3)	−28 (4)	+1 (4)
Fences within 30 km, exporters with a zone within 30 km	+4 (4)	+2 (8)	−19 (14)	+26 (16)

Panel A: entry into a six-digit product–destination pair, non-fenced exporters, 2002–06; fenced firm within 20 km exported the pair in the previous (next) year; firm × product × year and product × destination × year effects; standard errors clustered by firm. Panels B–D: equation (2), entry into a new four-digit product, entry years 2003–05, percent of each sample's baseline entry rate (0.146 percent in the full sample); standard errors clustered by nearest zone (36 clusters); p-values from $t(35)$. Province from the firm's registration code. Wild cluster bootstrap: null imposed, Rademacher weights, 9,999 draws. Panel E: no next-year term, entry years 2003–06, standard errors clustered by firm, which are about a quarter of those clustered by nearest zone; the 30 km row keeps only exporters with a zone within 30 km, so its reference group differs from that of panel B and it is not evidence about distance (distance rings: Appendix Table R2, panel D). [b] Products that fenced firms within 100 km first export in $t+1$ (not exported in $t-1$ or t), in a regression that also includes both full portfolios.

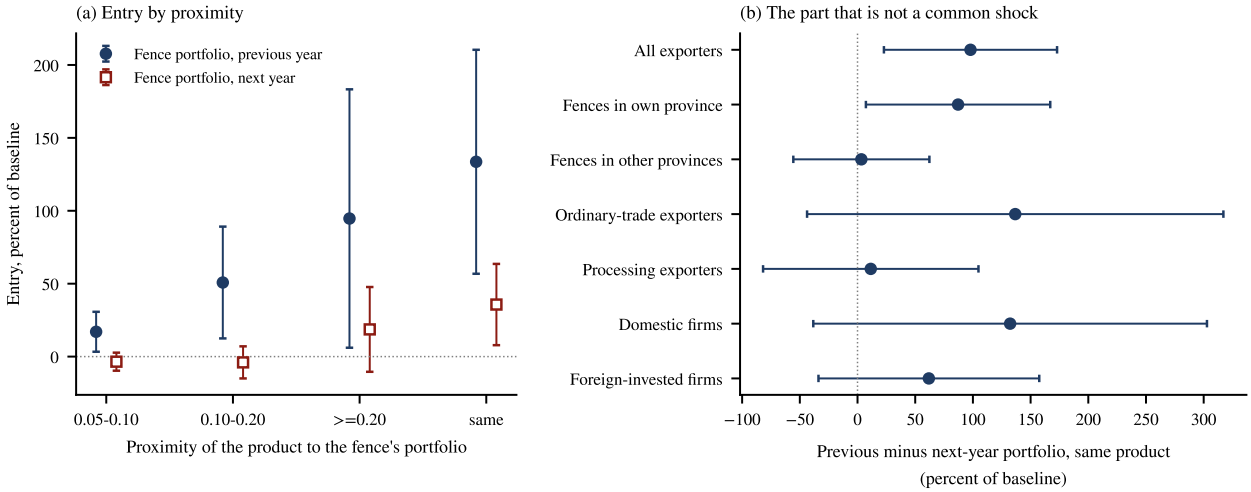


Figure I4: Entry of non-fenced exporters into new products, all splits. (a) By the product's proximity to what fenced firms within 100 km exported in the previous year and in the next year: percent of the baseline entry rate. (b) Previous-year minus next-year portfolio for the same product, by location of the fenced firms and type of exporter. Equation (2); 95 percent confidence intervals, clustered by nearest zone.

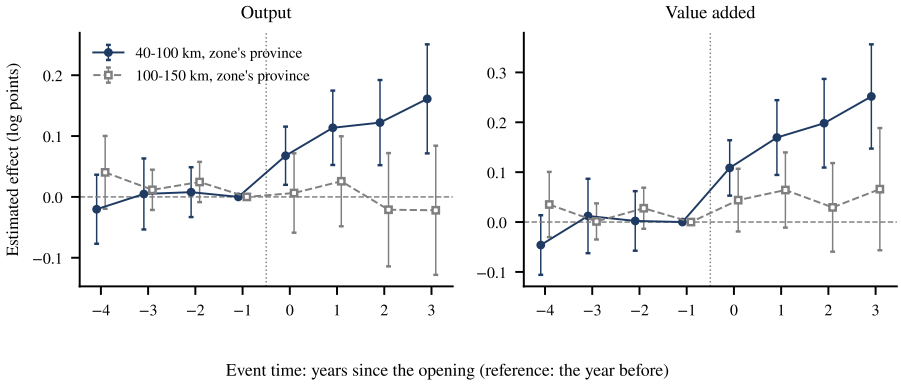


Figure I5: Paths in the hinterland and 100–150 km away inside the zone's province: equation (1) with one term per year; reference: the year before the opening; 95 percent confidence intervals.

J Estimation details

This appendix states how the paper's estimates are produced: the cells and the exposure variables, the comparison, the estimator, the quantities reported and their standard errors, and the checks of inference, timing and dating that the text refers to. The derivations and estimation details of the admission model are in Appendix T, and the checks with placebo zones and sensitivity to pre-trends in Appendix H.

J.1 Cells and exposure

The unit is a cell of 0.1 by 0.1 degrees, about 11 by 10 km at China's latitudes. The 19,149 cells are those in which at least one surveyed manufacturing firm is located in some year between 1998 and 2007. A cell's outcome in a year is the total over the surveyed manufacturing firms located in it (two-digit industries 13–43), excluding firms identified as inside a fence. A cell without firms in a year has outcome zero and stays in the panel, so every cell enters in every year.

Distances are great-circle distances between a cell's centroid and a zone's coordinates. Among the EPZs open by 2007, let g_c be the earliest opening year of a zone within 150 km of cell c and $z(c)$ the nearest zone among those opening in g_c; cells with no zone within 150 km have $g_c = \infty$. Of the 19,149 cells, 7,976 are exposed and 11,173 never are. The distance d_c to $z(c)$ places the cell in one of seven distance rings: 0–10 km (89 cells), 10–20 (188), 20–30 and 30–40 (610 together), 40–60 and 60–100 (3,132), and 100–150 km (3,957). A zone's province is the province of the cell nearest to its coordinates. The six distance groups $r(c)$ of equation (1) pool these rings as 0–10, 10–20 and 20–40 km regardless of province; 40–100 and 100–150 km inside the zone's province; and 40–150 km in another province. A cell keeps the group of its first zone in every year.

Two further sets of terms, collected in χ_{ct} in equation (1), keep every exposed cell-year out of the comparison. Years four and later after the first opening receive one indicator per distance group. When a second EPZ opens later, by 2007, closer to the cell than the lower edge of its distance ring, three indicators mark the year of that opening, the year after, and all later years; 1,429 cells have such a second exposure. Both sets are estimated jointly with the main terms; no cell-year is dropped.

J.2 The comparison

Each cell belongs to a matched stratum $q(c)$ defined by seven macro-regions (North, Northeast, East, Central, South, Southwest, Northwest), terciles of the log distance to the nearest of

23 major seaports, five bins of the number of surveyed firms in the cell in 1998 (at most one, 2–4, 5–19, 20–99, 100 or more) and four bins of its 1998 output (none, and terciles of positive output); 165 strata are populated, and $\lambda_{q(c),t}$ gives each its own year effects. The bins follow the distribution of manufacturing rather than its quantiles: 79 percent of cells had at most one surveyed firm in 1998, so terciles of firm density or output would split cells with identical values arbitrarily and separate little more than cells with two or more firms from the rest. Within the bins, cells near future zones and their matched comparison cells are close in baseline size (Table G1). The covariates $\boldsymbol{x}_c$ are the same three baseline measures and the electronics share of the cell's 1998 output, each standardized and interacted with year indicators for 1999–2007. The 1998 measures are computed from the surveyed firms that can be located by exact coordinates in that year; they rank cells before any opening and are not outcomes. Within a stratum, an exposed cell is compared with never-exposed cells and with exposed cells before their own openings, and the covariate terms absorb differences in trends associated with baseline size, port access and electronics within strata.

J.3 Estimator

Equation (1) is estimated by Poisson pseudo-maximum likelihood. The estimator solves

$$\sum_{c,t} \boldsymbol{z}_{ct}\big(y_{ct} - \hat{y}_{ct}\big) = \mathbf{0}, \qquad \hat{y}_{ct} = \exp\big(\hat{\alpha}_c + \hat{\lambda}_{q(c),t} + \boldsymbol{x}_c'\hat{\boldsymbol{\mu}}_t + \boldsymbol{d}_{ct}'\hat{\boldsymbol{\beta}} + \hat{\chi}_{ct}\big), \tag{J1}$$

where $\boldsymbol{z}_{ct}$ collects all regressors and fixed-effect indicators and $\boldsymbol{d}_{ct}$ the exposure indicators. It is consistent whenever the conditional mean is correctly specified, whatever the distribution of y_{ct} (Gourieroux, Monfort and Trognon 1984; Santos Silva and Tenreyro 2006). Three features make it the natural choice here. It handles zeros: half of the hinterland's cells had no manufacturing above the survey threshold in 1998. It does not require homoskedastic errors, which in log-linear regressions on totals would bias the coefficients. And its coefficients aggregate: $\exp(\beta_{re})$ is the ratio of a group's expected total at event time e to the total it would have had without the opening, so the level effects reported below add up across cells and groups. With cell effects, the Poisson model has no incidental-parameter bias in the slope coefficients (Hausman, Hall and Griliches 1984; Wooldridge 1999). The fixed effects are concentrated out by iteratively reweighted least squares with alternating projections. A cell, or a stratum-year, whose outcome is zero in every year carries no information about the coefficients once its own effect is included; the estimator sets such observations aside, which leaves the estimates unchanged.

J.4 Reported quantities and standard errors

The text reports four quantities, all functions of $\hat{\boldsymbol{\beta}}$ and its estimated covariance $\hat{V}$.

1. The *mean effect* of group r, $\bar{\beta}_r = \boldsymbol{v}_r'\hat{\boldsymbol{\beta}}$, where $\boldsymbol{v}_r$ puts $1/4$ on the four coefficients $\beta_{r0}, \ldots, \beta_{r3}$; its standard error is $(\boldsymbol{v}_r'\hat{V}\boldsymbol{v}_r)^{1/2}$. In percent, the effect is $100\,(e^{\bar{\beta}_r} - 1)$.

2. The *level change* of group r over the first four years,

$$\Delta_r = \sum_{e=0}^{3} \sum_{(c,t):\, r(c)=r,\ t-g_c=e} y_{ct}\big(1 - e^{-\hat{\beta}_{re}}\big), \tag{J2}$$

   the observed total minus the total predicted without the opening. Its standard error follows from the delta method with gradient $\partial\Delta_r/\partial\beta_{re} = \sum y_{ct}\, e^{-\hat{\beta}_{re}}$ over the same cell-years.

3. *Sums* of level changes across groups, used to add up the province's distance groups, and *differences* of mean effects between groups, with standard errors from the joint covariance of all the coefficients involved.

4. *Ratios to fenced exports*: a level change of value added divided by the exports of the fenced firms of the same zones in the same years.

Standard errors are cluster-robust by prefecture (329 clusters), with the usual small-sample correction. Prefectures are the governments that hold zones, approve land and collect most local taxes; clustering by prefecture allows arbitrary correlation across the cells of a prefecture and across years.

J.5 Inference with few treated prefectures

Some estimates rest on few treated prefectures: the second wave has 29 cells within 10 km, in nine prefectures. When few clusters are treated, cluster-robust standard errors can understate the sampling variability (MacKinnon and Webb 2017), so for the headline estimates we also compute p-values from the score wild cluster bootstrap with the null hypothesis imposed (Kline and Santos 2012), which is valid for nonlinear models such as ours. For a hypothesis $H_0 : \boldsymbol{v}'\boldsymbol{\beta} = 0$ (a mean effect or a difference of two), we write the model so that $\theta = \boldsymbol{v}'\boldsymbol{\beta}$ is the coefficient of a single regressor a_{ct}, estimate the model with $\theta = 0$ imposed, and form the efficient score of each prefecture p,

$$S_p = \sum_{(c,t)\in p} \tilde{a}_{ct}\big(y_{ct} - \tilde{\mu}_{ct}\big), \tag{J3}$$

where $\tilde{\mu}_{ct}$ are the restricted fitted values and $\tilde{a}_{ct}$ is a_{ct} net of its projection on all other regressors and fixed effects, weighted by $\tilde{\mu}_{ct}$. The statistic is $T = \sum_p S_p / (\sum_p S_p^2)^{1/2}$. Each bootstrap draw multiplies every prefecture's score by an independent weight ω_p from the six-point distribution of Webb (2023), $T^* = \sum_p \omega_p S_p / (\sum_p \omega_p^2 S_p^2)^{1/2}$, and the p-value is the share of 9,999 draws with $|T^*| \geq |T|$. Because the scores are evaluated at the restricted estimate, no model is re-estimated in the bootstrap. Table J1 compares the analytic and bootstrap p-values. The bootstrap widens the uncertainty of the estimates that rest on few prefectures

Table J1: Analytic and wild-bootstrap p-values for the headline estimates

Group	Outcome	Estimate (s.e.)	Treated prefectures	p-value		
				Analytic	Bootstrap (Webb)	Bootstrap (Rademacher)
0–10 km, second wave	New establishments	−0.41 (0.11)	9	<0.001	0.023	0.021
0–10 km, second wave	Output	+0.08 (0.06)	9	0.201	0.275	0.276
0–10 km, second wave	Value added	+0.16 (0.12)	9	0.184	0.266	0.262
0–10 km, second wave	Employment	+0.04 (0.05)	9	0.451	0.479	0.475
0–10 km, all zones	Output	+0.15 (0.03)	25	<0.001	<0.001	0.001
0–10 km, all zones	Value added	+0.21 (0.05)	25	<0.001	0.001	0.002
0–10 km, all zones	Employment	+0.18 (0.05)	25	<0.001	0.005	0.005
0–10 km, all zones	New establishments	−0.08 (0.11)	25	0.459	0.476	0.465
40–100 km, same province	Output	+0.11 (0.03)	119	<0.001	0.002	0.002
40–100 km, same province	Value added	+0.18 (0.04)	119	<0.001	<0.001	<0.001
0–20 km, thick − thin base	Supplier industries' output	+0.22 (0.09)	24	0.015	0.030	0.029

Equation (1). Analytic: cluster-robust standard error by prefecture and normal p-value. Bootstrap: score wild cluster bootstrap with the null imposed (Kline and Santos 2012), Webb six-point weights (Rademacher weights in the last column), 9,999 draws. Treated prefectures: prefectures with at least one cell in the group (for a difference, in either group).

without changing a conclusion of the paper. The fall in new domestic establishments around the second wave, with nine treated prefectures, has an analytic p-value of <0.001 and a bootstrap p-value of 0.023 (0.021 with Rademacher weights), and its production outcomes stay far from significance. The pooled output gain within 10 km keeps a p-value of <0.001 under both, and the supplier-base contrast moves from 0.015 to 0.030. The hinterland's output and value added, which rest on many prefectures, keep bootstrap p-values of 0.002 and <0.001.

J.6 Event paths and tests before the opening

The paths replace the post-opening terms of equation (1) with one indicator for each event time $k = -4, \ldots, 3$ except $k = -1$, plus one indicator for $k \geq 4$; the year before the opening and the years more than four years before it form the reference. The pre-opening test is the Wald test that the three coefficients at $k = -4, -3, -2$ are jointly zero, with the cluster-robust

covariance (χ^2 with three degrees of freedom; Appendix Table L3).

J.7 Alternative comparisons

Four alternatives to the matched comparison are reported. The plain comparison has cell and year effects only (Appendix K). The bridge adds the matched-stratum year effects and the baseline covariates one at a time (Appendix Table L1). The not-yet-treated comparison replaces the never-exposed cells with cells within 150 km of special customs zones that open after 2007 (later EPZs, comprehensive bonded zones, bonded ports and bonded logistics parks), with distance-group × year effects; a stricter version uses only the 241 cells within 20 km of such zones and more than 150 km from any zone open by 2007. Binary designs give exposure terms to one distance group only and leave every other cell in the comparison (Appendix Table L2).

J.8 Entry near the fence under each comparison

The pooled effect on new domestic establishments near the fence depends on the comparison more than any other estimate in the paper, and Table J2 traces it step by step. With cell and year effects alone, entry within 20 km falls by −0.21 (0.10). The fall survives macro-region × year effects and matched strata built on terciles of baseline size (−0.20 (0.10)). It disappears once the comparison also holds baseline size fixed, either through baseline covariates × year (−0.08 (0.10)) or through strata built on bins of baseline firm counts and output. The pooled fall is therefore a difference between places of different baseline size, which in the 2000s had different entry paths, and not an effect of the zones. The fall around the second wave, whose surroundings had flat paths before the opening, is not of this kind: it holds in the design of the paper and net of its own path before the opening (Section 4.1). Firms with exact coordinates give the same pattern.

J.9 Opening dates

For 23 of the 59 zones, and 21 of the 52 zones open by 2007, no customs acceptance year is recorded and the opening year is the approval year plus two (Appendix A). To check that the estimates do not rest on these imputed dates, we keep every cell and give the cells whose first zone has an imputed date their own exposure terms, so that the reported coefficients come only from zones with an official date (Table J3). Zones with an official date carry most of every distance group, and their estimates are those of the pooled sample. For each headline the official-date estimate has the sign of the pooled one and lies within one standard

Table J2: New domestic establishments near the fence under each comparison

	All located firms		Firms with exact coordinates	
Comparison	0–10 km	0–20 km	0–10 km	0–20 km
Cell and year effects	−0.31 (0.13)	−0.21 (0.10)	−0.33 (0.10)	−0.30 (0.10)
+ macro-region × year	−0.23 (0.13)	−0.23 (0.10)	−0.26 (0.10)	−0.29 (0.10)
+ port-distance × density class × year	−0.21 (0.14)	−0.16 (0.11)	−0.24 (0.10)	−0.24 (0.10)
and baseline covariates × year	−0.03 (0.14)	+0.01 (0.12)	−0.06 (0.11)	−0.07 (0.11)
Matched strata (terciles) × year	−0.20 (0.13)	−0.20 (0.10)	−0.19 (0.11)	−0.23 (0.11)
and baseline covariates × year	−0.08 (0.12)	−0.08 (0.10)	−0.05 (0.10)	−0.09 (0.11)
Matched strata (bins) × year	−0.08 (0.12)	−0.08 (0.09)	−0.03 (0.12)	−0.11 (0.10)
and baseline covariates × year (this paper)	−0.08 (0.11)	−0.07 (0.09)	−0.04 (0.12)	−0.10 (0.10)

Equation (1) with new domestic establishments as the outcome; mean of the first four years after the first opening for cells within 10 and within 20 km of the fence. Rows starting with "+" or "and" add the named controls to the row above; the matched-strata rows replace the region and port-distance effects with stratum × year effects. Terciles: matched strata built on terciles of baseline size (macro-region × port-distance tercile × terciles of 1998 firm count and output). Bins: the strata of this paper (Section 3). Exact coordinates: firms located by their own coordinates rather than by the centroid of their town or county. Standard errors clustered by prefecture.

Table J3: Zones with an official opening date

	Output	Value added	Employment	New domestic establishments	Cells
A. Zones with an official opening date					
0–10 km, zones of 2001–02	+0.17 (0.04)	+0.22 (0.05)	+0.34 (0.06)	−0.23 (0.15)	35
0–10 km, zones of 2003–05 (second wave)	+0.08 (0.07)	+0.24 (0.11)	+0.03 (0.06)	−0.31 (0.13)	21
0–10 km, zones of 2006–07	+0.16 (0.03)	+0.14 (0.06)	+0.02 (0.07)	−0.15 (0.23)	18
10–20 km	+0.01 (0.04)	+0.07 (0.05)	+0.03 (0.05)	−0.03 (0.11)	159
20–40 km	−0.03 (0.05)	+0.01 (0.06)	+0.03 (0.05)	−0.15 (0.12)	535
40–100 km, zone's province	+0.12 (0.04)	+0.18 (0.04)	+0.08 (0.04)	+0.09 (0.09)	2297
100–150 km, zone's province	+0.00 (0.04)	+0.05 (0.04)	−0.00 (0.04)	+0.04 (0.08)	2334
40–150 km, other provinces	+0.01 (0.06)	+0.01 (0.07)	−0.10 (0.08)	+0.07 (0.24)	1237
B. All zones (opening year imputed for 21 of 52)					
0–10 km, zones of 2001–02	+0.16 (0.04)	+0.23 (0.05)	+0.29 (0.06)	+0.08 (0.21)	40
0–10 km, zones of 2003–05 (second wave)	+0.08 (0.06)	+0.16 (0.12)	+0.04 (0.05)	−0.41 (0.11)	29
0–10 km, zones of 2006–07	+0.16 (0.03)	+0.13 (0.06)	−0.01 (0.07)	−0.01 (0.18)	20
10–20 km	+0.05 (0.05)	+0.10 (0.05)	+0.02 (0.05)	−0.05 (0.11)	188
20–40 km	−0.02 (0.05)	+0.03 (0.06)	+0.03 (0.05)	−0.09 (0.12)	610
40–100 km, zone's province	+0.12 (0.03)	+0.18 (0.04)	+0.09 (0.04)	+0.10 (0.09)	2709
100–150 km, zone's province	−0.01 (0.04)	+0.04 (0.04)	−0.00 (0.04)	+0.04 (0.07)	3010
40–150 km, other provinces	+0.00 (0.06)	−0.01 (0.06)	−0.10 (0.07)	+0.07 (0.23)	1370

Equation (1) with separate exposure terms for cells whose first zone has an official customs acceptance year and for cells whose first zone's opening year is imputed (approval year plus two); only the former are shown, next to the estimate that pools both. Mean of the first four years after the opening; standard errors clustered by prefecture.

error of it: new domestic establishments around the second wave −0.31 (0.13), output in the hinterland +0.12 (0.04) and value added +0.18 (0.04). The few cells around zones with imputed dates give noisy estimates of their own, with no common direction.

J.10 The second wave without any one prefecture

The second wave consists of 29 cells within 10 km, in nine prefectures, so a single city could drive its estimates. Removing the second-wave cells of one prefecture at a time, the effect on new domestic establishments stays between −0.49 and −0.37 and the effect on output between 0.04 and 0.11 (Table J4). No prefecture carries the fall in entry or hides a change in production.

Table J4: The second wave within 10 km, leaving out one prefecture at a time

Sample	New domestic establishments	Output	Second-wave cells
All nine prefectures	−0.41 (0.11)	+0.08 (0.06)	29
Without Qinhuangdao	−0.43 (0.11)	+0.07 (0.06)	25
Without Lianyungang	−0.38 (0.11)	+0.07 (0.06)	26
Without Wuhu	−0.42 (0.15)	+0.09 (0.06)	25
Without Qingdao	−0.41 (0.11)	+0.08 (0.06)	27
Without Zhengzhou	−0.37 (0.12)	+0.11 (0.05)	27
Without Beihai	−0.43 (0.11)	+0.08 (0.06)	27
Without Chongqing	−0.39 (0.13)	+0.04 (0.06)	26
Without Xi'an	−0.49 (0.11)	+0.04 (0.07)	24
Without Urumqi	−0.37 (0.11)	+0.09 (0.07)	25

Equation (1) with separate exposure terms by wave for cells within 10 km; each row removes from the sample the second-wave cells within 10 km of the zone in one prefecture. Mean of the first four years after the opening; standard errors clustered by prefecture.

J.11 Firm-level measures

A *new establishment* is a firm whose first year in the survey is after 1998 and whose founding year is no earlier than one year before it. *Revenue productivity* (TFPR) is estimated separately for each two-digit industry: log sales are regressed on log employment, log fixed capital (original value) and log intermediate inputs after removing industry-year means, and TFPR is the residual, so it measures a firm relative to the average of its industry and year. As a revenue measure it combines physical efficiency and price (Foster, Haltiwanger and Syverson 2008). Entrants are ranked among all new domestic establishments of 1999–2007 by this measure, and their counts by fifth form the cell-level outcomes of Section 4.1; size fifths

rank entrants by output at entry within industry and year. The average wage is estimated by stacking the wage bill and employment of each cell-year in one Poisson regression with cell × outcome and year × outcome effects and every exposure term interacted with an indicator for the wage bill; the interaction is the proportional change in the wage bill relative to employment, with the joint covariance.

J.12 Supplier industries and the supplier-base contrast

Industries are grouped by their links to the fence through the 2002 input–output table (Appendix B): the fence's own industry (electronics); supplier industries, whose direct input requirement per unit of the fence's output, weighted by the fence's national export composition, is in the top quartile, electronics excluded; user industries, in the top quartile of purchases from the fence's industries; and unrelated industries, the rest. A zone's supplier base is the 1998–2000 output of supplier industries in the cells within 40 km of it, and zones with a cell within 20 km are divided into terciles of it. The supplier-industry and unrelated-industry outcomes are built from the survey firms with at least 8 employees, located at village level, the firm file used for the input–output analysis; the other cell outcomes use all surveyed firms. Table J5 reports two checks of the contrast in Section 5: whether thick- and thin-base zones' supplier industries already diverged before the openings, and whether a split by the baseline output of *unrelated* industries produces the same contrast. Before the openings, supplier

Table J5: The supplier-base contrast: before the openings, and a split by unrelated industries

		Supplier-industry output		Unrelated-industry output	
Split	Group	Before	After	Before	After
Supplier split	thin	−0.02 (0.04)	−0.15 (0.08)	−0.05 (0.08)	+0.02 (0.10)
	middle	+0.03 (0.07)	−0.05 (0.10)	−0.04 (0.05)	+0.12 (0.11)
	thick	−0.05 (0.07)	+0.08 (0.06)	−0.00 (0.06)	+0.09 (0.06)
	thick minus thin	−0.02 (0.08)	+0.24 (0.09)	+0.05 (0.09)	+0.07 (0.11)
Unrelated split	thin	−0.13 (0.11)	−0.28 (0.15)	+0.13 (0.07)	+0.25 (0.09)
	middle	+0.06 (0.06)	+0.11 (0.06)	−0.11 (0.02)	+0.09 (0.12)
	thick	−0.02 (0.06)	+0.02 (0.05)	−0.01 (0.04)	−0.00 (0.07)
	thick minus thin	+0.11 (0.13)	+0.30 (0.15)	−0.14 (0.07)	−0.25 (0.10)

Equation (1); cells within 20 km of the zone, by tercile of the zone's baseline supplier base (rows "supplier split") or of the baseline output of unrelated industries around it (rows "unrelated split"). Before: mean of the coefficients at −4, −3, −2 years (reference: the year before the opening) in the event-path version of the equation; after: mean of the first four years after the opening in the same regression. Standard errors clustered by prefecture.

industries around thick- and thin-base zones moved together: the difference of their mean

lead coefficients is −0.02 (0.08) ($p = 0.76$), and the contrast appears only after the opening (+0.24 (0.09) in the same regression). The second check is less favorable to a reading that singles out suppliers. Zones with a thick supplier base are also zones with a thick industrial base: the two rankings place 20 of the 29 zones in the same tercile (rank correlation 0.85). Splitting zones by the baseline output of unrelated industries gives a similar contrast in supplier industries (+0.30 (0.15)), while unrelated industries fall where their own base was thick (−0.25 (0.10)), the mean reversion expected when zones are split by an outcome's own baseline. What the data show is that the fence's supplier industries held up around zones in thick industrial places and declined around zones in thin ones, the opposite of what mean reversion would produce in a split by their own baseline, and that unrelated industries show no such contrast under the supplier split (+0.07 (0.11) in the same regression). They cannot separate a thick supplier base from a thick industrial base.

K The plain comparison

With cell and year effects alone, cells within 10 km show no production gain (output +0.04, 0.05) and a fall in new domestic establishments of 0.31 (0.13), and the second wave contracts (domestic output −0.16, firms −0.36); the hinterland grows as under the like-with-like comparison (output +0.19). Three facts hold under both comparisons: no gain within 10 km attributable to the openings, the shift of entry away from the least productive and smallest entrants, and an unchanged average wage.

L Comparisons, design and pre-opening tests

A binary design that treats only the cells within 20 km of a zone puts the growing hinterland into its comparison group and lowers the estimate within 10 km by 0.03 to 0.06; a binary design that treats every cell within 150 km averages the response within 10 km and in the hinterland and cannot show where the response is. The second-exposure terms change the estimates by at most 0.04.

Table K1: Manufacturing outside the fence by distance: cell and year effects only

	0–10 km	10–20 km	20–40 km
A. Production			
Output	+0.04	+0.05	−0.01
	(0.05)	(0.05)	(0.05)
Value added	+0.05	+0.07	+0.02
	(0.05)	(0.06)	(0.05)
Fixed capital	+0.02	+0.10	−0.04
	(0.06)	(0.07)	(0.07)
Profits	−0.05	−0.08	−0.03
	(0.10)	(0.08)	(0.07)
Taxes	−0.03	−0.02	+0.02
	(0.07)	(0.06)	(0.07)
B. Labor			
Employment	+0.09	+0.10	+0.14
	(0.10)	(0.07)	(0.06)
Wage bill	+0.08	+0.08	+0.05
	(0.07)	(0.06)	(0.07)
C. Firms and exports			
Number of firms	+0.05	+0.12	+0.08
	(0.09)	(0.07)	(0.06)
New domestic establishments	−0.31	−0.15	−0.16
	(0.13)	(0.12)	(0.14)
Domestic exits	+0.31	−0.05	−0.18
	(0.12)	(0.08)	(0.09)
Export deliveries	+0.10	+0.08	+0.03
	(0.05)	(0.10)	(0.07)
Exporting firms	+0.01	+0.02	−0.02
	(0.07)	(0.07)	(0.08)
Cells	89	188	610

As Appendix Table I1, with year effects common to all cells and no baseline covariates.

Table K2: Cells within 10 km under the two comparisons

0–10 km from the zone	Cell and year effects	Like with like
Output	+0.04 (0.05)	+0.15 (0.03)
Value added	+0.05 (0.05)	+0.21 (0.05)
Employment	+0.09 (0.10)	+0.18 (0.05)
New domestic establishments, all	−0.31 (0.13)	−0.08 (0.11)
smallest fifth at entry	−0.49 (0.21)	−0.53 (0.24)
four smaller fifths	−0.42 (0.14)	−0.23 (0.13)
largest fifth	−0.08 (0.16)	+0.26 (0.16)
Second wave: domestic output	−0.16 (0.03)	+0.05 (0.06)
Second wave: domestic firms	−0.36 (0.10)	−0.19 (0.07)
Pay per worker	−0.00 (0.04)	–
40–100 km, zone's province: output	+0.19 (0.04)	+0.11 (0.03)

Cells within 10 km (last row: 40–100 km, zone's province). Size fifths: quintiles of output at entry within industry and year. Second wave: zones opened in 2003–05; domestic firms excluding census additions. Average wage under the like-with-like comparison: 0.00 (0.04).

Table L1: From the plain to the like-with-like comparison, one control at a time

	0–10 km	40–100 km, same province		100–150 km, same province	
	Output	Output	Employment	Output	Employment
Year effects (baseline)	+0.04 (0.05)	+0.19 (0.04)	+0.24 (0.06)	+0.05 (0.04)	+0.15 (0.05)
Macro-region × year	+0.01 (0.04)	+0.14 (0.04)	+0.09 (0.04)	+0.00 (0.04)	+0.00 (0.04)
Port-distance tercile × year	+0.04 (0.05)	+0.19 (0.04)	+0.24 (0.06)	+0.05 (0.04)	+0.15 (0.05)
Baseline density tercile × year	+0.10 (0.05)	+0.18 (0.04)	+0.24 (0.06)	+0.03 (0.04)	+0.15 (0.05)
Baseline output tercile × year	+0.09 (0.05)	+0.20 (0.04)	+0.24 (0.06)	+0.04 (0.04)	+0.15 (0.05)
Matched stratum × year	+0.14 (0.03)	+0.11 (0.03)	+0.10 (0.04)	−0.02 (0.04)	−0.01 (0.04)
Matched stratum and baseline covariates × year	+0.15 (0.03)	+0.11 (0.03)	+0.08 (0.04)	−0.02 (0.04)	−0.01 (0.04)
plus geography × year	+0.14 (0.04)	+0.09 (0.04)	+0.12 (0.03)	−0.03 (0.04)	+0.02 (0.04)
Comparison = 241 cells near zones opened after 2007 and geography × year, provinces pooled; columns 3–6 refer to 60–100 and 100–150 km					
Year effects	+0.04 (0.05)	+0.16 (0.04)	+0.20 (0.06)	+0.06 (0.04)	+0.17 (0.05)
Baseline and geography covariates × year	+0.06 (0.04)	+0.06 (0.04)	+0.03 (0.04)	−0.08 (0.04)	−0.05 (0.04)

Equation (1); rows change only the year effects and the covariates. The last two rows restrict the comparison group to the 241 cells within 20 km of zones that open after 2007 and more than 150 km from any zone open by 2007, with distance groups pooled over provinces. With all cells near later zones (and their not-yet-treated years) as comparison and ring × year effects, domestic output 40–100 km away in the zone's province rises by 0.13 (0.04) and value added by 0.18 (0.07).

Table L2: Multi-ring versus binary designs (like-with-like comparison)

	Output	Value added	Employment	New dom. establishments
Distance groups, each with own terms (the paper)	+0.09 (0.03)	+0.14 (0.04)	+0.08 (0.04)	−0.07 (0.09)
Binary: 0–20 km treated, all other cells comparison	+0.06 (0.03)	+0.08 (0.04)	+0.04 (0.04)	−0.07 (0.08)
without second-exposure terms	+0.06 (0.03)	+0.08 (0.04)	+0.05 (0.04)	−0.03 (0.08)
Binary: within 150 km treated, never-exposed comparison	+0.04 (0.03)	+0.10 (0.03)	+0.03 (0.03)	+0.05 (0.06)
Memo: 40–100 km, zone's province (the paper)	+0.12 (0.03)	+0.19 (0.04)	+0.08 (0.04)	+0.10 (0.09)

First row: the paper's design with 0–10 and 10–20 km pooled. Binary rows: only the named group carries exposure terms; all other cells, including those 20–150 km from a zone after its opening, enter the comparison.

Table L3: Joint test that the three leads are zero (p-values), like-with-like comparison

	Output	Value added	Employment	Firms	New dom. establishments
0–10 km	0.20	0.17	0.00	0.37	0.86
10–20 km	0.70	0.16	0.74	0.13	0.40
40–100 km, zone's province	0.89	0.47	0.69	0.91	0.22
100–150 km, zone's province	0.10	0.19	0.08	0.33	0.08

Equation (1) with one term per year from four years before to three years after the opening, reference: the year before; Wald test of the leads at −4, −3, −2 with the clustered covariance.

Table M1: Every outcome and distance group

	0–10	10–20	20–40	40–100 SP	100–150 SP	40–150 O
Output	+0.15	+0.05	−0.02	+0.11	−0.02	−0.00
	(0.03)	(0.05)	(0.05)	(0.03)	(0.04)	(0.06)
Value added	+0.21	+0.10	+0.03	+0.18	+0.04	−0.01
	(0.05)	(0.05)	(0.06)	(0.04)	(0.04)	(0.06)
Fixed capital	+0.11	+0.05	−0.08	+0.15	+0.02	−0.06
	(0.04)	(0.05)	(0.08)	(0.06)	(0.03)	(0.05)
Employment	+0.18	+0.02	+0.03	+0.08	−0.01	−0.10
	(0.05)	(0.05)	(0.05)	(0.04)	(0.04)	(0.07)
Wage bill	+0.19	+0.02	−0.00	+0.05	−0.14	−0.10
	(0.04)	(0.05)	(0.06)	(0.04)	(0.06)	(0.08)
Number of firms	+0.06	+0.05	−0.00	+0.05	+0.03	−0.04
	(0.05)	(0.05)	(0.05)	(0.03)	(0.04)	(0.10)
Profits	+0.18	+0.04	+0.05	+0.16	+0.04	+0.04
	(0.08)	(0.07)	(0.08)	(0.06)	(0.05)	(0.06)
Taxes	+0.16	+0.04	+0.04	+0.19	+0.01	−0.02
	(0.05)	(0.06)	(0.07)	(0.06)	(0.04)	(0.07)
Export deliveries	+0.15	+0.06	+0.00	+0.07	−0.02	+0.14
	(0.08)	(0.07)	(0.07)	(0.05)	(0.06)	(0.08)
Exporting firms	+0.11	+0.06	+0.01	+0.03	+0.03	+0.03
	(0.05)	(0.06)	(0.07)	(0.05)	(0.06)	(0.09)
New domestic establishments	−0.08	−0.05	−0.09	+0.10	+0.04	+0.07
	(0.11)	(0.11)	(0.12)	(0.09)	(0.07)	(0.23)
Domestic exits	+0.13	−0.09	−0.19	−0.09	−0.02	+0.06
	(0.11)	(0.09)	(0.09)	(0.07)	(0.07)	(0.12)
Output, domestic firms	+0.10	+0.01	−0.05	+0.11	+0.00	−0.03
	(0.05)	(0.07)	(0.07)	(0.04)	(0.05)	(0.05)
Output, foreign-invested firms	+0.12	+0.08	−0.01	+0.10	−0.05	+0.12
	(0.06)	(0.07)	(0.06)	(0.06)	(0.07)	(0.13)

Equation (1). SP: zone's province; O: other provinces. Mean of the first four years after the opening; standard errors clustered by prefecture in parentheses.

Table M2: Adding up, census additions, the zone's own cells, contrasts and the average wage

A. Value added added up over the zone's province, per RMB of fenced exports (joint standard errors)			
	Near field 0–40 km	Hinterland 40–150 km	Whole province
	+0.81 (0.65)	+1.83 (0.69)	+2.56 (1.29)
B. Firms added by the 2004 census excluded			
	Output	Value added	Employment
0–10 km	+0.14 (0.03)	+0.20 (0.05)	+0.17 (0.05)
40–100 km, zone's province	+0.10 (0.03)	+0.17 (0.04)	+0.08 (0.04)
C. Within 10 km: cells within 5 km of the zone and cells 5–10 km away			
	Output, all firms	Output, foreign-invested	Output, domestic
0–5 km	+0.12 (0.07)	+0.04 (0.10)	+0.04 (0.06)
5–10 km	+0.16 (0.05)	+0.18 (0.11)	+0.11 (0.05)
D. Differences between distance groups and pay per worker			
	Output	Value added	
40–100 km minus 0–10 km	−0.03 (0.04)	−0.03 (0.05)	
40–100 km minus 10–20 km	+0.07 (0.05)	+0.09 (0.06)	
40–100 km: zone's province minus other province	+0.12 (0.07)	+0.19 (0.07)	
Pay per worker, 0–10 km	+0.00 (0.04)		

Panel A: sums of the level effects of all distance groups of the zone's province (0–10, 10–20, 20–40, 40–100, 100–150 km and the rest of the province beyond 150 km, timed by the province's first opening; comparison: provinces without an EPZ by 2007), divided by fenced exports; standard errors from the joint clustered covariance. Panel B: firms first recorded in 2004 but founded before 2003 removed in every year. Panel C: cells within 5 km of the zone, where fenced firms missed by the customs match would sit, and cells 5–10 km away. Panel D: differences of mean post-opening coefficients with the joint covariance; average wage from a stacked Poisson regression of the wage bill and employment.

M The hinterland and robustness

N The hinterland: composition, firms, buyers and robustness

This appendix gives the estimates behind the statements of Sections 5.3 and 4.4 that have no table of their own. Unless a note says otherwise, results are for the ring, the cells 40 to 100 km from the first-opening zone inside its province, from equation (1): "after" is the mean of the coefficients for the first four years after the opening, "before" the mean of the coefficients for the three years before it, and "after minus before" their difference.

N.1 Robustness, balance and composition of the ring

Table N1 reports the response of the ring net of the leads as the comparison is tightened (Panel A), how the ring differs from the cells it is compared with before the openings (Panel

B) and what the ring consists of (Panel C). Adding the baseline industry mix leaves the output response at 0.10 to 0.11. Adding the size, rank and urban core of the city keeps 74 to 85 percent of it, and province × year effects keep 87 percent; with the finer distance controls added, the output response falls to 57 percent of the baseline and the value-added response to 69 percent.

N.2 Who grows

Table N2 classifies the firms of the ring by who controls their paid-in capital (Panel A) and by founding cohort (Panel B). Foreign, private and other legal-person firms hold 48 percent of the ring's output before the openings and account for 67 percent of the summed contributions; foreign firms founded before 1995 account for 19 percent. Table N3 shows when each type responds (Panel A), splits foreign-invested and domestic firms into electronics and other industries (Panel B) and places the firms relative to development zones (Panel C). Firms inside development zones listed in the catalog hold 6 percent of the ring's output before the openings and do not grow; the firms around them carry 71 percent of the summed contributions.

N.3 Product classes

Table N4 splits the output response by product class. Ten of the twelve classes respond by 0.06 to 0.23, general and special-purpose machinery is flat, and energy, mining and other industries, which hold 2 percent of output, fall with a large standard error. No class accounts for more than 22 percent of the gain.

N.4 Inside the incumbent firms

Table N5 follows the firms that were in the 2000 survey. Among all of them, sales, fixed capital, employment and intermediate inputs rise together and profit rises about as fast as sales (Panel A). In Panel B the change in each ratio is smaller than the standard error of its numerator, except the share of new products among all firms, which falls.

N.5 Buyers and local demand

Table N6 asks who buys the extra output (Panels A and B) and whether the demand of the ring's own cities rises (Panel C). Household goods rise by 0.17 and goods bought by firms by 0.09; within household goods, food rises as much as the income-elastic categories, which

Table N1: The response 40–100 km away under finer comparisons, balance of the ring, and its composition

A. Response of the ring, after minus before the opening (log points)

	Output		Value added	
	Estimate	Relative to baseline	Estimate	Relative to baseline
Baseline				
Matched stratum and baseline covariates × year	+0.14 (0.04)	100%	+0.20 (0.05)	100%
Finer baseline controls and geography				
+ decile of baseline output × year	+0.14 (0.04)	101%	+0.20 (0.05)	100%
+ deciles of distance to capital and megacity × year	+0.15 (0.04)	108%	+0.23 (0.05)	113%
+ decile of distance to city core, pre-opening growth × year	+0.15 (0.04)	106%	+0.22 (0.05)	109%
Baseline industry mix				
+ dominant product class × year	+0.13 (0.04)	90%	–	–
+ twelve class shares × year (instead of the dominant class)	+0.14 (0.04)	100%	–	–
Size, rank and urban core of the cell's city				
+ decile of prefecture size × year	+0.12 (0.04)	85%	–	–
+ provincial-level rank × year	+0.12 (0.04)	86%	–	–
+ urban-core indicator × year	+0.12 (0.04)	83%	–	–
Province × year effects in place of matched stratum × year effects				
with prefecture size, rank and urban core × year	+0.10 (0.04)	67%	–	–
with the output and distance deciles and pre-opening growth × year	+0.06 (0.04)	44%	+0.12 (0.05)	59%

B. Ring cells and never-exposed cells of the same strata, before the openings

	Ring	Comparison	Standardized difference	Output-weighted
Output in 1998 (log)	2.29	1.12	+0.51	+0.61
Number of firms in 1998 (log)	0.60	0.19	+0.49	+1.58
Distance to the nearest port (log km)	5.31	6.22	−0.94	−1.73
Distance to the provincial capital (log km)	4.60	5.27	−1.03	−0.58
Distance to the nearest megacity (log km)	6.01	6.55	−0.75	−2.03
Distance to the prefecture's core (log km)	3.75	3.93	−0.21	+0.58
Employment growth, 1999–2000 (log points)	0.22	0.15	+0.07	+0.08
Prefecture size: log of its 1998–2000 output	11.27	9.81	+1.18	+1.44
Capital, provincial-level or sub-provincial city (share)	0.26	0.09	+0.47	+0.37
Largest-output cell of its prefecture (share)	0.01	0.02	−0.06	−1.27

C. What the ring consists of

	Cells	Share of cells
All cells of the ring	2,709	100%
Outer counties of the host prefecture	877	32%
Neighboring prefectures of the same province	1,832	68%
No manufacturing above the survey threshold in 1998	1,375	51%
A development zone started by 2000 within 8 km	236	9%

Panel A: Equation (1) for output and value added of manufacturing firms outside the fence (annual survey of industrial firms, 1998–2007); the group is the cells 40–100 km from the first-opening zone inside its province, and every other distance group keeps its own exposure terms. Entries are the mean of the coefficients for the first four years after the opening minus the mean of the coefficients for the three years before it, with the standard error of the difference clustered by prefecture; "relative to baseline" divides by the first row. The baseline has year effects by matched stratum (macro-region and terciles of port distance, baseline firm density and baseline output) and baseline firm density, output, electronics share and port distance interacted with year, as in the main equation. Each later row adds the terms named to the baseline; within a group they accumulate, except that the class shares replace the dominant class. Deciles are taken over all cells of the design; pre-opening employment growth (1999–2000) enters linearly. The last group replaces the matched-stratum × year effects by province × year effects. Industry-mix, size, rank and core rows were estimated for output only. Panel B: means over the cells of the ring and over the never-exposed cells of the same matched strata, before the openings; the standardized difference is the difference of means divided by the pooled standard deviation, and the output-weighted difference weights cells by their 1998–2000 output with the same standard deviation. Panel C: numbers of 0.1-degree cells; a development zone is a national or provincial zone other than an export processing or bonded zone, within 8 km of the cell center.

Table N2: Who grows in the hinterland: control of paid-in capital and founding cohort

A. Output by control of paid-in capital (log points)					
	Before	After	After minus before	Output share	Share of gain
Foreign	+0.01 (0.10)	+0.19 (0.14)	+0.18	26%	42%
Private	+0.13 (0.09)	+0.33 (0.13)	+0.20	12%	21%
Other legal persons	+0.18 (0.10)	+0.36 (0.12)	+0.18	10%	16%
State	−0.04 (0.08)	+0.02 (0.10)	+0.06	28%	15%
Collective	+0.26 (0.07)	+0.29 (0.10)	+0.03	24%	6%
All firms	+0.06 (0.04)	+0.21 (0.06)	+0.14	100%	–
B. The same, by founding cohort and control					
	Before	After	After minus before	Output share	Share of gain
Founded before 1995					
Foreign	−0.02 (0.10)	+0.18 (0.12)	+0.20	16%	29%
Private	−0.06 (0.08)	+0.08 (0.11)	+0.13	5%	6%
Other legal persons	+0.17 (0.15)	+0.27 (0.20)	+0.10	3%	3%
State	−0.05 (0.11)	+0.05 (0.12)	+0.10	25%	22%
Collective	+0.24 (0.08)	+0.25 (0.11)	+0.01	19%	1%
Founded in 1995 or later					
Foreign	−0.01 (0.18)	+0.08 (0.20)	+0.08	10%	7%
Private	+0.20 (0.13)	+0.45 (0.17)	+0.25	7%	15%
Other legal persons	−0.02 (0.09)	+0.15 (0.14)	+0.18	6%	10%
State	+0.46 (0.23)	+0.49 (0.26)	+0.03	3%	1%
Collective	+0.20 (0.13)	+0.33 (0.14)	+0.13	5%	6%

Equation (1) for cell-year output of the firms of each type, annual survey of industrial firms, 1998–2007; cells 40–100 km from the first-opening zone inside its province; standard errors of the two means clustered by prefecture (the difference is shown without a standard error). Control is assigned from the first survey year with capital data by the largest component of paid-in capital: foreign (registered with Hong Kong, Macau, Taiwan or foreign capital, or foreign capital of at least 25 percent), state, collective, private (individual capital the largest, or registered as private with no larger component) and other legal persons (legal-person capital the largest; the owner is not observed). Firms without capital records, under 0.01 percent of output, are not shown. Founded before 1995 includes firms with a missing founding year. Output share: share of the ring's output in the cell-years before each cell's opening. Share of gain: output share times the difference, as a fraction of the sum over the types shown; the products add up only approximately to the total, because the coefficients are not additive.

are imprecise. None of the indicators in Panel C rises by more than 1.64 standard errors; registered population is the only one that moves by more than that, and it falls.

N.6 Intermediaries, product-market event study and the hinterland scenario

Table N7 gives further evidence for Sections 5.3 and 4.4. New trading intermediaries fall within 10 km of a zone and rise 10 to 20 km away, while exports outside the fence do not change (Panel A). Other Chinese exporters' sales of the product-markets of fenced firms

Table N3: Timing of the response, electronics, and location relative to development zones

A. Timing: coefficient by year relative to the opening					
	Output of firms controlled by			Births of private establishments	
	Foreign	Private	Other legal persons	All	In or near zones
Four years before	+0.10 (0.07)	−0.08 (0.08)	+0.03 (0.07)	−0.28 (0.13)	−0.40 (0.14)
Three years before	+0.04 (0.08)	−0.04 (0.04)	−0.00 (0.06)	−0.19 (0.11)	−0.30 (0.09)
Two years before	+0.01 (0.03)	−0.01 (0.02)	−0.01 (0.03)	−0.13 (0.09)	−0.12 (0.10)
One year before (reference)	0	0	0	0	0
Year of the opening	+0.10 (0.04)	+0.11 (0.05)	+0.18 (0.05)	+0.02 (0.10)	−0.00 (0.11)
One year after	+0.16 (0.05)	+0.18 (0.06)	+0.12 (0.08)	−0.19 (0.11)	−0.21 (0.11)
Two years after	+0.27 (0.07)	+0.15 (0.07)	+0.17 (0.08)	−0.16 (0.17)	−0.24 (0.13)
Three years after	+0.37 (0.09)	+0.22 (0.08)	+0.25 (0.10)	+0.08 (0.15)	−0.01 (0.14)
Four years after	+0.38 (0.10)	+0.20 (0.10)	+0.16 (0.11)	−0.25 (0.14)	−0.16 (0.14)
Joint test that the three earlier years are zero (p-value)	0.482	0.723	0.943	0.123	0.002
B. Foreign-invested and domestic firms, electronics and other (log points)					
	Before	After	After minus before		
Output					
Foreign-invested, electronics	−0.76 (0.39)	−0.75 (0.50)	+0.01		
Foreign-invested, other industries	+0.04 (0.09)	+0.21 (0.12)	+0.17		
Domestic, electronics	−0.22 (0.26)	+0.49 (0.27)	+0.70		
Domestic, other industries	+0.11 (0.05)	+0.22 (0.06)	+0.10		
Export value					
Foreign-invested, electronics	−0.55 (0.49)	−0.66 (0.60)	−0.11		
Foreign-invested, other industries	+0.00 (0.09)	+0.12 (0.13)	+0.12		
Domestic, electronics	+0.55 (0.59)	+1.10 (0.60)	+0.55		
Domestic, other industries	+0.22 (0.13)	+0.23 (0.15)	+0.01		
C. Output by location relative to development zones (log points)					
	Before	After	After minus before	Output share	Share of gain
Inside a catalogued development zone	−0.10 (0.09)	−0.15 (0.21)	−0.05	6%	−3%
Around a catalogued zone, outside its core	+0.10 (0.05)	+0.23 (0.08)	+0.12	44%	51%
Names a zone, not placed in a catalogued zone	+0.07 (0.24)	+0.43 (0.36)	+0.36	6%	20%
Not located, names no zone	+0.05 (0.08)	+0.13 (0.11)	+0.08	43%	32%
Located outside all zones, names no zone	n.e.	n.e.	n.e.	2%	n.e.

Equation (1); cells 40–100 km from the first-opening zone inside its province; annual survey of industrial firms, 1998–2007; standard errors clustered by prefecture. Panel A: one coefficient for each year relative to the opening instead of the four-year mean, reference the year before the opening, same design; output of the firms of each type of control (defined in Table N2) and counts of new establishments, 1999–2007: domestic firms first appearing within a year of their founding that are registered as private, of which "in or near zones" are those inside a development zone listed in the catalog, within three zone radii plus 3 km of one, or reporting a zone name or address (zones as defined in Panel C); the last row is the p-value of the joint test that the coefficients for four, three and two years before the opening are zero. Panel B: before, after and their difference as in Table N2, without a standard error for the difference; foreign-invested firms are registered with Hong Kong, Macau, Taiwan or foreign capital; electronics is communication equipment, computers and other electronic equipment (two-digit industry 40); export value is the export delivery value reported to the survey. Panel C: output of firms by their place relative to development zones, with output share and share of gain as in Table N2; zones are the national and provincial development zones of the 2006 catalog with their approved areas, a firm is inside a zone if its village-level coordinates lie within the radius of a disc of that area plus 2 km of the zone's center and around it if within three radii plus 3 km, and firms that lack such coordinates or lie outside these discs but name a zone in their name or address are placed in the third row; n.e.: the Poisson estimate for this group (2 percent of output) is unstable, with coefficients above 10, and is not reported.

Table N4: Output response by product class

A. Output by product class (log points)					
	Before	After	After minus before	Output share	Share of gain
Chemicals, plastics, rubber and pharmaceuticals	+0.07 (0.05)	+0.24 (0.08)	+0.17	17%	21%
Textiles, clothing and leather	+0.08 (0.06)	+0.19 (0.11)	+0.12	16%	13%
Metal smelting and rolling	−0.12 (0.12)	+0.07 (0.11)	+0.19	10%	14%
Electronics and communication equipment	−0.61 (0.20)	−0.39 (0.25)	+0.21	10%	15%
Food and beverages	+0.20 (0.06)	+0.36 (0.08)	+0.16	9%	11%
Electrical machinery, appliances and instruments	+0.06 (0.08)	+0.18 (0.10)	+0.12	7%	6%
General and special-purpose machinery	+0.04 (0.07)	+0.02 (0.09)	−0.01	7%	−1%
Light manufacturing (wood, paper, furniture, toys)	+0.24 (0.10)	+0.35 (0.12)	+0.11	6%	5%
Motor vehicles and other transport equipment	+0.02 (0.11)	+0.24 (0.14)	+0.22	6%	9%
Building materials, other non-metallic minerals	+0.23 (0.07)	+0.36 (0.10)	+0.13	5%	5%
Metal products	+0.15 (0.11)	+0.26 (0.13)	+0.12	5%	4%
Energy, mining and other	+1.06 (0.36)	+0.86 (0.54)	−0.20	2%	−3%
B. Output by relation to the fence, construction and land use (log points)					
	Before	After	After minus before	Output share	Share of gain
Relation to the fence					
Same industry as the fence (electronics assembly)	−0.82 (0.29)	−0.46 (0.36)	+0.35	7%	17%
Inputs to assembly (parts, packaging, molds)	+0.17 (0.11)	+0.23 (0.14)	+0.07	9%	4%
Neither	+0.08 (0.04)	+0.21 (0.06)	+0.13	84%	79%
Construction and public works					
Tied to construction	+0.18 (0.08)	+0.27 (0.09)	+0.09	7%	4%
Not tied to construction	+0.06 (0.04)	+0.20 (0.06)	+0.15	93%	96%
Land use					
Land-intensive bulk industry	+0.02 (0.06)	+0.18 (0.08)	+0.17	22%	22%
Not land-intensive	+0.11 (0.04)	+0.27 (0.06)	+0.16	78%	78%

Equation (1) for cell-year output of the firms of each class, annual survey of industrial firms, 1998–2007; cells 40–100 km from the first-opening zone inside its province; standard errors of the two means clustered by prefecture (the difference is shown without a standard error). Each firm is assigned a class from its own four-digit industry name in the latest survey year with a report, by an automated classifier that saw the name, its three- and two-digit parent names and the most frequent self-reported products of firms with that name, but not the firm's location or treatment; firms whose name is not in the map take the most common class of mapped firms in their two-digit industry. Output share: share of the ring's 1998–2000 output. Share of gain: output share times the difference, as a fraction of the sum over the rows of the same group. Panel B: the classifier also marked each name as the fence's own industry, an input to assembly, tied to construction or public works, or a land-intensive bulk industry.

Table N5: Cell totals and ratios of the firms present in 2000

A. Cell totals of the firms present in 2000 (log points)

	All firms			Non-state firms		
	Before	After	After minus before	Before	After	After minus before
Sales	+0.06 (0.05)	+0.19 (0.07)	+0.12	+0.04 (0.07)	+0.16 (0.10)	+0.12
Output	+0.06 (0.05)	+0.16 (0.07)	+0.11	+0.03 (0.07)	+0.14 (0.10)	+0.11
Employment	+0.01 (0.04)	+0.09 (0.07)	+0.08	−0.07 (0.05)	−0.05 (0.08)	+0.02
Fixed capital	−0.04 (0.08)	+0.15 (0.08)	+0.18	+0.01 (0.05)	+0.11 (0.08)	+0.10
Intermediate inputs	+0.04 (0.05)	+0.13 (0.07)	+0.09	+0.04 (0.07)	+0.12 (0.10)	+0.08
Wage bill	−0.01 (0.06)	+0.05 (0.08)	+0.06	−0.06 (0.07)	−0.03 (0.09)	+0.03
Profit (positive part)	+0.30 (0.17)	+0.44 (0.18)	+0.14	+0.14 (0.09)	+0.37 (0.13)	+0.24
Loss	−0.48 (0.22)	−0.41 (0.24)	+0.06	−0.63 (0.21)	−0.60 (0.24)	+0.02
Subsidy income	+0.15 (0.13)	+0.37 (0.18)	+0.22	+0.56 (0.32)	+0.84 (0.37)	+0.28
Taxes	+0.14 (0.09)	+0.32 (0.11)	+0.18	+0.07 (0.07)	+0.24 (0.10)	+0.17
Interest paid	−0.00 (0.07)	+0.09 (0.10)	+0.09	+0.04 (0.09)	+0.12 (0.13)	+0.08
Liabilities	−0.05 (0.05)	+0.11 (0.08)	+0.16	−0.03 (0.08)	+0.03 (0.13)	+0.06
New-product output	+0.13 (0.20)	−0.04 (0.24)	−0.17	−0.21 (0.23)	−0.42 (0.33)	−0.21
Export value	+0.17 (0.11)	+0.19 (0.13)	+0.03	−0.04 (0.06)	−0.04 (0.12)	−0.01

B. Change in ratios (log points), with the numerator's standard error as benchmark

	All firms			Non-state firms		
	Change	Standard error	Change over standard error	Change	Standard error	Change over standard error
Profit to sales	+0.02	0.18	+0.10	+0.11	0.13	+0.89
Subsidy income to sales	+0.09	0.18	+0.54	+0.16	0.37	+0.42
Taxes to sales	+0.06	0.11	+0.53	+0.05	0.10	+0.49
Interest paid to liabilities	−0.06	0.10	−0.63	+0.02	0.13	+0.15
Liabilities to fixed capital	−0.03	0.08	−0.34	−0.04	0.13	−0.31
New-product output to output	−0.28	0.24	−1.17	−0.32	0.33	−0.98
Export value to output	−0.08	0.13	−0.60	−0.11	0.12	−0.94
Output to sales (inventory build-up)	−0.02	0.07	−0.24	−0.02	0.10	−0.15
Wage bill to employment (pay per worker)	−0.02	0.08	−0.27	+0.01	0.09	+0.12

Equation (1) for cell-year sums of each account item over the firms in the 2000 annual survey of industrial firms that can be located, held fixed and assigned to their 2000 cell, in every year they report (1998–2007); the same firms are followed in exposed and comparison cells. Non-state firms are those not controlled by the state or a collective by paid-in capital (control as defined in Table N2). Values are truncated at zero; profit is the positive part and loss is reported separately. Panel A: standard errors of the two means clustered by prefecture (the difference is shown without a standard error). Panel B: a ratio's change is the difference between the changes (after minus before) of the numerator and the denominator; no standard error is computed for that difference, so the standard error of the numerator's after-opening coefficient is shown as a benchmark, and the last column divides the change by it. Ratios in the order shown: profit, subsidy income and taxes over sales; interest paid over liabilities; liabilities over fixed capital; new-product output and export value over output; output over sales; wage bill over employment.

rise when fenced firms first export them, but they were already rising before (Panel B). Panel C converts the response of the hinterland into value added per RMB of fenced exports (Section 4.4).

O Relocation records and where movers go

Growth in the provincial hinterland (Section 4.4) could be activity that moved there from the zone's neighborhood or from elsewhere in the province, which is the question of Table G4. Two facts about registered firms bear on it from the data side: how much of the fence's

Table N6: Buyers and demand in the ring's cities

A. Output by buyer (log points)					
	Before	After	After minus before	Output share	Share of gain
Household goods	+0.12 (0.06)	+0.32 (0.08)	+0.20	22%	30%
Goods bought by firms (intermediates, equipment)	+0.01 (0.05)	+0.12 (0.06)	+0.11	59%	48%
Construction and public works	+0.15 (0.08)	+0.31 (0.09)	+0.16	7%	9%
Mixed or unclassified	+0.12 (0.09)	+0.27 (0.12)	+0.15	13%	14%
All output	+0.06 (0.04)	+0.21 (0.06)	+0.14	100%	–
Export value, all firms	+0.12 (0.12)	+0.23 (0.13)	+0.11		
B. Household goods by category (log points)					
	Before	After	After minus before	Output share	Share of gain
Food, beverages and tobacco	+0.22 (0.07)	+0.38 (0.09)	+0.16	8%	9%
Clothing and footwear	+0.04 (0.08)	+0.19 (0.16)	+0.15	6%	6%
Furnishings and household appliances	+0.23 (0.13)	+0.35 (0.18)	+0.12	2%	2%
Vehicles	−0.20 (0.32)	+0.26 (0.33)	+0.46	2%	6%
Recreation and consumer electronics	−0.06 (0.17)	+0.07 (0.21)	+0.13	3%	3%
Personal care	−0.29 (0.22)	+0.59 (0.29)	+0.88	0%	3%
Other household goods	+0.40 (0.25)	+0.52 (0.30)	+0.12	1%	0%
C. Urban districts of the ring's cities (log points)					
	Before	After	After minus before	Prefectures	
Retail sales	−0.00 (0.03)	−0.00 (0.05)	−0.00 (0.02)	281	
Retail sales per head	+0.02 (0.03)	+0.03 (0.05)	+0.01 (0.02)	278	
Average wage, non-private sector	+0.02 (0.02)	+0.04 (0.03)	+0.01 (0.01)	278	
Household savings per head	−0.03 (0.04)	−0.03 (0.04)	−0.00 (0.03)	278	
Wage bill, non-private sector	−0.06 (0.02)	−0.08 (0.04)	−0.03 (0.02)	278	
Urban employment, non-private sector	−0.04 (0.04)	−0.05 (0.05)	−0.01 (0.02)	278	
Registered population	−0.02 (0.01)	−0.03 (0.02)	−0.01 (0.01)	278	
GDP	+0.04 (0.03)	+0.06 (0.05)	+0.02 (0.02)	278	
Wholesale sales, large firms	+0.18 (0.15)	+0.21 (0.20)	+0.03 (0.10)	278	
Number of large wholesale firms	+0.15 (0.07)	+0.31 (0.12)	+0.16 (0.10)	278	
Employment in wholesale and retail trade	−0.19 (0.19)	−0.16 (0.20)	+0.03 (0.05)	241	
Road freight	+0.06 (0.10)	+0.19 (0.12)	+0.14 (0.10)	278	
Total freight	+0.04 (0.07)	+0.10 (0.10)	+0.06 (0.05)	278	
Road passengers	+0.02 (0.06)	+0.03 (0.09)	+0.01 (0.04)	278	

Panels A and B: equation (1) for cell-year output (export value in the last row of Panel A) of the firms of each type, annual survey of industrial firms, 1998–2007; cells 40–100 km from the first-opening zone inside its province; standard errors of the two means clustered by prefecture (the difference is shown without a standard error); output share is of the ring's output in the cell-years before each cell's opening and share of gain is as in Table N2 (in Panel A, the shares of household goods are sums over the categories of Panel B). Buyers are assigned from each firm's industry name by the automated classifier described under the product-class table; household goods are mapped to consumption categories by a fixed table written before the estimation. Panel C: city statistical yearbooks, urban districts of prefectures, 2000–2010, raw values; ordinary least squares on logs with prefecture effects, region (east, central, west) × year effects and log 2000 GDP × year; each prefecture takes the distance group, jurisdiction and opening year of its largest cell (by 1998–2000 output); the ring group is prefectures whose largest cell lies 40–100 km from the first-opening zone inside its province, the comparison is prefectures never within 150 km; event-time terms as in equation (1); the difference has its own standard error; standard errors clustered by prefecture.

Table N7: Intermediaries, other exporters' sales of the product-markets of fenced firms, and the hinterland scenario

A. Intermediaries and exports outside the fence, after minus before (log points)

	Any province			Zone's province		Other provinces
	0–10 km	10–20 km	20–40 km	40–100 km	100–150 km	40–150 km
New intermediaries (first export year)	−0.51 (0.26)	+0.78 (0.36)	+0.00 (0.25)	−0.23 (0.23)	−0.04 (0.21)	−0.30 (0.23)
Active intermediaries	−0.05 (0.12)	+0.34 (0.13)	+0.02 (0.10)	+0.03 (0.10)	−0.08 (0.08)	+0.02 (0.11)
Exports of intermediaries	+0.06 (0.13)	+0.21 (0.12)	+0.03 (0.12)	+0.20 (0.12)	+0.04 (0.09)	+0.07 (0.12)
Exports of producing firms	−0.01 (0.11)	−0.14 (0.12)	−0.14 (0.10)	−0.07 (0.07)	−0.05 (0.06)	−0.07 (0.08)
All exports outside the fence	−0.06 (0.09)	−0.06 (0.09)	−0.12 (0.08)	+0.03 (0.07)	−0.08 (0.06)	−0.08 (0.08)

B. Other exporters' sales of the fences' product-markets, by year relative to entry (log points)

	Sales by other Chinese exporters			Sales by all
	All	Processing	Ordinary trade	exporters
Four or more years before	−0.19 (0.11)	−0.37 (0.16)	+0.01 (0.04)	−0.17 (0.11)
Three years before	−0.09 (0.05)	−0.17 (0.08)	−0.01 (0.02)	−0.07 (0.05)
Two years before	−0.06 (0.03)	−0.12 (0.04)	−0.01 (0.01)	−0.05 (0.02)
One year before (reference)	0	0	0	0
Year of first export	+0.54 (0.16)	+0.72 (0.21)	+0.23 (0.09)	+0.57 (0.17)
One year after	+0.24 (0.08)	+0.37 (0.11)	+0.08 (0.04)	+0.28 (0.08)
Two years after	+0.24 (0.08)	+0.38 (0.12)	+0.08 (0.05)	+0.26 (0.08)
Three years after	+0.21 (0.09)	+0.35 (0.14)	+0.06 (0.04)	+0.23 (0.09)
Four or more years after	−0.03 (0.16)	+0.07 (0.22)	−0.01 (0.07)	+0.01 (0.15)
Joint test that the three earlier years are zero (p-value)	0.049	0.015	0.810	0.145
Product-markets entered by a fence	11,939			

C. Hinterland scenario for the national account (billion RMB per year)

	Share of baseline value added	Value added gained
No effect of the zones	0.00	0.0
Strictest comparison (province × year effects and finer controls)	0.12	39.8
Baseline comparison	0.20	66.4
Baseline value added of the ring, mean 1998–2000	332	

Panel A: equation (1) for cell-year exports recorded in customs transactions, 2000–2007, of firms outside the fence, with the baseline comparison; each column is a distance group of cells with its own exposure terms; entries are after minus before with the standard error of the difference clustered by prefecture. An intermediary is a firm whose name contains a word for import-export, trade or foreign economic relations; new intermediaries are firms whose first export year, from 2001, is the year in question. Panel B: unit is a six-digit product × destination × year, 2000–2007; a pair is treated in the first year of 2001–2007 in which a fenced firm exports it, for pairs with no fenced exports in 2000, and the comparison is pairs of the same products that China exported outside the fences in 2000 and no fenced firm has entered; Poisson regression of the pair's sales by all other Chinese firms (columns 1–3: all, processing, ordinary trade) or by all firms (column 4) on event-time indicators, with pair, four-digit product × year and destination × year effects; the ends are binned and the year before entry is the reference; standard errors clustered by four-digit product; the last rows give the p-value of the joint test that the three earlier coefficients are zero and the number of product-markets entered. Panel C: share of baseline value added times the baseline value added of the cells of the ring (survey value added outside the fence, mean of 1998–2000); the shares are the after-minus-before response of value added of Panel A of Table N1 under the strictest comparison (0.12) and under the baseline comparison (0.20), and zero.

output is traceable to firms that were already active in the host prefecture, and where the firms that relocate go. Neither measures how much activity moved; both show what the firm records can and cannot reveal. The relocation records are described in Appendix B.6.

Firms already in the host prefecture. For each zone with customs records of fenced firms, we take the exports of every firm registered inside the fence in operating years 0–3 and ask whether the firm was already active in the host prefecture in the two years before the opening. A fenced exporter counts as already active if its name, after legal-entity suffixes, country qualifiers, brackets, and spaces are removed, matches the name of a firm that exported through customs or appeared in the manufacturing survey in the host prefecture in those two years; or if the link between customs and survey records ties it to a survey firm present in the host prefecture before the opening, or to another customs registration of the same firm that exported before the opening. A firm that moves into the fence appears under a different customs code, because the code records the special-supervision status of the registered location (Appendix B.3), so the name and survey links, not the customs code, identify it. Weighted by fenced exports across the 24 zones with customs records of fenced firms, 5.6 percent of fenced exports in the first four operating years come from firms that were already active in the host prefecture. The match misses local firms that were neither exporters nor above the survey's reporting threshold before they moved, and firms that changed their names when they moved; for such firms 5.6 percent is a lower bound. The records thus trace a small part of the fence's exports to firms already in the host prefecture; they do not set an upper limit on it.

Where relocating firms go. Table O1 estimates where relocating firms go. A destination in another prefecture of the same province is chosen with a relative probability of about 0.024 compared with a destination at the same distance and of the same size in the firm's own prefecture (choice sets of 150 km, municipalities excluded), and cross-province moves are rarer still. The estimate changes little with 100 or 200 km choice sets, with confirmed moves only, and with the municipalities included. Legal entities rarely cross prefecture borders. The growth of the hinterland is as large in neighboring prefectures as in the outer counties of the host prefecture (Table G3), and registered firms that move across a prefecture border are rare, which makes it unlikely that moves of registered firms are the main source of the growth in neighboring prefectures.

Two limits apply. The model conditions on moving, so it says where movers go, not how many firms move; and a move across a provincial border that breaks the firm identifier is recorded as an exit and an entry, not as a move, so the records capture within-province moves more completely than cross-province moves. The records therefore support the statement

that registered firms rarely relocate across prefecture borders, not the statement that no activity relocated.

Table O1: Where relocating firms go: destination choice across prefecture borders

Choice set radius (km)	Sample	Other prefecture, same province γ_P	e^{γ_P}	Other province, γ_C	e^{γ_C}	Log distance	Moves
150	All candidate moves	−3.84 (0.29)	0.021	−7.68 (0.65)	0.0005	−1.35 (0.11)	6,233
150	Confirmed moves	−3.77 (0.32)	0.023	−7.88 (0.66)	0.0004	−1.07 (0.11)	3,250
150	Excluding municipalities	−3.75 (0.29)	0.024	not identified	–	−1.41 (0.12)	4,879
100	All candidate moves	−3.90 (0.33)	0.020	−7.11 (0.68)	0.0008	−1.35 (0.12)	6,195
100	Confirmed moves	−3.88 (0.36)	0.021	−7.28 (0.69)	0.0007	−1.05 (0.11)	3,221
100	Excluding municipalities	−3.82 (0.33)	0.022	not identified	–	−1.41 (0.12)	4,841
200	All candidate moves	−3.88 (0.28)	0.021	−7.91 (0.65)	0.0004	−1.36 (0.11)	6,246
200	Confirmed moves	−3.86 (0.30)	0.021	−8.18 (0.66)	0.0003	−1.07 (0.11)	3,259
200	Excluding municipalities	−3.78 (0.28)	0.023	not identified	–	−1.42 (0.11)	4,891

Notes: Conditional-logit destination choice of relocating firms, estimated as Poisson regressions with mover fixed effects (Guimarães et al. 2003). Choice sets are all counties with above-scale firms within the stated radius of the origin. e^{γ} is the probability of choosing a destination in another prefecture of the same province (another province), relative to a destination at the same distance and of the same size in the origin prefecture. Regressions also include log destination firm count. When no relocation in the sample crosses a province border, γ_C is not identified. Standard errors clustered by origin prefecture in parentheses.

P Entry by size, ownership and exports

Table P1: New domestic establishments by size at entry under four definitions (cell and year effects)

	Q1	Q2	Q3	Q4	Q5
New establishments, output within industry-year (text), 0–10 km	−0.49 (0.21)	−0.48 (0.19)	−0.38 (0.20)	−0.44 (0.15)	−0.08 (0.16)
New establishments, employment, all industries pooled, 0–10 km	−0.28 (0.21)	−0.43 (0.19)	−0.00 (0.21)	−0.56 (0.22)	−0.32 (0.12)
All first appearances in the survey, output within industry-year, 0–10 km	−0.37 (0.29)	−0.04 (0.16)	−0.01 (0.16)	+0.08 (0.15)	+0.16 (0.13)
New establishments, exact-coordinate file only, output within industry-year, 0–20 km	−0.45 (0.18)	−0.23 (0.17)	−0.10 (0.15)	−0.36 (0.14)	−0.34 (0.08)

Equation (1) with cell and year effects. The last row uses only firms with exact coordinates and pools 0–20 km.

Ranking entrants by employment across all industries places labor-intensive industries in the upper fifths and changes the largest fifth (−0.32, 0.12); ranking by output within industry and year compares each entrant with its industry peers. Firms that first appear in the survey long after their founding show no decline under any measure.

Q Factor prices: wages and industrial land

This appendix describes how the wage and land-price outcomes are measured, how land prices are adjusted for quality, and where the coverage of each source ends.

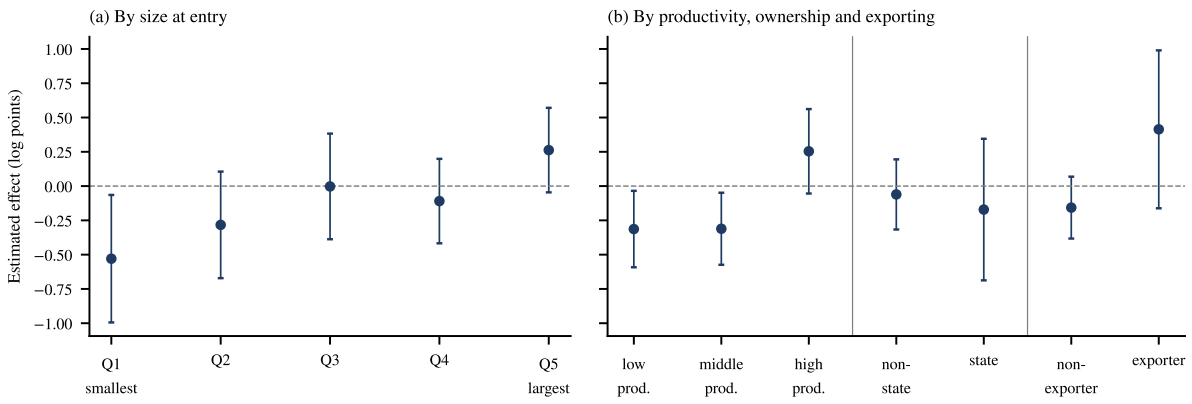


Figure P1: New domestic establishments within 10 km after the first opening: (a) by size at entry (quintiles of output within industry and year); (b) by productivity at entry, ownership and export status. Equation (1); 95 percent confidence intervals. Differences by ownership and export status are not significant.

Q.1 Wages

The city-level average wage is the prefecture's total manufacturing wage bill divided by its total employment in the firm survey (payable wages over the number of employees), computed from all surveyed firms; it is an employment-weighted average rather than an average of firm-level wages. Municipalities are treated as single cities. The series is nominal and covers 1998–2007. A second series, the average wage of staff and workers from the city statistical yearbooks (whole-prefecture scope, 2003–2019), is used as a check.

Two gaps limit what can be measured. The wage bill variable is missing for every firm in 2008 and 2009, so the firm survey cannot measure wages after 2007; and the postcode field used for sub-city geography is missing in 2000 and 2008–2010. Wage effects for zones that opened after 2007 are therefore measurable only in yearbook data at the city level.

Q.2 Industrial land prices

Source. Land prices come from parcel-level transaction records of the China Land Market Network, 2000–2021, with price, area, use, transfer channel, coordinates, floor-area ratio, land grade, and lease term. A parcel is industrial if its stated use is industrial. The price outcome is the log price per hectare of industrial parcels transferred through market channels; commercial and residential parcels are measured separately. Parcels priced more than fifty times the median of their city, year, and use are dropped.

Quality adjustment. Because parcels enter the data only when they are sold, the average transaction price mixes changes in the price of comparable land with changes in which land

the government chooses to sell. We residualize log prices on parcel characteristics,

$$\ln p_j = \gamma_1 \ln \text{area}_j + \gamma_2 \,\text{FAR}_j + \gamma_3 \,\text{grade}_j + \gamma_4 \,\text{lease}_j + \gamma_5 \,\text{newquota}_j + \delta_{\text{year}(j)} + u_j, \qquad \text{(Q1)}$$

with year fixed effects only, so that spatial price differences, which carry the treatment effect, are not absorbed. The residual $\hat{u}_j$ is the constant-quality log price, aggregated to cell × year. Standard errors in the second stage are not corrected for estimation of γ. The land test of Section 4.1 (Figure Q1) does not use these residuals: its price regressions control directly for the form of sale and the log of parcel area.

Coverage. Disclosure of land transactions became mandatory with the tender, auction, and listing rules of 2007: the number of cities with priced industrial transactions rises from 227 in 2006 to 340 in 2007. Around zones approved before 2007, coverage jumps from under a fifth before approval to more than two-thirds after, which would bias any comparison that spans the break. Land-price comparisons therefore use transactions from 2007 onward. All five parcel characteristics in equation (Q1) are observed for 29 percent of paid industrial parcels (167,863 of 574,400), so the quality-adjusted sample is narrower than the raw price sample. Coordinates for most parcels are address-level, which supports comparisons at kilometer scale but not finer.

Q.3 Land around later zones

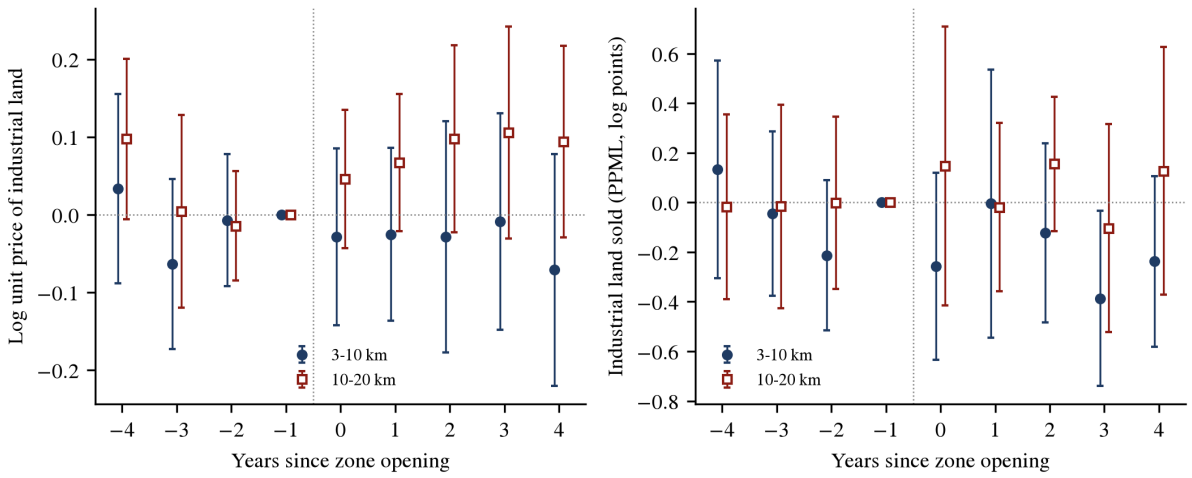


Figure Q1: Industrial land around special customs zones opened on new sites in 2008–2016 (49 zones): log price per hectare of parcels sold at auction (left) and area of industrial land sold per 0.1-degree cell and year (right, Poisson), 3–10 and 10–20 km from the zone relative to 40–100 km in the same prefecture and year. Prefecture × year effects; price regressions control for the form of sale and log parcel area; reference: the year before the opening; 95 percent confidence intervals, clustered by zone. Mean effect over years 0–3, 3–10 km: price −0.02 (0.06), area −0.19 (0.16).

Part IV. Learning

R Ideas: samples, definitions and inference

Imitation. Non-fenced exporters with known locations, a 10 percent sample of firms; every six-digit product–destination pair the firm did not serve in $t-1$ among its candidate set; entry years 2002–06 (with the next-year term, 2002–05: 1.62 million observations). Exposure rings by distance to the nearest fenced firm that exported the pair (0–20, 20–50, 50–100, 100–200 km). Without the next-year term, the 20–50 and 50–100 km rings are +1.2 and +1.4 percentage points; with it they fall to +0.8 and +0.8, and the next-year terms are +1.3 and +0.5.

Capability. Candidate products are four-digit products the firm has not exported since 2000 (the first year of the customs records), so a product exported before 2000 can count as an entry. Proximity is $\phi_{pq} = N_{pq}/\max(N_p, N_q)$, where N_{pq} counts non-fenced exporters of both products in 2000–02. Without the next-year term (entry years 2003–06, 3.32 million observations, baseline 0.141 percent), the gradient is 12, 39, 97 and 144 percent of the baseline (firm-clustered standard errors 2, 5, 11, 11; nearest-zone-clustered 7, 20, 47, 40). With fenced portfolios and portfolios of development zones without EPZ status entered together, the fence's gradient is 10, 35, 91 and 138 percent, and the other zones' −6, −14, −28 and +1.

Table R1: Related-product entry by exporter type: same product

	Previous year	Next year	Difference	p-value
Ordinary-trade exporters	+156 (93)	+19 (16)	+137 (92)	0.15
Processing exporters	+105 (43)	+93 (15)	+11 (48)	0.81
Domestic firms	+168 (91)	+36 (19)	+132 (87)	0.14
Foreign-invested firms	+150 (43)	+89 (23)	+62 (49)	0.21

Equation (2) estimated separately for each group; percent of each group's baseline entry rate; standard errors clustered by nearest zone (36 clusters); p-values from $t(35)$. Trade regime from the firm's exports in $t-1$: ordinary-trade exporters have no processing exports. Ownership from the firm's customs code. Differences between groups are not tested.

Inference. Exposure varies by zone, product and year, so the main tables cluster by the nearest of the 36 zones. Clusters are unequal: the largest (Guangzhou) holds 17 percent of entries and the three largest 38 percent; p-values use $t(35)$, and for the key differences the text also reports p-values from the wild cluster bootstrap with the null imposed (Rademacher weights, 9,999 draws), which does not rely on equal cluster sizes (MacKinnon and Webb

Table R2: Related-product entry: comparison group, inference, timing and distance

Panel A. Comparison group: same product and proximity of 0.20 or more (percent of each sample's baseline)				
	Previous year	Next year	Difference	p-value
All exporters (Table 3): same product	+134 (39)	+36 (14)	+98 (38)	0.015
proximity 0.20 or more	+95 (45)	+19 (15)	+76 (36)	0.043
Exporters with a zone within 100 km in $t-1$: same product	+86 (29)	+36 (21)	+51 (28)	0.080
proximity 0.20 or more	+57 (34)	+31 (18)	+26 (28)	0.37
Panel B. Inference: difference, all exporters (percent of baseline)				
	Same product	p-value	0.20 or more	p-value
Nearest zone (36 clusters)	+98 (38)	0.015	+76 (36)	0.043
Groups of zones linked within 100 km (23 clusters)	+98 (37)	0.015	+76 (42)	0.085
Groups of zones linked within 200 km (12 clusters)	+98 (36)	0.020	+76 (45)	0.12
Registration province (31 clusters)	+98 (38)	0.015	+76 (42)	0.079
Nearest zone and product (two-way)	+98 (39)	0.017	+76 (36)	0.044
Province and product (two-way)	+98 (38)	0.015	+76 (42)	0.079
Panel C. Timing: same product (percent of baseline)				
	All exporters	p-value	Zone within 100 km	p-value
Previous year, still exported by fenced firms in t	+142 (39)	<0.001	+92 (29)	0.003
Previous year, dropped by fenced firms in t	+96 (47)	0.049	+63 (35)	0.080
Next year	+32 (14)	0.031	+33 (21)	0.13
Still exported minus next year	+110 (38)	0.006	+59 (27)	0.040
Dropped minus next year	+64 (45)	0.17	+30 (34)	0.39
Still exported minus dropped	+46 (23)	0.056	+29 (20)	0.16
	Previous year	Year t	Next year	
Previous, current and next-year portfolios together	+119 (37)	+31 (12)	+21 (14)	
Panel D. Distance from the zone: same product, exporters with a zone within 150 km (percent of baseline)				
	Previous year	Next year	Difference	p-value
0–30 km	+23 (36)	+48 (40)	−24 (55)	0.66
30–60 km	+43 (50)	−16 (41)	+60 (40)	0.15
60–100 km	+141 (66)	+35 (25)	+106 (55)	0.062
100–150 km	−2 (24)	−9 (25)	+7 (28)	0.81

Equation (2), entry into a new four-digit product, entry years 2003–05, percent of each sample's weighted baseline entry rate (0.146 percent for all exporters, 0.171 percent for exporter-years with a zone within 100 km in $t-1$, 0.152 percent for exporters with a zone within 150 km); standard errors in parentheses, clustered by nearest zone unless panel B says otherwise; p-values from $t(G-1)$, where G is the number of clusters. Panel B: zones are grouped by single linkage of zone centers within 100 or 200 km. Panel C: the previous-year same-product indicator is split by whether fenced firms of zones within 100 km export the product in year t; the last row adds the year-t portfolio's bins to equation (2). Panel D: exposure to the zones in each ring around the exporter, each ring with its own previous- and next-year bins, in one regression. The sample of panels C and D is rebuilt with the construction and random draws of the main sample and has the same observations.

2017). Two-way clustering by firm and zone equals clustering by zone, because each firm belongs to one zone. Province is the registration province in the firm's customs code; for each zone, the province of all its fenced firms agrees.

S Learning: Design and Further Results

This appendix gives the full design behind Section 5.3: what fenced firms hold and what neighbors could observe, the definitions and sample of equation (2), the three checks of the comparison, the timing test and distance, where and for whom the association holds, the persistence of entries, and how large the association is. Further tables are in Appendix R.

S.1 What Fenced Firms Hold

Fenced firms are foreign assemblers in a few products. In 2006, 99 percent of the 588 fenced firms were foreign-invested and 97 percent wholly foreign-owned, and automatic data-processing machines (Harmonized System code 8471) made up 61 percent of fenced exports. Their customs operations were new: 94 percent of fenced firms, followed across changes of customs identifier, had no customs record outside a fence before they entered one.

Few neighboring exporters were directly exposed to the markets that fenced firms serve. New fenced firms enter markets that fenced firms elsewhere already serve far more often than markets served by fenced firms in their own customs district. Of the product-destination markets that new fenced firms entered in 2006, 94.5 percent were already served by a fenced firm somewhere in China and 16 percent by an incumbent fenced firm in the same customs district. Direct exposure was rare for non-fenced exporters: of the product-destination entries open to them, 0.17 percent had a fenced firm within 20 km exporting the same product to the same destination in the previous year, and 5.3 percent had one anywhere in China.

S.2 Design of the Related-Products Test

In equation (2), E_{ipt} marks entry and ℓ indexes four mutually exclusive proximity bins (0.05–0.10, 0.10–0.20, 0.20 or more, and the product itself, which takes precedence over the others), with proximity below 0.05, or no zone within 100 km, as the reference. The proximity $\kappa_{ip,t-1}$ is the largest proximity between p and any product exported in $t-1$ by the fenced firms of zones whose center, the mean location of their fenced firms, lies within 100 km of i: the previous-year portfolio. $\kappa_{ip,t+1}$ is the same for exports in $t+1$, the next-year portfolio. Proximity between two products is the smaller of the two conditional probabilities that an exporter of one also exports the other, as in Hidalgo et al. (2007) but computed from

firm-level co-exporting among non-fenced exporters in 2000–2002, before the window. The terms c_m are effects for bins m of $o_{ip,t-1}$, the largest proximity between p and the products i exported in $t-1$, and α_{it} and α_{pt} are firm-year and product-year effects. The comparison is therefore between products of the same firm in the same year, and between firms for the same product in the same year. Exporters with no zone within 100 km are in the reference group: they help estimate the product-year effects against which the others are compared. We estimate the equation by weighted least squares, weighting sampled non-entries by the inverse of their sampling rate, and report each b in percent of the weighted entry rate, 0.146 percent.

The next-year portfolio serves as a timing test. Regional specialization or demand that moves fenced firms and their neighbors together should make the next-year portfolio predict a neighbor's entry as well as the past portfolio does; capability acquired from what fenced firms already make should run from the past portfolio. The two portfolios overlap, so the test compares their incremental predictive power rather than separating two independent shocks. Neither portfolio nor the fixed effects of equation (2) absorb shocks specific to a province, product and year.

The sample is a quarter of the non-fenced, non-intermediary exporters with known locations near 36 zones, with all entries and 3 percent of non-entries (reweighted); entry years are 2003–05, and standard errors are clustered by the nearest zone (36 clusters).

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The sample is a quarter of the non-fenced, non-intermediary exporters with known locations near 36 zones, with all entries and 3 percent of non-entries (reweighted); entry years are 2003–05, and standard errors are clustered by the nearest zone (36 clusters).

S.3 Checks

Three checks qualify this reading (Appendix Table R2). The first concerns the comparison. Exporters with no zone within 100 km are in the reference group and help set the product-year effects. Among exporter-years with a zone within 100 km in the previous year, 55 percent of the weighted observations and 64 percent of the entries, the same-product asymmetry is 51

(28) percent of that sample's baseline rate ($p = 0.080$), about half the full-sample estimate: the next-year coefficient is unchanged and the previous-year coefficient falls. The asymmetry for proximity of 0.20 or more, 26 (28), is not statistically significant there. About half of the full-sample asymmetry therefore reflects product trends that differ between places near zones and places far from them. Clustering by groups of nearby zones, by province, or two-way by zone and product leaves the full-sample same-product asymmetry statistically significant (p from 0.015 to 0.020).

The second check concerns the timing test, which rests on the products that fenced firms stop exporting. In the same-product bin, products in the previous-year portfolio but not in the next-year one are 0.38 percent of the observations, with 1,321 entries, and 85 percent of the products in a previous-year portfolio are still in it a year later. A handover of products, such as a foreign buyer moving orders from a fenced firm to a firm outside the fence or a firm moving a product line out of the fence, would also make the previous-year portfolio predict entry, but only through the products that fenced firms drop. When the previous-year portfolio is split by whether fenced firms still export the product in year t, the association is 142 (39) percent of the baseline for the products they keep and 96 (47) for the products they drop, and the products they keep exceed the next-year portfolio by 110 (38; $p = 0.006$). The association does not run only through handovers. With the year-t portfolio added, the coefficients on the previous, current and next-year portfolios are 119 (37), 31 (12) and 21 (14).

The third check concerns distance. With exposure separated into rings around the zones, each with its own previous- and next-year terms, among exporters with a zone within 150 km, the same-product asymmetry is −24 (55) within 30 km, 60 (40) at 30–60 km, 106 (55; $p = 0.062$) at 60–100 km and 7 (28) at 100–150 km. The point estimates place the association 30 to 100 km from the zones rather than next to them, the range in which the hinterland grows (Section 4.4), which points to diffusion within the region more than to learning between neighbors.

S.4 Where, for Whom, Persistence and Size

Where and for whom. The point estimate of the asymmetry is largest among exporters registered in the same province as the fenced firms (Figure 4b; Appendix Table I3, panel C). For the same product, the past portfolio of fenced firms in the same province within 100 km is associated with entry higher by 141 percent of the baseline rate, against 54 percent for their next-year portfolio, an asymmetry of 87 (41). For fenced firms in other provinces at the same distances the two are 27 and 24 percent, an asymmetry of 3 (30).

The two asymmetries differ by 84 (50) percent of the baseline rate, which is not statistically significant ($p = 0.10$; bootstrap 0.11): the evidence places the asymmetry within provinces but does not show that it stops at their borders.

Point estimates of the asymmetry are larger for ordinary-trade and domestic exporters than for processing and foreign-invested ones, but the subgroup differences are imprecise (Appendix Table R1). The splits in this subsection and the next were chosen after the main specification.

The pattern is consistent with diffusion mediated by geography or by provincial institutions, which the data cannot tell apart; registration provinces need not be where decisions are made. The province also bounds the hinterland's growth (Section 4.4), but the firms differ: the point estimate for firms that were not exporting when first recorded is larger than for exporting firms (Table G3, panel B), so we do not take the shared boundary as evidence of a shared mechanism.

Entry, scale and persistence. The gradient is present among entries that last as well as among entries that do not (Appendix Table I3, panel D). For entries still exported the next year, the same-product gradient is 197 percent of the baseline, and the past portfolio exceeds the next-year portfolio by 117 percent of the baseline rate (45).

Conditional on entry, which selects on success, entries into fenced firms' own products start 0.23 (0.10) larger in first-year export value; the difference in the probability of exporting the product again the next year, 3.7 percentage points (2.3), is imprecise.

How large? Taken at face value, the asymmetry corresponds to about 5,500 additional entries into firm-product pairs a year in the estimation sample (standard error 2,000), 15 percent of its new-product entries, or about 22,000 a year among all the exporters the sample represents. This is an association, not a count of entries that fenced firms caused, and the data do not show whether these entries raised total exports. Among exporters with a zone within 100 km the asymmetry, and with it the count, is about half as large.

The number of exporting firms rises by 0.11 (0.05) in logs within 10 km (Table G2), where the rise in production comes mainly from zones placed in places already growing, and changes by 0.03 (0.05) 40 to 100 km away. Counts of exporters and entries into firm-product pairs are different margins, so neither estimate settles whether fenced firms raised exports around them.

Part V. The Model

T The Admission Model

T.1 Functional Forms

Profits. With demand $q = D_\ell p^{-\sigma}$ in location ℓ and marginal cost $c_\ell/\varphi_{j\ell}$, the profit-maximizing price is $p = \frac{\sigma}{\sigma-1} c_\ell/\varphi_{j\ell}$, revenue is $r_{j\ell} = \sigma \pi_{j\ell}$ and operating profit is (3) with $A_\ell = \frac{D_\ell}{\sigma}\big(\frac{\sigma c_\ell}{\sigma-1}\big)^{1-\sigma}$. If productivity is Pareto with shape θ, profits are Pareto with tail $\zeta = \theta/(\sigma-1)$.

Profits across locations follow equation (4) of Arkolakis et al. (2018),

$$\Pr\big(\pi_{jE} \le x_E, \pi_{jP} \le x_P, \pi_{jN} \le x_N\big) = 1 - \Big[\sum_\ell \big(T_\ell\, x_\ell^{-\zeta}\big)^{1/(1-\rho)}\Big]^{1-\rho}, \tag{T1}$$

with $T_\ell \propto A_\ell^\zeta$ and $\rho \in [0,1)$; at $\rho = 0$ a firm does well in at most one location. The parent zone's signal is the form of Khandelwal et al. (2013, footnote 28),

$$s_j = \tau z_j + \sqrt{1-\tau^2}\,\xi_j, \tag{T2}$$

where z_j is the normal score of the firm's productivity rank within its industry and year and ξ_j is standard normal noise. The share of productivity fifth b in the high tier $s_j \ge \bar{s}$ is

$$h_b(\bar{s},\tau) = 5\int_{(b-1)/5}^{b/5} \Phi\Big(\frac{\tau\,\Phi^{-1}(q) - \bar{s}}{\sqrt{1-\tau^2}}\Big)\,dq. \tag{T3}$$

With guaranteed employment, workers' location shares and expected utility are those of Fajgelbaum et al. (2019),

$$L_\ell = \bar{L}\,\frac{(B_\ell w_\ell)^\varepsilon}{\sum_k (B_k w_k)^\varepsilon}, \qquad U = \Gamma\big(1-\tfrac{1}{\varepsilon}\big)\Big[\sum_k (B_k w_k)^\varepsilon\Big]^{1/\varepsilon}. \tag{T4}$$

With Pareto profits, no re-entry and equal plots, a rule that ranks applicants by profit and admits the share r costs, in units of the operating profit of all applicants,

$$\mathcal{D} = \frac{1-r}{\zeta} + r - r^{1-1/\zeta} \approx \frac{(\zeta-1)(1-r)^2}{2\zeta^2}, \tag{T5}$$

second order in $1-r$, while random rationing within a tier costs $(1-r)/\zeta$, first order.

Re-entry. Let D_{low} be the number of applicants of the low fifths turned away within 10 km and $n^0_{\ell,\text{low}}$ their entry into location ℓ without the zone. If a share δ_ℓ of the turned-away

enters ℓ,

$$\Delta \ln n_{\ell,\text{low}} - \Delta \ln n_{\ell,\text{top}} \approx \delta_\ell \, D_{\text{low}} / n^0_{\ell,\text{low}}. \tag{T6}$$

T.2 Proofs

Throughout, the signal and profit of an applicant are jointly distributed in the same way whichever area it chooses, profits are in the Pareto region of (T1) at the fixed cost F, and the area within 10 km is small in the prefecture.

Proposition 1. (a) For any set $\mathcal{A}$ of areas, let

$$T_{\mathcal{A}} = \Big[\sum_{k \in \mathcal{A}} T_k^{1/(1-\rho)} \Big]^{1-\rho}, \qquad \psi_\ell^{\mathcal{A}} = \frac{T_\ell^{1/(1-\rho)}}{\sum_{k \in \mathcal{A}} T_k^{1/(1-\rho)}}.$$

Applied to equation (4) of Arkolakis et al. (2018), their Lemma 1 gives

$$\Pr\Big(\max_{k \in \mathcal{A}} \pi_{jk} \le x, \ \ell = \arg\max_{k \in \mathcal{A}} \pi_{jk} \Big) = \psi_\ell^{\mathcal{A}} \left(1 - T_{\mathcal{A}} x^{-\zeta} \right),$$

so the chosen area is independent of the maximal profit, which is Pareto with shape ζ and scale $T_{\mathcal{A}}$. Entry into area ℓ is $J \psi_\ell T_{\mathcal{A}} F^{-\zeta}$, proportional to ψ_ℓ. (b) Since the chosen area is independent of profit and the signal is distributed alike across areas, the applicants within 10 km are a representative sample of the pool's distribution of productivity and signal, scaled by $\exp(\nu)$. Let $z = \Phi^{-1}(q)$ for an applicant at productivity rank q. By (T2), $s \mid z \sim N(\tau z, 1 - \tau^2)$, so the applicant is in the high tier with probability $\Phi\big((\tau z - \bar{s})/\sqrt{1-\tau^2}\big)$; averaging over the ranks of fifth b gives h_b in (T3). The high tier is admitted and the low tier with probability e, so the admitted share of fifth b is $h_b + (1 - h_b)e$ and entry of that fifth changes by $\nu + \ln[h_b + (1 - h_b)e]$. A fifth with $h_b = 0$ falls by $\nu + \ln e$ whatever its rank. As $\tau \to 1$, h_b is one above $\Phi(\bar{s})$ and zero below it; with $e = 0$ this is a hard cutoff on productivity. As $\bar{s} \to \infty$ only one tier remains and the admitted share is e in every fifth. (c) With $\rho = 0$, (T1) becomes $1 - \sum_\ell T_\ell x_\ell^{-\zeta}$, and for any two areas E and k, $\Pr(\pi_{jE} > x_E, \pi_{jk} > x_k) = 0$ in the Pareto region. An applicant with $\pi_{jE} \ge F$ therefore has $\pi_{jk} < F$ in every other area and stays out when turned away. With $\rho > 0$ the joint survival is positive, and some applicants turned away enter elsewhere. For the mirror, let $n^0_{\ell b}$ be the entry of fifth b into area ℓ without the zone and D_b the applicants of fifth b turned away within 10 km. With the area's own change ν_ℓ, common to its fifths, entry after the opening is $n^0_{\ell b} e^{\nu_\ell} + \delta_\ell D_b$, so $\Delta \ln n_{\ell b} = \nu_\ell + \ln(1 + \delta_\ell D_b e^{-\nu_\ell} / n^0_{\ell b}) \approx \nu_\ell + \delta_\ell D_b / n^0_{\ell b}$ when re-entrants are few relative to the area's entry. The top fifth lies in the high tier, so $D_5 \approx 0$. Averaging over fifths 1–3 and subtracting fifth 5 removes ν_ℓ and gives (T6), with $D_{\text{low}} / n^0_{\ell,\text{low}}$ the average of $D_b / n^0_{\ell b}$; the

estimator in Section 7 uses the ratio of totals, which equals this average when the ratios are similar across the low fifths.

Representative applicants. Proposition 1 uses two properties. First, by Lemma 1 of Arkolakis et al. (2018), the location a firm chooses is independent of the profit it earns there. Second, the productivity fifth and the signal depend on the firm only through its productivity rank within its industry and year in the location where it produces. A firm's rank there is a function of its maximal profit relative to the profits of the industry-year in that location, and the maximal profit is Pareto with the same tail in every location, so the distribution of ranks among the firms that choose the parent zone is the distribution in the pool. Applicants to the parent zone are then a representative draw of the pool's productivity fifths, scaled by $\exp(\nu)$. The property fails if the opening changes the relative attractiveness of the zone by productivity, as complementarity between productivity and place does in Gaubert (2018). We model that failure as a log-linear tilt $\kappa(b-3)$ in the entry of fifth b; Section 7.2 reports its estimate and the price of the fence it implies.

The price as a change. Let $V_R(K)$ be the value admitted under rule R with land K, $V^*(K)$ the value admitted when K is allocated in order of value per hectare, and $G(K) = V^*(K) - V_R(K) \geq 0$. The applicants who obtain land at K_0 but not at K_1 forgo $V_R(K_0) - V_R(K_1)$, net of their value elsewhere, which is the sum in (9). Adding and subtracting V^* gives $V_R(K_0) - V_R(K_1) = [V^*(K_0) - V^*(K_1)] + [G(K_1) - G(K_0)]$. The step in entry identifies the left side, because it compares admissions after the fence with admissions before it. Under random rationing within tiers before and after, the admission probability of fifth b falls by the factor P_b of (7), and the turned-away of fifth b are a random draw of its applicants.

Proposition 2. The multiplier. With plots of size d_j, the constraint counts the expected hectares of each tier, $\mathbb{E}[d_j \mid c]$, and the problem is linear in (a_H, a_L) with one resource constraint, so tiers are admitted in order of expected value per hectare; with plots of equal size this is the order of expected value. Problem (5) is linear in (a_H, a_L) with one resource constraint. The Lagrangian $\sum_c a_c n_c(\mathbb{E}[v_j \mid c] - \Lambda\bar{a}) + \Lambda\bar{K}$ with $a_c \in [0,1]$ gives the first-order conditions of Fu (2014, equations 13–14): $a_c = 1$ when $\mathbb{E}[v_j \mid c] > \Lambda\bar{a}$, $a_c = 0$ when it is below, and $a_c \in [0,1]$ when equal, with $\Lambda \geq 0$ and complementary slackness. When the land binds and $\mathbb{E}[v_j \mid s_j]$ rises with s_j, which holds because $z \mid s$ rises in s in the likelihood-ratio order and v rises in z, the high tier is admitted and $\Lambda = \mathbb{E}[v_j \mid L]/\bar{a}$ at the low tier. A fall in $\bar{K}$ lowers e and, once $e = 0$, rationing reaches the high tier and Λ rises to its expected value per hectare; plots freed by exits raise e in the following year. With a continuum of applicants and a continuous signal this is the supply-and-demand lemma of Azevedo and Leshno (2016). *Cutoff and charge.* With the signal ranking applicants by profit, the cutoff

admits $\pi_{jE} \geq \bar{\pi}$ with $\Pr(\pi \geq \bar{\pi} \mid \pi \geq F) = (\bar{\pi}/F)^{-\zeta} = r$. A charge λ admits $(1-\lambda)\pi_{jE} \geq F$, the same set when $(1-\lambda)^{\zeta} = r$; a minimum size admits sales $\sigma\pi_{jE} \geq \sigma\bar{\pi}$, again the same set. With $\rho = 0$, no applicant turned away enters elsewhere under any of the three, so entry, composition and output coincide. The three differ in rent: the charge collects λ times the operating profit of the admitted. In units of the operating profit of all potential entrants without the claim, $\int_F^{\infty} \pi \, dG$, the admitted earn $(\bar{\pi}/F)^{1-\zeta} = r^{1-1/\zeta} = r/(1-\lambda)$, so the charge transfers $r\lambda/(1-\lambda)$. The cutoff and the minimum size collect nothing. Only a charge levied per hectare on the transaction moves the transaction price.

Proposition 3. Welfare is the sum of the surplus of potential entrants and incumbents, workers' welfare in money, the receipts of governments and the profits of the province's exporters. Compare each with the prefecture without the zone, holding prices outside it fixed. Each applicant turned away forgoes its value net of fixed costs and gains its value elsewhere; summing gives $-\mathcal{C}$. Each incumbent that loses its plot forgoes its value net of its value elsewhere; summing gives $-\mathcal{I}$. Each admitted establishment pays the transfer the rule collects above the administered price, and the parent zone receives it. Workers' welfare changes by the workers' problem (6); the first term is paid by employers, whose profits fall by the same amount to first order, so only the wedge term remains. The fence's taxes are a receipt of governments; with the share $1 - \alpha_F$ of the fence's activity located elsewhere in the country without the zone, the national receipt is $\alpha_F T_F$, and the same share scales the fence's jobs. An agglomeration shift η raises the profits of establishments within 10 km by $\eta\Pi_E$ to first order. The exporters' gain is that of Section 6.2. Summing, transfers cancel and the remaining terms are (10). Each is a change in a quantity times a wedge, the form of Harberger (1964): in a market without a wedge a small change in quantity has no first-order effect on welfare. *Decomposition.* The efficient allocation chooses a set of applicants to maximize their total value subject to the land. With plots small relative to $\bar{K}$, admitting applicants in order of value per hectare solves this problem. No rule that uses the same land therefore turns away less value. *First and second order, equation* (T5). With Pareto profits above F, equal land per establishment and a signal that ranks by profit, the applicants turned away have profits in $[F, \bar{\pi})$ with $\bar{\pi} = Fr^{-1/\zeta}$. Their profit net of the fixed cost, as a share of all operating profit, $\zeta F/(\zeta - 1)$ per entrant, is

$$\mathcal{D} = \frac{\zeta - 1}{\zeta} \int_1^{r^{-1/\zeta}} (x - 1)\, \zeta x^{-\zeta - 1} \, dx = \frac{1 - r}{\zeta} + r - r^{1-1/\zeta}.$$

Expanding $r^{1-1/\zeta}$ to second order in $1 - r$ gives $\mathcal{D} \approx (\zeta - 1)(1 - r)^2/(2\zeta^2)$; the derivative at $r = 1$ is zero. A signal unrelated to profit, or random rationing of a whole tier, turns away a

random share $1-r$, whose profit net of fixed costs is the share $1-(\zeta-1)/\zeta = 1/\zeta$ of their operating profit, so the cost is $(1-r)/\zeta$. Under a charge, the burden on potential entrants is $\mathcal{C}+\mathcal{D} = r^{1-1/\zeta} - r + \mathcal{D} = (1-r)/\zeta$.

Proposition 1, retention. An incumbent of fifth b is reviewed with probability χ. If reviewed, it keeps its plot for sure in the high tier and with probability e in the low tier, so it loses the plot with probability $(1-e)(1-h_b)$. The share of fifth b that keeps its plots is $1-\chi(1-e)(1-h_b)$. When the review uses current rather than past productivity and ranks are correlated ϕ over time, $s \mid z_0 \sim N(\tau\phi z_0, 1-\tau^2\phi^2)$ for an incumbent classified by its earlier rank z_0, and h_b is evaluated at $\tau\phi$; the simulation checks of the replication files confirm both forms. (b) By (4), an incumbent that loses its plot relocates if $\pi_{jo} \geq F+m$. If an incumbent's profit elsewhere rises with its current profit, the displaced that relocate are the largest of the displaced. Review displaces incumbents of every size within the low tier, so relocating incumbents come from the upper part of the size distribution of the low tier.

Exporters' gain (Section 6.2). With export profits Pareto with tail ζ above the fixed cost, the number of exporters is proportional to $f^{-\zeta}$, so $\Delta \ln N = -\zeta \Delta \ln f$. Net export profits are $V^X = N\left(\mathbb{E}[\pi^X \mid \pi^X \geq f] - f\right) = Nf/(\zeta-1) \propto f^{1-\zeta}$, and base exports are $X_0 = \sigma N_0 f_0 \zeta/(\zeta-1)$, so $V_0^X/X_0 = 1/(\sigma\zeta)$ and $\Delta V^X/X_0 = (e^{-(\zeta-1)\Delta\ln f}-1)/(\sigma\zeta)$. The incumbent exporters keep their gross profits and save $f_0(1-e^{\Delta\ln f})$ each, which in units of X_0 is $(\zeta-1)(1-e^{\Delta\ln f})/(\sigma\zeta)$ since $N_0 f_0/X_0 = (\zeta-1)/(\sigma\zeta)$; the remainder is the surplus of new exporters. The gain is a welfare gain of the province's exporters if the province is small, fixed costs use real resources, profits accrue in the province, and the fixed cost is the only channel.

Workers (Section 6.4). The two closures differ in what clears the labor market. Under flexible wages, $q_\ell = 1$, location shares are (T4), and a change in labor demand moves the wage by the inverse of ε in a small location; the marginal worker is indifferent, so new jobs carry no gain. Under rigid wages, employment is labor demand, the share of applicants hired is $q_\ell = L_\ell^d/L_\ell^s$, and expected utility in ℓ is $q_\ell B_\ell w_\ell u_{i\ell} + (1-q_\ell)B_O w_O u_{iO}$; the location shares are no longer of the Fréchet form, but the welfare effect of an additional job is the gain of the worker who fills it, $g_\ell w_\ell$. The account uses only that gain, so it does not need the location shares under queues. Let $\bar{L}$ workers choose among the areas and the outside option, with utility $B_\ell w_\ell u_{i\ell}$ and Fréchet tastes. Without a wedge, U in (T4) is each worker's expected utility; differentiating, $d\ln U = \sum_\ell (L_\ell/\bar{L})\, d\ln w_\ell$ (the change in amenities is zero), and in money $\bar{L}U\, d\ln U = \sum_\ell w_\ell L_\ell\, d\ln w_\ell$ when U is normalized to the average wage. Now let jobs in area ℓ be rationed, so that a worker who would choose ℓ obtains a job there only with some probability, and let the marginal worker who obtains one gain $g_\ell w_\ell$ over her outside option. A job added in ℓ goes to such a worker, so it raises welfare by $g_\ell w_\ell$, and

$dW^L = \sum_\ell w_\ell L_\ell \, d\ln w_\ell + \sum_\ell g_\ell w_\ell \, dL_\ell$. (b) Taking logs of the share in (T4) for area E and the outside option and differencing, $d\ln L_E - d\ln L_O = \varepsilon(d\ln w_E - d\ln w_O)$ when amenities do not change. (c) Employment is the sum of the employment of establishments: the fence's, the incumbents' and the entrants'. The rule removes the employment of the applicants turned away, except for those that enter elsewhere, which none do when $\rho = 0$.

T.3 The Later-Treated Comparison

The comparison cells lie at the same distances, by ring, from special customs zones that opened after the survey ends in 2007, so their surroundings are untreated in our years. Ring-by-year effects absorb the change common to treated and comparison cells at each distance. Outcomes are set to missing in cells that are neither within 150 km of a zone in our sample nor within 150 km of a later zone, so that every coefficient compares treated cells with later-treated cells at the same distance. The covariance of the moments stacks the prefecture-level influence functions of the separate Poisson regressions, scaled to match each regression's clustered variance.

T.4 Sorting by Productivity

The step does not follow from sorting. If the opening drew relatively more productive firms toward the zone, as complementarity between productivity and place would (Gaubert 2018), entry would tilt toward the top fifths as a ramp. Adding a tilt $\kappa(b-3)$ to the entry of fifth b gives $\kappa = 0.10$ and lowers J by only 0.6, far below the 3.84 that one more parameter would need. The data therefore do not ask for sorting, but they cannot exclude it, and it matters: with the tilt, the low tier is admitted with probability 0.71 and the price of the fence falls to 50 percent of the baseline. One further moment speaks against sorting. Complementarity would raise the sales of the most productive incumbents more than those of others, but the top fifth's sales per remaining incumbent change by 0.08 (0.11), the same as all incumbents' 0.08.

T.5 Assumptions behind the Welfare Results

Eight assumptions carry the numbers of Section 8. The main text states each where it is used; here we collect them with the direction in which the results move if each fails.

1. *No rationing before the openings.* The price of the fence is the change in the value turned away, so it does not depend on this assumption. The split between scarcity and

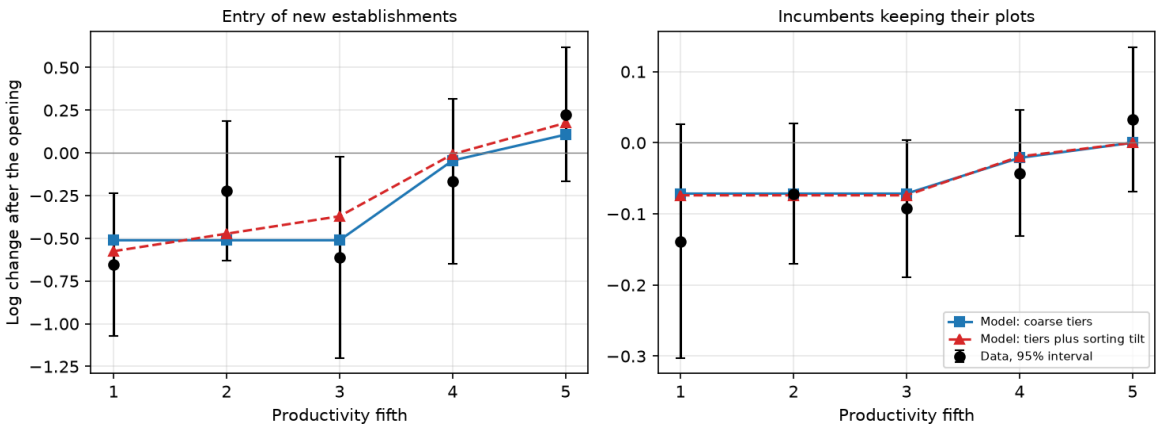


Figure T1: Entry and retention by productivity fifth: coarse tiers with and without a sorting tilt
Notes: As Figure 5; triangles add a tilt in entry across fifths, $\kappa(b-3)$, that sorting of more productive firms toward the zone would produce.

the rule does: if the parent zones already rationed, part of what we attribute to the rule predates the fence, and the rule's share is an upper bound.

2. *Applicants are representative of the pool by productivity.* Proposition 1 needs the opening to change how many firms apply, not which. If it also drew relatively more productive firms toward the zone, a tilt the data neither require nor exclude (Section 7.2), the low tier was turned away less often and the price of the fence would be about 1.8 billion yuan, 50 percent of the baseline.

3. *Gross values.* Values are gross of fixed costs, which we do not observe; net of them, the price is at least 3.2 billion yuan.

4. *Random rationing within the low tier.* If the parent zone ranked the low tier partly by value, the price would lie between the feasible allocation and the estimated rule, so the estimate is an upper bound given the tiers.

5. *Values of comparable entrants nationwide over four years.* Admitted establishments near the zones survive somewhat longer than the national cohort, and small entrants keep growing after age three; both would raise the price of every rule that ranks applicants at entry.

6. *Entry within 10 km measures admission to the parent zone.* Entry within 10 km but outside the parent zone, if not rationed, dilutes the step across fifths, which makes the low tier appear more often admitted than it was and the price of the fence too small.

7. *Where the turned-away went.* They did not enter in neighboring prefectures (at most 0.07 of them); entry elsewhere in the host prefecture cannot be ruled out and would lower the price, as the half re-entry scenario shows.

8. *Partial equilibrium.* Prices outside the prefecture, the pool of applicants and the fence's own activity are held fixed; the labor wedge and the share of the fence's activity new to the country are ranges, and the account is reported across them.